Distributional Dynamics[*]

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Abstract

We develop a method to estimate joint distributions at quarterly (business-cycle) frequency by combining heterogeneous microdata sources. The method recovers the full joint distribution even when surveys observing all variables jointly are infrequent and other sources observe only subsets. We treat distributions as functional time series, summarize their dynamics through latent factors, and estimate a Bayesian state-space model accommodating differences in survey frequency, coverage, and measurement. Combining U.S. microdata and macroeconomic time series, we construct the first quarterly estimates of the joint distribution of household consumption, income, and wealth from 1962 to 2024. Validation exercises demonstrate reliability despite sparse observations.

Keywords: *Consumption, income, and wealth inequality; Macroeconomic dynamics; Dynamic state-space model; Functional time-series data; Bayesian statistics*

JEL Classification: *E21, E32, E37, D31, C32, C55*

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[¶] The replication code and distributional time series are available at: **distributionaldynamics.com**.

1 Introduction

Macroeconomic dynamics emerge from the decisions of heterogeneous economic agents, including workers, households, and firms. A central development in modern macroeconomics has been to build increasingly rich microfoundations that account for differences in agents' economic circumstances and their implications for aggregate outcomes. In heterogeneous-agent models, macroeconomic dynamics depend on the joint distribution of individual states: Agents' positions along multiple dimensions shape how they respond to shocks and how these responses aggregate.[1]

This development in economic theory has moved ahead of measurement. In the spirit of Prescott (1986), progress in quantitative macroeconomics rests on continuous feedback between theory and measurement: theory identifies economically relevant objects, while measurement provides the empirical discipline for further theoretical progress. Heterogeneous-agent macroeconomics has reached such a point: Models characterize rich distributional states and their evolution at business-cycle frequency, while their empirical counterparts are typically observed only infrequently and incompletely. This measurement gap limits our ability to confront these theories with the distributional dynamics at the core of their mechanisms. Quantitative models are therefore often disciplined and evaluated largely through steady-state cross-sections, selected moments, or aggregate time series rather than the distributional dynamics.[2] This paper closes this gap.

We develop a novel method to estimate the *distributional dynamics* of the macroeconomy. The idea is to treat the joint distribution as a time series of functions and summarize its dynamics through low-dimensional factor structures embedded in a state-space model. This allows us to combine multiple sources of micro- and macro-data and recover the high-frequency evolution of the distribution even when the underlying microdata are sparse, observed at different frequencies, or cover only subsets of the variables of interest. The new method is general and therefore applicable to a broad class of joint distributions. Its basic data requirement is that the variables defining the joint distribution be observed together in at least one source over several, potentially infrequent, periods.

A second contribution addresses a distinct measurement challenge that arises when drawing on multiple microdata sources: different datasets typically measure the same underlying economic object using different sampling designs, definitions, and measurement concepts. Rather than assuming that these sources coincide, we treat them as complementary, imperfect measurements of a common underlying distribution and explicitly allow for sampling uncertainty, measurement error, and persistent or time-varying differences in operationalization. The novel method thereby produces *consensus estimates* of the underlying distributional dynamics from the heterogeneous information available to the researcher.

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[1] See, e.g., Kaplan, Moll, and Violante (2018), Bayer, Luetticke, Pham-Dao, and Tjaden (2019), Auclert (2019), and Bilbiie (2020), and, for the cross-sectional effects of policy, Mian, Straub, and Sufi (2020), Holm, Paul, and Tischbirek (2021), Andersen, Johannesen, Jørgensen, and Peydró (2023), and Bhandari, Evans, Golosov, and Sargent (2021). Kaplan and Violante (2018) and McKay and Wolf (2023) survey this literature.

[2] An exception is provided by carefully designed environments with block recursivity, where equilibrium objects can be characterized without keeping track of the full cross-sectional distribution (Menzio and Shi 2011). Our focus is the broad class of heterogeneous-agent models in which the distribution is part of the equilibrium state.

The paper's third contribution is to apply the method to construct high-frequency estimates of the *joint distribution* of consumption, income, and wealth for the United States, delivering the first quarterly estimates of this joint distribution from 1962Q3 to 2024Q1. We make these estimates publicly available at distributionaldynamics.com, alongside quarterly distributions of household balance-sheet items (real estate, mortgages, consumer credit, stocks, and pension assets), extending the type of information provided by the Distributional Financial Accounts (DFAs) (Batty et al. 2020). The website also provides the estimated factor structure, posterior draws, and basis coefficients, allowing researchers to construct distributional statistics of interest. The replication package also enables extensions to other economically distributionally relevant joint distributions over other economic agents, e.g., firms. Together, these resources provide new empirical targets for evaluating and developing quantitative theories of heterogeneity, and we plan to expand them over time.

Turning to the details of the method, we adopt the understanding that joint distributions are functional data and hence infinite-dimensional objects. Because describing their dynamics requires more than tracking a finite set of moments, it poses a statistical challenge. Our method, which builds on the contribution of Chang, Chen, and Schorfheide (2024), addresses this challenge by projecting these infinite-dimensional objects onto a finite set of basis functions and using functional principal component analysis to uncover a low-dimensional factor structure in their dynamics. The resulting factors form the distributional states of the state-space model. Importantly, the number of factors is not fixed by the method, but chosen for each application from the data.[3] Nothing in the method requires the number of retained factors to be small. If an application calls for a rich factor space, the method carries through unchanged, albeit at greater computational cost.

Heterogeneous-agent theory, however, provides a rationale for why rich distributional dynamics may admit a low-dimensional representation: household decisions respond to a limited set of aggregate equilibrium objects, including wages, asset returns, taxes, and transfers (Auclert, Bardóczy, Rognlie, and Straub 2021; Bayer, Born, and Luetticke 2024). These objects are closely tied to the macroeconomy, which itself admits a low-dimensional factor structure (Stock and Watson 2002). Reduced-form evidence also supports a close link between distributions and aggregate prices.[4] Consistent with these findings, we also find that a small set of aggregates achieves the best model fit, coupled with eight distributional factors that capture 99% of the business-cycle-frequency variation in the distributional data, with negligible approximation error relative to observed microdata.

Given this factor representation, the state-space model allows us to estimate the distributional dynamics using Bayesian techniques and link them to aggregate variables without imposing a fully structural macroeconomic model. Its measurement equation provides the flexibility needed to combine micro-datasets that measure the same underlying object but differ in sampling uncertainty, frequency, variable coverage, and measurement concepts. It also allows datasets containing only subsets of the variables of interest to inform the underlying joint distribution (Schorfheide and Song 2015; Durbin and Koopman 2012). The Kalman smoother then recovers high-frequency movements in the complete joint distribution, including periods in which direct microdata observations are unavailable.

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[3] For the selection of distributional and aggregate factors, one can rely on comparing several factor-selection criteria (Ahn and Horenstein 2013; Bai and Ng 2002; Freyaldenhoven 2022), which only requires one run of the estimation. We chose the number of factors based on comparing model fits via marginal data densities (MDD), which requires more estimations but is disciplined by the outcome of the model. Remarkably, we find that both approaches point toward the same number of factors, around 8-10 factors.

[4] Papers on the price-distribution nexus include Di Maggio, Kermani, and Majlesi (2020), Chodorow-Reich, Nenov, and Simsek (2021), and Kuhn, Schularick, and Steins (2020).

We apply our method to the joint distribution of household consumption, income, and wealth in the United States. Observing all three variables jointly is central to confronting heterogeneous-agent theory with the data. Households with similar current incomes can occupy very different positions in the wealth distribution, such as a young household with little accumulated wealth and a retired household with substantial wealth. Their different wealth buffers and adjustment margins may lead to very different consumption responses to the same shock (Carroll, Hall, and Zeldes 1992). More generally, economically relevant household types are characterized by combinations of states rather than individual marginals, as illustrated by the wealthy hand-to-mouth (Kaplan, Violante, and Weidner 2014). The joint distribution thus provides the empirical counterpart needed to characterize heterogeneous consumption dynamics and confront quantitative theories with their distributional predictions (Auclert 2019; Bilbiie 2020; Patterson 2023; Bilbiie et al. 2025).

Furthermore, the available U.S. microdata illustrate precisely the measurement problem addressed by the paper's second contribution. No single survey provides comprehensive information on consumption, income, and wealth at high frequency over a long time horizon. We combine the *Panel Study of Income Dynamics* (PSID), *Survey of Consumer Finances* (SCF), *Consumer Expenditure Survey* (CEX), *Survey of Income and Program Participation* (SIPP), and *Current Population Survey* (CPS), complemented by a comprehensive set of macroeconomic time series. The PSID satisfies the requirement that all variables defining the joint distribution are observed together over several periods, while the other datasets provide complementary information at different frequencies and with different variable coverage. The datasets also differ substantially in sample design, size, and measurement concepts. Our method combines this heterogeneous information to construct a consensus estimate of the quarterly joint distribution of consumption, income, and wealth, capturing changes in both the marginal distributions and their cross-sectional dependence.

Our method delivers estimates of the joint distributions even in periods without direct microdata observations. To assess its empirical performance and establish the reliability of the resulting estimates, we subject the procedure to several validation exercises. First, we show that the factor representation entails almost no loss of information relative to the observed microdata, preserving economically relevant features of the distributions, including their tails, in the estimation.. Second, we test whether the model can recover distributional data that are deliberately treated as unobserved, exploiting their comovement with aggregate variables. We demonstrate this for the consumption distribution in the CEX and the wealth distribution in the SCF. Third, we validate that the reported dynamics are indeed supported by the micro- and macrodata, as opposed to the model dynamics. Even when data are sparse, the modeled dynamics are in the neighborhood of other external sources, e.g., the DFAs.

In a final exercise, we use simulated data from a theoretical HANK model as a data generating process. In this setting, the true high-frequency distribution is known, allowing us to show that our method closely tracks its movements when supplied only with microdata samples of the size and frequency available in the U.S. data.

Given the encouraging validation results, we use the estimated dynamic factor model to estimate the contributions of the individual data sources and report, for each quarter, how much of it is anchored in observed data rather than inferred through the state-space model. This provides transparent information about the empirical provenance of the estimates, alongside measures of their uncertainty.[5]

In the last part of the paper, we return to the paper's initial motivation and use the new distributional time series to close the loop between theory and measurement. As a first use case, we provide empirical counterparts to the distributional objects that feature in theories of consumption dynamics in recessions. We ask whether consumption dynamics are similar across downturns by tracing changes in relative consumption along the joint distribution of income and wealth during the last three U.S. recessions. These objects arise naturally in heterogeneous-agent models, but until now their empirical counterparts have not been observed at business-cycle frequency over a long horizon spanning several recessions with markedly different underlying drivers. Our estimates therefore provide empirical targets against which theories of heterogeneous consumption dynamics can be evaluated.

The results reveal clear and substantial differences across recessions. The household groups experiencing the largest relative consumption changes differ across episodes and, importantly, cannot generally be identified from either income or wealth alone. Wealth *and* income differences were particularly important during the Dot-com recession; wealth more important around the Global Financial Crisis, and again the combination of income and wealth during the COVID-19 recession. The distributional transmission of aggregate shocks therefore appears strongly state- or shock-dependent: which households experience the largest relative consumption changes, and which dimensions of their economic position matter most, varies substantially across recessions. This complements Bilbiie, Galaasen, Gürkaynak, Mæhlum, and Molnár (2025), who show that automatic stabilizers render household heterogeneity largely irrelevant for aggregate dynamics on average in Norway. Our results for the U.S. suggest that such averages can mask substantial variation across episodes, consistent with the time- and state-dependent role of heterogeneity emphasized by McKay and Wolf (2023).

The new estimates also show that aggregate innovations play an important role in the distributional dynamics. They account for between a third and a half of the *high-frequency* fluctuations in the distributional factors, in the marginal distributions of consumption, income, and wealth, and in the dependence structure linking them. Aggregate dynamics therefore shape not only how these variables are distributed separately, but also how they co-move across households, complementing the structural findings of Bayer, Born, and Luetticke (2024). This latter dimension is particularly relevant in incomplete-markets economies: wealth provides a buffer against income risk, so households with similar incomes but different wealth holdings may differ substantially in their ability to self-insure (Krueger, Mitman, and Perri 2016). The joint distribution captures these differences and how their prevalence changes over the business cycle, information invisible in the marginals alone.

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[5] We also validate the choice of priors in the Bayesian estimation, particularly with respect to measurement error, and show that the state-space model is consistent with the sampling uncertainty of the observed distributional data at the sampling points. We further compare the predicted dynamics of the income and wealth distributions with those implied by the Distributional Financial Accounts methodology (DFA, see Batty et al. 2020) and those estimated by the World Inequality Database (Alvaredo et al. 2016; Piketty, Saez, and Zucman 2018).

The relative importance of aggregate and autonomous distributional innovations also differs across recessions. Historical decompositions across different NBER recessions show the relative importance of aggregates is higher, about 55 percent on average, so distributional dynamics track aggregate innovations more closely around recessions than over the cycle as a whole. More broadly, this first application already illustrates some of the new possibilities opened up by measuring distributional dynamics at business-cycle frequency, as these new measurements provide previously missing evidence to distinguish mechanisms, discipline quantitative models, and study how shocks and policies reshape the distribution of economic activity across heterogeneous agents.

The remainder of this paper is organized as follows: Section 2 provides an overview of the relevant literature. Section 3 develops our estimation method and Section 4 evaluates its quality. Section 5 presents two applications of the method. Finally, Section 6 concludes the paper. An appendix follows.

2 Literature

Relevant methodological papers. The paper most closely related to ours is the important contribution of Chang, Chen, and Schorfheide (2024), which develops a Bayesian state-space approach to estimate the coevolution of technology shocks, GDP, employment, and the marginal distribution of earnings. Their earnings distribution is observed in a single, data-rich source (the CPS, with 60,000 observations), at a single frequency, and without missing observations (1989–2017)—conditions that permit carrying the functional coefficients (to be explained in Section 3) directly as states. None of this carries over to our problem, which leads our work to differ in three respects.

First, our application considers the *joint* distribution of consumption, income, and wealth, together with a richer set of aggregates. For heterogeneous-agent macroeconomics, economically relevant information lies in the dependence structure—which balance sheets absorb an income loss, whose consumption responds to a wealth revaluation—and representing it raises the dimension of the functional coefficients from dozens to thousands. Carrying them as states becomes infeasible. Following Kneip and Utikal (2001) and Tsay (2016), we instead estimate the state-space model in a lower-dimensional factor space obtained through functional PCA (see also Meeks and Monti 2023, for a macroeconomic application).

Second, no single dataset continuously observes the joint distribution in our application. We combine five surveys that differ in variable coverage, frequency, sampling design/size, and measurement concepts, so missing observations and mixed frequencies are the rule rather than the exception—requiring the mixed-frequency measurement framework with dataset-specific, design-based error structures at the core of the method.[6]

Third, our application also delivers a new public data product: a quarterly panel of joint distributions for the United States from 1962 to 2024—as factor estimates, coefficients, or microdata—with posterior uncertainty and source attribution for every estimate, previously unavailable to researchers. The method is also made accessible, including the code to run the method, so users can plug in their own micro- and macrodata inputs and estimate their own joint distributions.

Another relevant contribution is Marcellino, Renzetti, and Tornese (2024), who nowcast the distribution of *total* household income, also from the CPS, using high-frequency macroeconomic and financial indicators: each annual distribution is represented by a kernel-smoothed log-quantile-density (Petersen and Müller 2016), reduced by a truncated Karhunen–Loève expansion, and predicted by mixed-data-sampling (MIDAS) regressions. The first two differences above apply here as well; in addition, we obtain now- and backcasts as smoothed states of the mixed-frequency state-space model of Schorfheide and Song (2015), rather than as regression predictions.

Our representation itself differs from both papers because it must accommodate a multivariate distribution and different datasets informing the *same* object. We represent that distribution using orthonormal polynomials. This choice has three important benefits. It turns the mathematical restrictions needed for a valid distribution into simple restrictions on the coefficients (again, explained later); it lets datasets with only partial information contribute to the same set of underlying coefficients as datasets with complete information; and it provides a natural way to compare distributions when extracting factors. Additionally, compared with methods based on log-densities or log-quantile densities, our rank-based representation keeps many economically relevant quantities linear and avoids kernel smoothing. This means that all of the available survey data can be combined within one linear filtering framework.

More broadly, our method builds on the literature that formulates temporal disaggregation in a state-space framework (see e.g., Harvey and Chung 2000) and on functional data methods in economics (see e.g., Kneip and Utikal 2001; Diebold and Li 2006; Chang, Kim, and Park 2016). The key advantage over (primarily static) methods such as Chow and Lin (1971) and Fernandez (1981) is that the state-space representation is inherently dynamic and lends itself to the Kalman filter for estimation and diagnostics.

Macroeconomic motivation. Our contribution relative to the growing body of work linking macroeconomic aggregates with distributional dynamics (see e.g., Baumeister et al. 2025; Koop et al. 2026; Ettmeier, Kim, and Schorfheide 2024) is to recover the dynamics of the full joint distribution—including its marginals and dependence structure—from multiple partially overlapping surveys at mixed frequencies. This provides the still-missing descriptions of the short-run dynamics of the consumption, income, and wealth distribution to the theoretical literature on heterogeneity and aggregate dynamics (Kaplan, Moll, and Violante 2018; Bayer et al. 2019; Bayer, Born, and Luetticke 2024) and extends the rapidly growing empirical work on the effects of policy shocks on marginal distributions.[7]

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[6] While there is a nascent literature (see, e.g., Ettmeier, Kim, and Schorfheide 2024) emphasizing the role of unit-level dynamics in properly addressing the mixed-frequency problem, we abstract from this consideration in the present study and follow the standard mixed-frequency approach.

Through our application, we also speak to a central debate on macroeconomic amplification from household inequality. Auclert (2019) and Bilbiie (2020) formalize the idea that amplification depends on the covariance between the marginal propensity to consume and the cyclicality of income. Patterson (2023) finds this covariance to be strongly positive for U.S. labor earnings, creating a powerful "matching multiplier"; recent evidence from Bilbiie et al. (2025) for Norway provides a counterpoint, showing that once disposable income is considered, automatic stabilizers render the covariance negligible over *average* business cycles. Our estimates show that distributional consumption dynamics differ *strongly* across recessions, suggesting that average business-cycle responses can mask substantial variation across episodes.

Distributional measurement. The paper also contributes to a large empirical literature, beginning with Piketty and Saez (2003), measuring trends and fluctuations in inequality. Existing high-frequency distributional datasets primarily focus on income or wealth, but do not provide the joint distribution of consumption, income, and wealth. Blanchet, Mouillet, Saez, and Zucman (2026) construct monthly income and wealth distributions for the United States; Smith, Zidar, and Zwick (2023), using administrative data and the capitalization method of Saez and Zucman (2016), provide high-frequency wealth estimates concentrated on the upper tail. The Distributional Financial Accounts (Batty et al. 2020) produce quarterly balance-sheet estimates since 1989 by combining SCF microdata with the Financial Accounts through Chow-Lin/Fernández-type models. Our state-space approach offers three advantages over these methods: it treats the underlying microdata as noisy samples of a time series of distribution functions, explicitly handling sampling uncertainty in a dynamic setting; it can combine multiple microdata sources with different operationalizations of the same economic concepts; and it recovers the *joint* distribution of consumption, income, and wealth at business-cycle frequency going back to the 1960s.

3 Method

This section describes a general method for generating high-frequency estimates of joint distributions of economic variables of interest over a large number of micro-units. The method proceeds in four estimation stages, followed by reconstruction, as summarized in Figure 1 and based on the idea that the dynamics of distributions are functional time-series data (Section 3.1). First, starting from microdata, we estimate and represent joint distributions by their quantile functions coupled with a copula, isolating marginal shapes from their dependence structure and projecting these functions onto Legendre polynomials (Section 3.2.1). Second, we reduce dimensionality using functional principal component analysis (PCA) (Section 3.2.2). Third, we formulate the dynamics of these latent distributional factors, along with aggregate factors, as a state-space model that handles mixed frequencies (Section 3.2.3). Fourth, we estimate the sys-

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[7] See, for example, Berger, Bocola, and Dovis (2023), Coibion et al. (2017), Cloyne, Ferreira, and Surico (2020), Holm, Paul, and Tischbirek (2021), Chang and Schorfheide (2024), and Bartscher et al. (2022). McKay and Wolf (2023) survey this literature.

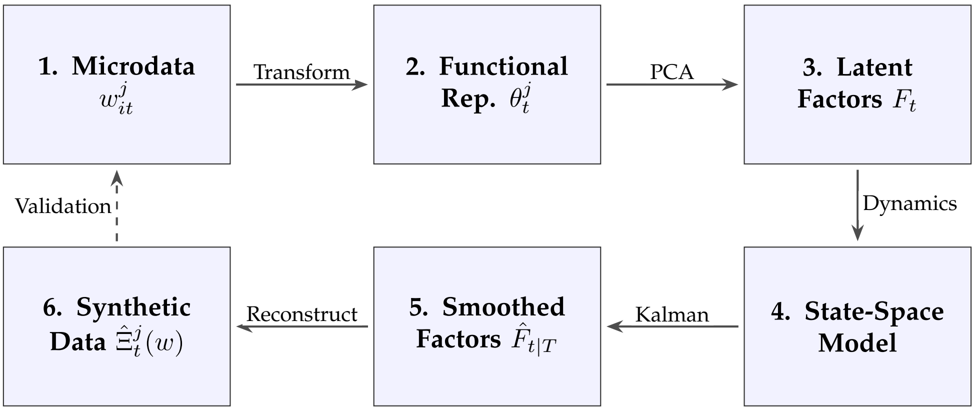

Figure 1: Visual Roadmap of the Methodology

tem using Bayesian techniques, explained in Appendix C, and recover the full distributional time series via the Kalman smoother. The reverse process constructs the high-frequency synthetic distributional data, explained in Appendix D. Section 3.3 describes how one can use the data and how one can use the method. In the following, we describe the method using the example of the joint distribution of consumption, income, and wealth, an important macroeconomic application.

3.1 Distributions as time series of functional data

We consider a sequence $\Xi_t(w)$ of multidimensional distribution functions, indexed by $t \in \mathbb{T} := \{1 \dots T\}$, defined over a d-dimensional *vector* $w \in \mathbb{R}^d$. In the case of our application, we have $d = 3$, where w is a vector of consumption, income, and wealth at the household level. In addition to this sequence of distribution *functions*, there is a sequence of real-valued vectors $Y_t \in \mathbb{R}^q$ of latent aggregate factors, measured with noise by a vector $\hat{Y}_t$ of stationary aggregate data (Section 4.1). Throughout, Y_t denotes the latent aggregate factors and $\hat{Y}_t$ their measurements. In the following exposition, we assume that $\hat{Y}_t$ is observed at all times $t \in \mathbb{T}$. The extension to missing observations in $\hat{Y}_t$ is standard.

From the distributions $\Xi_t(w)$, we observe only randomly drawn samples. We allow these samples to come from different sampling procedures or to have different operationalizations of the underlying theoretical variables. For example, the PSID and the SCF use different sampling procedures and slightly different concepts of wealth and income. We index each of the sampling procedures/datasets by $j = 1 \dots J$. All of these different datasets are typically not observed for all time periods. Instead, dataset j is only observed in a particular subset $\mathcal{T}^j \subset \mathbb{T}$. In addition, not all samples contain all variables of interest, but may contain only a subset $\mathcal{D}^j \subseteq \mathbb{D} := \{1, \dots, d\}$ of variables. For example, the Current Population Survey (CPS) provides only income information but neither wealth nor consumption. However, at least one dataset j must contain all variables of interest, i.e., $\mathcal{D}^j = \mathbb{D}$. In our application, this dataset is the PSID, which contains information on consumption, income, and wealth (for some years).

Our goal is to obtain estimates of the joint distribution functions, $\hat{\Xi}_t(w)$, $\forall t \in \mathbb{T}$, by efficiently combining information from the various related microdata sources and aggregate information,

Y_t. We assume that there is a time-series structure such that the density $d\Xi_t$ evolves according to the functional difference equation

$$d\Xi_{t+1} = G\left(d\Xi_t, Y_{t-\ell:t}\right) + \epsilon_{t+1}, \tag{1}$$

where $Y_{t-\ell:t}$ are observed lagged aggregate data.[8] G determines the dynamics of the system, and ϵ_t are the corresponding shocks to the functional equation.[9] A time-series structure of this kind arises, for example, in heterogeneous-agent New Keynesian (HANK) models (see e.g. Kaplan, Moll, and Violante 2018; Bayer, Born, and Luetticke 2024). It is assumed that Ξ_t is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^d$, so that a density $d\Xi_t$ exists; we additionally assume $d\Xi_t$ to be continuous. Moreover, we require that the operator G is Lipschitz-continuous in its first argument, a condition satisfied mechanically under the proposed linear model. We provide details below.

Viewing the J sampling procedures as capturing the same fundamental object Ξ_t but with some measurement error, ν_t^j, allows us to combine the data in a systematic way. This means that a dataset gives us an estimate

$$d\tilde{\Xi}_t^j = \int_{\mathbb{R}^{\mathbb{D}\setminus\mathcal{D}^j}} d\Xi_t + \nu_t^j \qquad \text{for} \quad t \in \mathcal{T}^j, \tag{2}$$

where the integral runs over the components of w unobserved in dataset j—those indexed by $\mathbb{D} \setminus \mathcal{D}^j$—which are thereby integrated out.[10] The measurement error then captures sampling uncertainty, differences in sampling procedures, and differences in operationalization of economic concepts, all of which vary by dataset j.

3.2 Implementing the Estimation

Estimating (1) directly is not feasible because it is an infinite-dimensional nonlinear functional difference equation and, of course, we only observe samples of the distribution functions, not the functional data itself. Our innovation is to overcome this challenge by making it possible to estimate (1) using traditional Bayesian techniques and a Kalman-filter (Appendix C). This requires transforming (1) into a linearized (infinite-dimensional) state-space model (Section 3.2.3), which is estimable once we render it finite and of reduced dimensionality under an appropriate factor representation (Section 3.2.2). However, first, we need to operationalize the measurement of the distribution functions as they appear in (2) by transforming the microdata samples into estimates of the distribution functions themselves (Section 3.2.1). In doing so, we have to account for changes in the effective support of the distributions and deal with the unobservability of some of the micro-variables in some datasets. In the following, we cover these points, which ultimately define our estimation procedure.

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[8] We deliberately leave the law of motion of Y_{t+1} unspecified at this point; the system is completed in Section 3.2.3, where the aggregates follow their own transition equation and are neither assumed exogenous to ϵ_{t+1} nor to $d\Xi_t$.

[9] To ensure Ξ_{t+1} is a valid distribution: $\int G(d\Xi, \cdot)(w)\, dw = 1$, $\int \epsilon_t(w), dw = 0$, and $G(d\Xi, Y_{t-\ell:t})(w) \geq -\epsilon_{t+1}(w)$ (non-negativity). Non-negativity also applies to $d\tilde{\Xi}$ and thus restricts ν in (2) below.

[10] For clarification, the tilde ($d\tilde{\Xi}$) is to denote that this is an estimate from the data vs. an estimate generated from the model, which will be denoted by a hat ($d\hat{\Xi}$).

3.2.1 Transforming the microdata

Handling changes in scale One challenge in working with distributional data is that the magnitude of the variables of interest in w, and thus the support of $\Xi(w)$, changes over time. The population support of w may well be regarded as fixed; what varies is the *effective* support—the region containing all but a negligible fraction of the mass. Secular growth in incomes and wealth shifts and dilates the distribution, and the realized extremes in any finite sample expand or contract the range over which Ξ_t can be estimated. This is particularly important when considering long time-horizons.

This variation in the support poses two problems. First, over time, any finite-dimensional representation of Ξ_t requires a common domain. If *instead* one would estimate Ξ_t on time-varying domains, the finite dimensional description could no longer be posed consistently in the form of (1)—our object of interest. Second, across datasets *within* a period, sampling designs differ in their coverage of the tails, so dataset-specific estimates are defined on dataset-specific domains and cannot be interpreted as noisy measurements of a common underlying object, as the measurement equation (2) requires.

We deal with this in two ways. First, to deal with level changes, we rescale the vector w observed for individual i in micro-dataset j at time t, w_{it}^{j}, by its dataset- and time-specific mean $\bar{w}_t^j$.[11] Second, to deal with changes in the width of the effective support (e.g., due to changes in inequality), we change domain from levels to ranks. Assigning each household its rank in the cross-section—the poorest household always occupying the lowest rank and the richest the highest, regardless of the levels involved—maps every measure into the unit interval, a domain that is constant over time and across datasets by construction.

Two objects then summarize the distribution without loss of information. The mapping from ranks back to levels—the quantile function, i.e., the inverse of the marginal cumulative distribution function—recovers each marginal distribution, one function per measure. The joint distribution over ranks across measures—by definition, a copula—captures their dependence. Together, the quantile functions and the copula contain the same information as Ξ_t itself, but in contrast to Ξ_t, all of these functions are defined on the fixed domains $[0,1]$ and $[0,1]^d$: their supports are time-invariant by construction. Sklar's theorem (Sklar 1973) states this formally: any multivariate distribution function can be decomposed into its marginals and a copula:

$$\Xi_t(w) \;=\; C_t\big(\Xi_{1t}(w_1), \ldots, \Xi_{mt}(w_m), \ldots, \Xi_{dt}(w_d)\big), \tag{3}$$

where Ξ_{mt} denotes the marginal distribution of measure $m \in \mathbb{D}$ and the copula $C_t : [0,1]^d \to [0,1]$ is itself a distribution function with uniform marginals—unique under our maintained assumption that Ξ_t is absolutely continuous. The copula isolates the dependence structure (e.g., whether high-income households also tend to hold high wealth) from the shapes of the individual distributions (e.g., how skewed income is), and it is invariant to strictly increasing transformations applied componentwise to w—in particular, the inverse-hyperbolic-sine treatment of the marginals introduced below leaves the copula unchanged.

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[11] Estimating relative to dataset-specific time means also lets us flexibly align the synthetic distributions with per-capita or per-household aggregate targets: we simply rescale using the desired aggregate target rather than the dataset mean. A "consensus" business-cycle component can be obtained by rescaling with average fixed effects and omitting any trend.

For our application, the copula is

$$\begin{aligned} C_t(u_1, \ldots, u_d) &= P_t(U_1 \leq u_1, \ldots, U_m \leq u_m, \ldots U_d \leq u_d) \\ &= P_t\left(w_1 \leq \Xi_{1t}^{-1}(u_1), \ldots, w_m \leq \Xi_{mt}^{-1}(u_m), \ldots, w_d \leq \Xi_{dt}^{-1}(u_d)\right) \qquad \forall t \in \mathbb{T} \end{aligned} \tag{4}$$

where $U_m \sim U[0,1]$ for $m \in \mathbb{D}$ are the uniform marginals generated by taking the probability integral transform of each component m s.t. $U_m = \Xi_m(w_m) \sim U[0,1]$. The second line highlights the quantile functions or the inverse transform of the univariate CDFs, $\Xi_{mt}^{-1}(u_m)$, where:

$$\Xi_{mt}^{-1}(u_m) = \inf\{w_m \in \mathbb{R} : \Xi_m(w_m) \geq u_m\} \qquad \forall t \in \mathbb{T}, m \in \mathbb{D}. \tag{5}$$

Finally, for C_t to be a copula, the constraint must hold for all $m \in \{1 \ldots d\}$ that when integrating out all but one dimension, i.e., setting all $u_j = 1$ except for $j = m$ (the marginal distribution), the copula is identical to the value of the marginal distribution:

$$C_t(1, \ldots u_m, \ldots 1) = u_m. \tag{6}$$

Dealing with the infinite-dimensionality of the distribution functions The second challenge in our endeavor is dealing with the infinite-dimensionality of the distribution functions, while maintaining a functional approach. To do so, we will rely on a series estimator for both the copula and quantile functions. Series estimators project the functions of interest onto some infinite-dimensional space of polynomials, namely the space of square-integrable functions $\mathcal{L}^2$.

In the following, we will work with the copula density dC_t, which contains the same information as C_t under absolute continuity and provides several advantages that we explain later. Treat copula densities dC_t and (transformed) quantile functions Ξ_{mt}^{-1} as elements of $\mathcal{L}^2$, and project them onto a space of shifted orthonormal Legendre polynomials $\{Q_o : o \in \mathbb{N}_0\}$ for some order o, satisfying

$$\int_{[0,1]} Q_o(x)\, Q_k(x)\; dx = \delta_{ok}, \tag{7}$$

with $\delta_{ok} = 1$ if $o = k$ (reflecting the normalization condition) and 0 otherwise by orthogonality. Classical Legendre polynomials are defined on $[-1, 1]$, so the shift is to the same domain as the copula densities and quantile functions—in the support $[0, 1]$. The orthonormal shifted Legendre polynomials Q_o themselves are defined on $[0, 1]$ by

$$Q_o(x) = \sqrt{2o+1}\, L_o(2x - 1), \tag{8}$$

where L_o are the classical Legendre polynomials defined by

$$L_0 = 1, \qquad L_1(x) = x, \qquad (o+1)L_{o+1}(x) = (2o+1)\, x\, L_o(x) - oL_{o-1}(x). \tag{9}$$

One concern is that the approximation method could be poorly chosen, as the underlying population functions are themselves not completely square-integrable. For example, empirical evidence for distributions of consumption, income, and wealth points to tail behaviour that violates the $\mathcal{L}^2$ condition (see, e.g., Vermeulen 2018).[12] For the copula density, the same issue arises—density at the tails may fail to lie in the fulldomain space $\mathcal{L}^2([0,1]^d)$ for copulas exhibiting non-zero tail dependence e.g., the Gumbel copula.

To ensure that these underlying functions lie in $\mathcal{L}^2$, rendering the projection apropos, we first treat the data and apply the inverse-hyperbolic-sine transformation on them. This regularizes the upper tail of the distribution. This treatment is similar in nature to working with log densities, as in Chang, Chen, and Schorfheide (2024).[13] We then estimate our quantile functions on these transformed values. Appendix A provides evidence that by using the transformed percentile functions, the projection performs well at capturing variation in the tails. For the copula reconstruction (after having estimated the model), we impose a *local* $\mathcal{L}^2$ condition on a trimmed domain $[0,1]^d$, for some very small $\varepsilon = 10^{-6}$. We assume that inside that interior cube, $dC_t(\mathbf{u})$ is bounded and hence square-integrable.[14] Appendix A also addresses the treatment of measures (e.g., stocks) in the presence of non-negligible zero-mass.

Altogether, this means the following representation:

$$\Xi_{mt}^{-1}(u_m) = \sum_{o\in\mathbb{N}_0} \xi_{mot} Q_o(u) \tag{11}$$

$$dC_t(u_1, u_2, ...u_d) = \sum_{(o_1,\ldots,o_d)\in\mathbb{N}_0^d} \kappa_{(o_1,\ldots,o_d),t} \prod_{m=1}^{d} Q_{o_m}(u_m), \quad u_m \in [0,1] \tag{12}$$

where both functions are represented as infinite sums of polynomials over the order of the polynomials $o \in \mathbb{N}_0$. Because of orthonormality (7) and using (11), for some fixed o, we obtain:

$$\int_0^1 \Xi_{mt}^{-1}\, Q_o(u)du = \int_0^1 \sum_{k\in\mathbb{N}_0} \xi_{mkt} Q_k(u)\, Q_o(u)du = \sum_{k\in\mathbb{N}_0} \delta_{ko}\xi_{mkt} = \xi_{mot}. \tag{13}$$

This means that the coefficients can be obtained as the inner products of the functions and the Legendre polynomials of some order o. The same holds true for the copula density:

---

[12]
$$\int_a^b \left[Q(p)\right]^2 dp = \int_a^b \Big[\underbrace{x_m\,(1-p)^{-1/\alpha}}_{\text{Pareto quantile}}\Big]^2 dp = x_m^2 \int_a^b (1-p)^{-2/\alpha}\, dp. \tag{10}$$

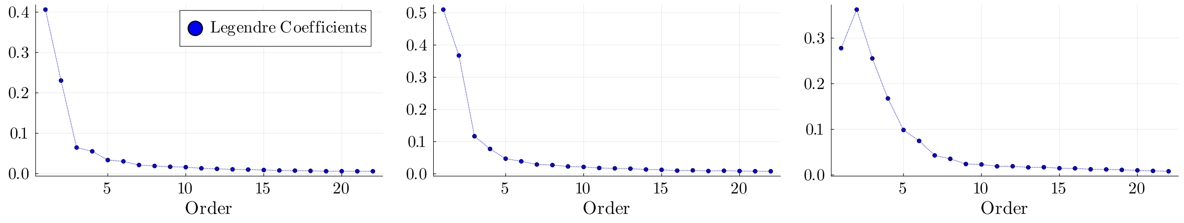
In particular, if the uppertail of the income distribution on $p \in [a,b]$ follows a Pareto law with shape α, then its quantile function is $Q(p) = x_m(1-p)^{-1/\alpha}$. This lies in $\mathcal{L}^2[0,1]$ (i.e. $\int_a^b Q(p)^2\,dp < \infty$) if and only if $\alpha > 2$. When $\alpha \le 2$, the integral diverges and the second moment does not exist.

[13] For notational convenience, we use $\Xi_{mt}^{-1}(u_m)$ as the treated quantile function.

[14] A simulation study shows that our chosen estimator minimizes the mean integrated squared error relative to other nonparametric approaches such as, beta-kernel smoothing, empirical copulas, and checkerboard methods, especially when fitting bi- and trivariate copulas with moderate to strong tail dependence, which may be the case here. See Bakam and Pommeret (2023) for more details.

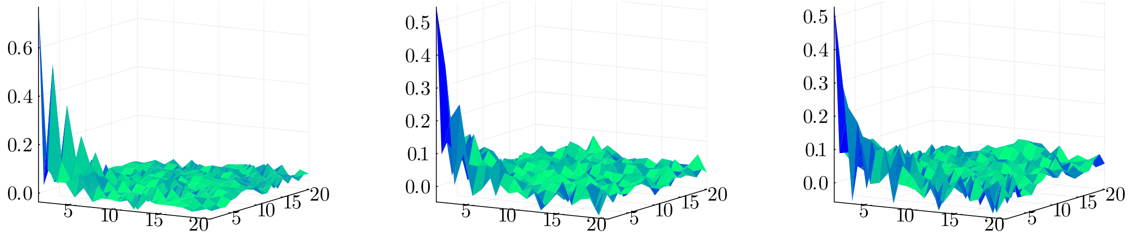
Figure 2: Legendre Coefficients Across the Distributional Data

Legendre Coefficients across Quantile Functions



(a) Consumption (b) Income (c) Wealth

Legendre Coefficients across Copulas



(d) Consumption & Income (e) Consumption & Wealth (f) Income & Wealth

Notes: The first row shows the coefficients (dots) on the Legendre polynomials from estimating the quantile function, by polynomial order. The second row shows the coefficients (surface) on the Legendre polynomials from estimating the copula density, by the polynomial orders of the two variables. Data are from the 2019 PSID.

$$\int_{[0,1]^d} \prod_{m=1}^{d} Q_{o_m}(u_m)\, dC_t\ du_1, \ldots, du_d = \kappa_{(o_1,\ldots,o_d),t}. \tag{14}$$

For the estimation of these coefficients in practice, we rely on the uniformity of ranks and that ranks are within $[0,1]$ and replace the inner product by sample averages:[15]

$$\hat{\xi}_{mot} := N_j^{-1} \sum_i w_{mit} Q_o(u_{mit}) \tag{15}$$

$$\hat{\kappa}_{(o_1,\ldots,o_d),t} := N_j^{-1} \sum_i \left(\prod_{m=1}^{d} Q_{o_m}(u_{mit}) \right) \tag{16}$$

where u_{mit} is the rank of w_{mit}, the latter being the sample analogue of Ξ_{mt}^{-1} for observation i.

In the estimation, we truncate the sums in (11) and (12) at a given maximal order O and by orthonormality of the polynomials, the kept coefficients are not affected by the truncation. The coefficients in our case indeed decrease rapidly with each polynomial as we show in Figure 2. The figure plots the coefficient value as a function of the order of its respective polynomial term. At around order 10, the coefficients all become very small.[16] Evidence in Appendix A supports this, showing a truncation to $O = 11$ captures the variation observed in the data well.

Are there advantages to using the quantile function here? An alternative in the functional-data literature is to transform the distribution into an unconstrained function space before projecting—log-densities, as in Chang, Chen, and Schorfheide (2024), or the log-quantile-density transformation (Petersen and Müller 2016; Marcellino, Renzetti, and Tornese 2024). Estimation typically requires kernel smoothing, unlike our rank-based coefficients—but avoids imposing non-negativity and monotonicity constraints in a linear model. We find in the end the cost of staying in quantile coordinates is small: the only constraint on the quantile function, monotonicity, we verify ex post by evaluating the Legendre approximation on a fine grid (10,000 points), only finding violations in measures where there is a flat region in the support (e.g., bottom wealth components), with rearrangement of quantiles (Chernozhukov, Fernández-Val, and Galichon 2010) available as a remedy if needed, but avoiding the numerical stability issues of the log-transformation of densities around tails.

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[15] Survey weights are disregarded here for notational convenience but applied in the application

[16] A small coefficient in absolute value does imply the contribution of the corresponding polynomial to be small in an $\mathcal{L}^2$ sense.

Handling partial observability with the copula density A key advantage and contribution of the procedure concerns a difficulty that any application combining multiple data sources faces: the microdata may not always contain the entire vector w, but only a subset. When w is incompletely observed, we cannot estimate the full d-dimensional copula directly—only copulas with the unobserved dimensions integrated out. For the copula density, these lower-dimensional copulas are exact *slices* of the higher-dimensional coefficient array, so partially observing datasets remain informative about the full d-dimensional copula. The representation in the form of (Legendre) polynomials makes this precise. Integrating out the "missing" dimension d must return the lower-dimensional density,

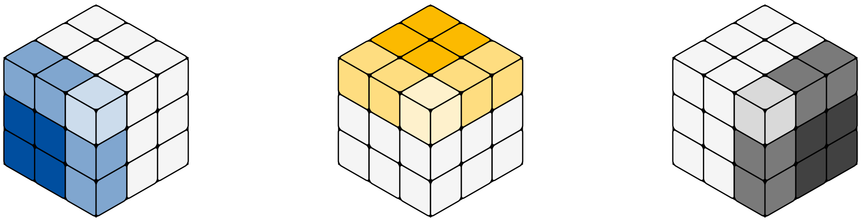
$$\int_0^1 dC(u_1,\dots,u_d)\,du_d \overset{!}{=} dC(u_1,\dots,u_{d-1}). \tag{17}$$

Writing out the integrand using (12) and making use of the fact that the order-zero (shifted) Legendre polynomial $Q_0 \equiv 1$ integrates to one and all higher orders to zero, we obtain

$$\begin{aligned}\int_0^1 dC(u_1,\dots,u_d)\,du_d &= \int_0^1 \sum_{o_1}\cdots\sum_{o_d} \kappa_{(o_1,\dots,o_d)} \prod_{m=1}^{d} Q_{o_m}(u_m)\,du_d \\ &= \sum_{o_1}\cdots\sum_{o_{d-1}}\sum_{\mathbf{o_d}} \kappa_{(o_1,\dots,o_{d-1},\mathbf{o_d})}\left(\prod_{m=1}^{d-1} Q_{o_m}(u_m)\right)\int_0^1 Q_{\mathbf{o_d}}(u_d)\,du_d \\ &= \sum_{o_1}\cdots\sum_{o_{d-1}} \kappa_{(o_1,\dots,o_{d-1},0)}\left(\prod_{m=1}^{d-1} Q_{o_m}(u_m)\right). \end{aligned} \tag{18}$$

In words, the polynomial coefficients of the lower-dimensional copula density are identical to the leading "slice" of the higher-dimensional copula density. This makes it straightforward to combine datasets with different variable coverage. For example, although the SCF does not observe consumption, its joint observations of income and wealth can still inform the income–wealth dependence within the full joint distribution of consumption, income, and wealth. Using the copula distribution function rather than its density would lose this simple property: each lower-dimensional coefficient would depend on many higher-dimensional coefficients, making the relationship considerably more complicated and introducing approximation errors after truncation. Figure 3 also illustrates another advantage over the distribution. Some coeffi-

Figure 3: Geometric Representation of Partially Observed Copula Coefficients



(a) Only Variables 1& 3 (b) Only Variables 2& 3 (c) Only Variables 1& 2

Notes: Figure shows three cubes. A cube can be interpreted as an array of copula coefficients $\kappa^{j}_{(o_1,\ldots,o_d),t}$ for some dataset j at time t, for $d = 3$. Each cube corresponds to a scenario where one variable is missing in the estimation of the copula density. The light edge denotes the (0,0,0) coordinate. For each scenario, the white boxes are coefficients we cannot estimate. The slightly colored boxes correspond to the immutable coefficients, which have fixed values independent of data. The darker colored boxes are scenario specific and correspond to (time-varying) coefficients that need to be estimated.

cients (shown as white cubelets) are fixed by the requirement that each variable has a uniform marginal distribution. These coefficients are known in advance, never estimated, and are enforced exactly at every step—not true for the distribution. As a result, every estimated copula density automatically satisfies the uniform-margins condition.

3.2.2 Dealing with the Curse of Dimensionality

Vectorizing the coefficients for each time period, t, leaves us with sequences of coefficients, $\theta^j_t = (\xi^j_{mot}, \ldots, \kappa^j_{(o_1,\ldots,o_d),t})$, for each cross-sectional dataset, j. For example, with $d = 3$ dimensions in our application—consumption, income, and wealth and using polynomials up to order eleven ($O = 11$) to organize the data—the copula density would be represented by a vector with $(O+1)^d - (d \times O + 1) = 1694$ free entries for each time point and $d \times O + 1 = 34$ fixed. These fixed entries are not included in the estimation. In addition, there would be $d \times (O+1) = 36$ coefficients of the polynomials (including a constant) representing the quantile functions, which we collect in θ^j_t as well.

This example makes apparent that even for a modest polynomial order, the dimensionality of $\theta^j_t \in \mathbb{R}^N, N = (O+1)^d + (d-1)$ for a dataset with d variables is large. Equally important is the frequency of missing observations across the N entries; in our application, the data structure is irregular and includes many gaps. Taken together, this makes it impractical to specify and estimate a time-series model directly on the raw θ^j_t coefficients. For this purpose, we postulate (and then estimate) a dynamic factor model for θ. The *orthonormal* representation offers further advantages: The variance (over time) of a coefficient is proportional to its contribution to fluctuations of the function (in the $\mathcal{L}^2$ sense). Put simply: The polynomial coefficient provides a useful form of standardization that has a natural metric and allows us to uncover the factor structure behind the time-series changes in the distributions. This factor structure finally allows us to overcome the curse of dimensionality in the distributional data.

For this purpose, all (variable) coefficients of the polynomial representation of dC^j_t (and separately of the quantiles) are horizontally concatenated:

$$\boldsymbol{\theta}^j = \begin{bmatrix} \theta^j_{1,1} & & \theta^j_{1,T} \\ & \ddots & \\ \theta^j_{N,1} & & \theta^j_{N,T} \end{bmatrix}.$$

Define $\boldsymbol{\theta}$ as the matrix obtained by horizontally concatenating the individual blocks $\boldsymbol{\theta}^j$, retaining only those time periods for which every coefficient is available.[17] A PCA (a singular value decomposition) is then performed, which nonparametrically reduces the dimensionality of the data (Breitung and Eickmeier 2006; Chen, Er, and Wu 2005). For the PCA, we define $\tilde{\boldsymbol{\theta}}$, which is the standardized representation of $\boldsymbol{\theta}$. The standardization is done by distribution object $\zeta \in \{1, \ldots, d+1\}$ (d quantile functions and a copula).[18] We standardize by object (and not by coefficient) because the polynomial coefficients represent standardized contributions to the object's structure (e.g., the quantile function or copula). As a result, additional standardization within the same object is unnecessary, since their relative magnitudes are naturally balanced by the properties of the polynomial basis. This leaves us with the standardized observation $\tilde{\theta}^j_{nt}$ of coefficient n in data source j at time t.

The PCA of $\tilde{\boldsymbol{\theta}}$ provides us with a projection matrix $\hat{\Gamma} \in \mathbb{R}^{N\times r}$, with full column rank r, that projects $r \ll N$ factors into the N dimensional distributional data. Throughout, we normalize so that each factor has unit in-sample variance, with the loadings $\hat{\Gamma}$ absorbing the scale.[19]

More specifically, we decompose $\tilde{\boldsymbol{\theta}}$ into latent orthogonal factors $\begin{bmatrix} \hat{f}'_\Gamma & \hat{f}'_\gamma \end{bmatrix}'$ divided into "important" and "unimportant" factors according to their contribution to the total variance (measured by their singular value). Their respective time-invariant loadings are $\begin{bmatrix} \hat{\Gamma} & \hat{\gamma} \end{bmatrix}$, for $\hat{f}_\Gamma \in \mathbb{R}^{r\times T_c}$ and $\hat{f}_\gamma \in \mathbb{R}^{(T_c - r)\times T_c}$ for T_c number of time periods with observed data. This decomposition is unique up to the scale (and sign) of each factor, which allows us to normalize so that all factors have unit variance, with the loadings absorbing the inverse rescaling. Altogether, we have:

$$\tilde{\boldsymbol{\theta}} = \begin{bmatrix} \hat{\Gamma} & \hat{\gamma} \end{bmatrix} \begin{bmatrix} \hat{f}_\Gamma \\ \hat{f}_\gamma \end{bmatrix} \tag{19}$$

where $\hat{f}_\Gamma$ represents the r important factors, which capture almost all of the variation in the data, and $\hat{f}_\gamma$ the $T_c - r$ less important factors, which can be interpreted as measurement noise.[20]

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[17] Before the concatenation, each coefficient is detrended using an HP-filter. This takes care of dataset-specific effects. We store the information needed to transform $\tilde{\boldsymbol{\theta}}^j$ back to the originally observed objects to obtain source-specific predictions. For example, the income quantiles (that would be one object ζ) in the SCF and the PSID (two sources j) may be permanently different due to differences in sample design and operationalization.

[18] The normalization of coefficient n in object ζ in dataset j is given by $\tilde{\theta}^j_{nt} = \left(\frac{\theta^j_{nt} - \mu^j_n}{\sigma_{\zeta(n)}} \right)$ where μ^j_n is the coefficient-specific mean and $\sigma_{\zeta(n)}$ is the standard deviation of *all* coefficients (rows) pertaining to object ζ. This removes dataset- and coefficient-specific fixed effects, μ^j_n. This can capture, e.g., permanent differences in sampling procedure and operationalization. The order of operations is: each coefficient series is HP-detrended within dataset j (see the previous footnote), then demeaned, then scaled. The demeaning is stated for explicitness only—in sample, HP residuals sum to zero, so $\mu^j_n = 0$. The stored trend, mean, and scale are reapplied in reverse order to map model output back to source-specific observables.

[19] Concretely, with P the orthonormal principal axes and Λ the diagonal matrix of the corresponding eigenvalues (the sample variances of the principal components) in decreasing order, we set $\hat{\Gamma} := P\Lambda^{1/2}$ and $\hat{f}_\Gamma := \Lambda^{-1/2}P'\tilde{\boldsymbol{\theta}}$: the product $\hat{\Gamma}\hat{f}_\Gamma$ is unchanged, each factor has unit variance, and the scale of the loadings carries the explanatory weight of each factor. The priors on the factor innovations reference this unit-variance convention (Appendix C).

This step delivers two objects: the dimension r of the factor space and the estimated loadings $\hat{\Gamma}$, whose columns constitute the functional basis (see Kneip and Utikal 2001; Tsay 2016). In our application, each factor can capture joint movements in the distributions of consumption, income, and wealth and their dependence structure, rather than tracking a single distributional statistic. The dimension r determines the size of the state-space model set up in the next subsection. We select r based on a comparison of marginal data densities of the estimated state-space model across r (Appendix B)—a comparison that is meaningful precisely because varying r leaves the observation vector unchanged and thus likelihoods comparable.[21] The factor time paths themselves are *not* taken from the PCA: they are latent states, estimated by the Kalman smoother using all datasets—including those excluded from the PCA because of partial coverage—with $\hat{\Gamma}$ held fixed as the linear map between the high-dimensional coefficients $\tilde{\boldsymbol{\theta}}_t$ and the factors f_t. Holding $\hat{\Gamma}$ fixed is part of the modular estimation strategy discussed in Appendix C; Section 3.2.3 makes the mapping precise.

The estimation of $\hat{\Gamma}$ requires time observations where every coefficient is observed, yet some data sources will not satisfy this criterion *at all*—for example, the SCF never measures consumption. Observations from such incomplete sources are therefore excluded from the PCA that yields $\hat{\Gamma}$. In Appendix B, we consider alternative estimators that accommodate partial unobservability in $\tilde{\boldsymbol{\theta}}^j$. Following standard practice, we select among these alternatives the estimator with the highest marginal data density.

3.2.3 Factor State Space Model and Measurement

With this preprocessing of the data, we can turn to specifying the state-space model that captures the evolution of the r latent distributional factors. Due to the possible mixed-frequency nature of the data, we postulate a mixed-frequency state space model as in Schorfheide and Song (2015), with the following augmented state vector

$$F_t = \left[f_t' \; f_{t-1}' \; \ldots \; f_{t-p+1}' \right]' \in \mathbb{R}^{rp \times 1}, \tag{20}$$

where F_t collects the r latent distributional factors f_t, in addition to its $p - 1$ lags. For the application of our methodology in Section 5, we set $p = 4$, with the high-frequency data of interest being quarterly in order to allow for annual aggregation in measurement when a dataset collects a sample over an entire year. To facilitate the reading across sections, we provide the example of the quarterly case, though the methodology easily generalizes to other frequencies.

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[20] By working with an approximate factor model i.e., our principal components estimator, we do not force the idiosyncratic (measurement error) terms to be cross-sectionally independent; they may display weak correlation across series. Consistency, however, requires that this residual dependence remain limited in the Bai-Ng sense (no eigenvalue of the idiosyncratic covariance can diverge with N). We therefore adopt the standard “weak cross-sectional dependence” assumptions used by Bai (2003), Bai and Ng (2006), and, more recently, Freyaldenhoven (2022), which rule out pervasive or strong common shocks beyond those captured by the retained factors; however, we retain components that explain 99% of the total variation, suggesting that any remaining idiosyncratic error is quantitatively small and therefore unlikely to contain an additional pervasive factor. We confirm this by inspecting the largest *residual* eigenvalue and perform standard cross-correlation tests, finding no evidence of strong common factors (shocks) in the residuals, hence, supplying evidence of weak cross-dependence in our context.

[21] For our application, the marginal data density rises monotonically from three to eight factors (Table 4) and worsens thereafter. Selection via other criterion (Freyaldenhoven 2022; Bai and Ng 2002; Ahn and Horenstein 2013) also point to the same neighborhood as the MDD, around 8-10 factors. We therefore select $r = 8$ for our application, which explains 99% of the business-cycle-frequency variation.

State Equation Stacking the lags directly with the q aggregate factors Y_t[22] in the state gives the following representation

$$\begin{bmatrix} F_t \\ Y_t \end{bmatrix} = \Phi \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \epsilon_t = \begin{bmatrix} \Phi_{FF} & \Phi_{FY} \\ \Phi_{YF} & \Phi_{YY} \end{bmatrix} \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \Omega) \tag{21}$$

and for the case of quarterly data

$$F_{t-1} = \begin{bmatrix} f_{t-1} \\ f_{t-2} \\ f_{t-3} \\ f_{t-4} \end{bmatrix}, \Phi_{FF} = \begin{bmatrix} A & 0 & 0 & 0 \\ I_r & 0 & 0 & 0 \\ 0 & I_r & 0 & 0 \\ 0 & 0 & I_r & 0 \end{bmatrix}, \Phi_{FY} = \begin{bmatrix} B \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Phi_{YF} = \begin{bmatrix} C & 0 & 0 & 0 \end{bmatrix}, \Phi_{YY} = D,$$

where distributional factors f_t are assumed to follow a VAR(1) process, summarized by the law of motion matrix $A \in \mathbb{R}^{r \times r}$. In addition, they are assumed to co-move with aggregate factors Y_t, whose relationship is summarized by loading matrix $B \in \mathbb{R}^{r \times q}$ and $C \in \mathbb{R}^{q \times r}$. The second block row of (21) thus specifies the data-generating process of the aggregate factors: $Y_t = Cf_{t-1} + DY_{t-1} + \epsilon_{Y,t}$. In particular, Y is not assumed exogenous to the distribution: it responds to the lagged distributional factors through C, which is estimated and left unrestricted, and its innovations may be contemporaneously correlated with those of the distributional factors through Ω.

The aggregate factors are principal components and hence contemporaneously orthogonal; we additionally restrict their autoregressive matrix $D \in \mathbb{R}^{q \times q}$ to be diagonal. Each aggregate factor follows its own autoregression. This is a parsimony restriction—orthogonality does not rule out lagged cross-predictability—and it restricts neither the factors' effect on the distributional factors (B), the feedback from the distribution (C), nor the correlation of their innovations (Ω), through which cross-factor dynamics can still operate. All other matrices (A, B and C) are left unrestricted. As in Schorfheide and Song (2015), the remaining rows of Φ deliver the identities $f_{t-l} = f_{t-l}$ for $l = 1, ..., p-1$—hence the I_r in the example of a quarterly estimation.

While it is customary to restrict the innovations ϵ_t to be uncorrelated, both serially and between factors, so that Ω is a diagonal matrix, we do estimate both the variances and covariances of Ω. This allows for a shock structure where innovations to the distributional factors can be *contemporaneously* correlated with innovations to the aggregate factors. This is also how the linearized system accommodates within-period comovement between distribution and aggregates: no variable dated t enters the right-hand side of (21)—the system is strictly first order—and any contemporaneous comovement is captured by the off-diagonal blocks of Ω as correlated innovations, without imposing a causal direction.

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[22] Recall that the set of aggregate factors at time t depends on an information set that captures both contemporaneous and past aggregate variation. More on the set of aggregates in Section 4.1.

Measurement Equation Since distributional factors are not directly observable, we complement the model with an observation equation for each dataset j:

$$\tilde{\theta}_t^j = H_t^j \left(\hat{\Gamma}^{MF,j} F_t + \nu_{F,t}^j \right), \quad \nu_{F,t}^j \sim \mathcal{N}\left(\mathbf{0}, \hat{\Sigma}_j^{1/2} S^j \hat{\Sigma}_j^{1/2\prime} \right) \tag{22}$$

This observation equation maps factors F_t, through the loading matrix $\hat{\Gamma}^{MF,j}$, to the standardized coefficients $\tilde{\theta}_t^j$, allowing for a Gaussian measurement error $\nu_{F,t}^j$. The selector matrix $H_t^j \in \{0,1\}^{N \times N}$ indicates whether (parts of) the distribution are observed in data source j at time t, following Durbin and Koopman (2012). For example, the CPS provides information on the income distribution but not on consumption, wealth, or their dependence structure. In quarters without CPS observations, the selector matrix omits the corresponding measurements. The Gaussian error term $\nu_{F,t}^j$ has a variance-covariance matrix decomposed into S^j, a diagonal scaling matrix with diagonal entries s_ζ^j varying by object ζ and $\hat{\Sigma}_j$, a positive semi-definite (covariance) matrix.

The matrix $\hat{\Gamma}^{MF,j}$ is constructed to take time aggregation into account[23]. Specifically, these matrices map the high-frequency factors to the lower-frequency observations. For example, when the frequency of interest is quarterly, as it is in the application:

$$\hat{\Gamma}_{\zeta(n)}^{MF,j} = \begin{cases} \frac{1}{4}\left[\hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}, \hat{\Gamma}_{\zeta(n)}\right] & \text{annual flow,} \\ \left[\hat{\Gamma}_{\zeta(n)}, \mathbf{0}, \mathbf{0}, \mathbf{0}\right] & \text{quarterly flow or stocks.} \end{cases}$$

In the first case, the distributional object corresponds to an *annual flow* (e.g. the PSID income / consumption quantile function). We then approximate the distribution of annual averages by the average distribution over four quarters and set $\hat{\Gamma}_{\zeta(n)}^{MF,j}$ as in case *annual flow*. For stock variables (e.g. any wealth object for which the date is known), the measurement loads exclusively on the contemporaneous factor block and the lagged blocks are set to zero i.e., $\hat{\Gamma}_{\zeta(n)}^{MF,j}$ as in case *quarterly flow*, for $\mathbf{0}$ the same size as $\hat{\Gamma}_{\zeta(n)}$. The same holds for the third case, *quarterly flows* (e.g., SIPP income, CEX), which again loads on the contemporaneous block only.

For the copula, since multiple variables are necessary for its construction, the different measurements of the variables could be mixed *within* e.g., annual income flows and stock wealth. In these cases, for some point in time t, the lowest frequency is chosen among the set of measurements as an approximation to the temporal aggregation.

Measurement Error The measurement error for the distributional data, $\nu_{F,t}^j$, is composed of sampling uncertainty and other errors that reflect the fact that a given dataset has its specific operationalizations of the common economic variables being measured. Differences in operationalizations can not only shift the level of a particular measurement (which we capture through fixed effects, see Footnote 18), but can also become differentially important over time.[24] In principle, to fully capture these time-varying differences, we would need to estimate a measurement error variance for each coefficient and dataset. However, this would be too large a number of parameters to estimate within the time-series model; a common problem when estimating approximate dynamic factor models with maximum likelihood (see, e.g., Bai and Li 2016). The standard approach in the literature uses quasi-maximum likelihood with EM algorithms (Doz, Giannone, and Reichlin 2012; Banbura and Modugno 2014; Barigozzi and Luciani 2019). EM is a local method: it converges to a stationary point that depends on its starting values (Wu 1983), and its guarantees hold only when initialized near the true parameters (Balakrishnan, Wainwright, and Yu 2017), which is hard to ensure in high-dimensional settings such as ours.

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[23] Here, we approximate the unit aggregate $F(N^{-1} \sum x_i)$ by the distributional aggregate $N^{-1} \sum_i \Xi_i(x_i)$.

[24] For example, the PSID and SCF differ in the way they ask respondents about their business wealth (Pfeffer, Schoeni, Kennickell, and Andreski 2016). PSID and CPS differ in the sampling unit, making household/family income sensitive to labor supply patterns (Gouskova, Andreski, and Schoeni 2010). Similarly, differences in the

For this reason, we do two things, the second of which will be discussed in Appendix C, when we cover our sampling procedure. First, we assume that the covariance of all measurement errors is generated by scaling the whitened directions of the sampling uncertainty (and in that sense preserves the dependency structure). Under this assumption, the matrix Σ_j can be estimated outside the time-series model using bootstraps or supplied replication weights to estimate the covariance from sampling uncertainty by data source j.[25]

This achieves differently the diagonality assumed in the aforementioned literature. This leaves N diagonal elements of S^j to be estimated within the time-series model. To reduce the parameter count further, we constrain these elements to vary only by dataset j and variable ζ, yielding just $d+1=4$ parameters per dataset. Consequently, each diagonal element s^j_ζ indicates by what factor the sampling-based variance for ζ must be inflated (or deflated) in dataset j; the standard error correspondingly scales with $\sqrt{s^j_\zeta}$. Therefore, the overall measurement error variance in (22) is $\hat{\Sigma}_j^{1/2} S^j (\hat{\Sigma}_j^{1/2})'$.

Thus, similarly to feasible generalized least squares, we transform observations $\tilde{\theta}^j_t$ by premultiplying the coefficients with the external estimate $\hat{\Sigma}_j^{-1/2}$[26]

$$\begin{aligned} \hat{\Sigma}_j^{-1/2}\tilde{\theta}^j_t &= H^j_t(\hat{\Sigma}_j^{-1/2}\,\hat{\Gamma}^{MF,j}F_t + \tilde{\nu}^j_{F,t}) \qquad \tilde{\nu}^j_{F,t} \sim \mathcal{N}(0, S^j) \\ \tilde{\tilde{\theta}}^j_t &= H^j_t(\tilde{\hat{\Gamma}}^{MF}_j F_t + \tilde{\nu}^j_{F,t}), \qquad \tilde{\hat{\Gamma}}^{MF}_j := \hat{\Sigma}_j^{-1/2}\hat{\Gamma}^{MF,j}. \end{aligned} \tag{23}$$

With the measurement error treatment in place, denoted by the additional tilde, we can stack the J measurement equations, and with the aggregates, arrive at the final measurement equation

$$\begin{bmatrix} \tilde{\tilde{\theta}}_t \\ \hat{Y}_t \end{bmatrix} = H_t \left(\begin{bmatrix} \tilde{\hat{\Gamma}}^{MF} & \mathbf{0} \\ \mathbf{0} & I_Y \end{bmatrix} \begin{bmatrix} F_t \\ Y_t \end{bmatrix} + \begin{bmatrix} \tilde{\nu}_{F,t} \\ \nu_{Y,t} \end{bmatrix} \right) \qquad \nu_{Y,t} \sim \mathcal{N}(\mathbf{0}, \Sigma_Y). \tag{24}$$

---

propensity to sample business owners between the different datasets make income sensitive to relative changes in business and labor income (Kim and Stafford 2000). Finally, the CEX and the PSID differ in the consumption categories covered in the survey, with the PSID being much coarser (Insolera, Simmert, and Johnson 2021).

[25] We draw bootstrap samples / use the supplied replication weights for each dataset j, $\{\tilde{\boldsymbol{\theta}}^j_{t,b}\}_{b=1}^B$, for each period t. Then, we demean the bootstrap samples b for each j and t and compute the average within-time variance-covariance matrix $\hat{\Sigma}_j$ pooling the demeaned bootstrap samples of the dataset j. If an object ζ is unobserved in dataset j, we set the covariance terms to zero and the diagonal elements to one to still be able to compute $\Sigma_j^{-1/2}$. To regularize $\hat{\Sigma}_j$ we add a small variance term $\lambda\,\bar{\sigma}^2_j\, I$. Here $\bar{\sigma}^2_j$ is the average bootstrap variance of the coefficients survey j observes and we set $\lambda = 0.05$. This preserves the ranking between measurements in terms of precision.

[26] The whitening is applied blockwise to the subvector observed in each period—i.e., $\hat{\Sigma}_j^{-1/2}$ is taken conformably with the selection H^j_t—so the transformation and the selection commute.

Note that the terms are free of the j index to indicate they have been vertically stacked and note the additional appendage for the aggregates Y_t. Measurements of the aggregate factors are observed with noise Σ_Y.

In sum, Equations (21) and (24) form a linear state-space system that can be estimated using standard Bayesian techniques (described in Appendix C) and a Kalman-filter-smoother. The posterior distribution allows us to capture parameter uncertainty and uncertainty about the latent factors given the parameters. Apply the Kalman-smoother to the posterior mode recovers the full path of latent factors $\{\hat{F}_t\}_{t=1}^T$.

Two-step estimation vs. a unified Bayesian approach. Our procedure is deliberately modular: we first summarize the microdata by estimating the distributional coefficients $\tilde{\boldsymbol{\theta}}_t^j$ and their uncertainty Σ_j, and then use these estimates as noisy observations in the state-space model. An alternative would be to estimate everything jointly by placing a household-level likelihood directly inside the state-space model. For the simpler case of independent cross-sectional data, Chang, Chen, and Schorfheide (2024) show that this distinction is largely one of implementation: the exact joint model is computationally difficult, while a practical approximation leads back to the same two-step approach, where period-by-period estimates enter the state-space model with uncertainty that shrinks as survey size increases. Our setting differs because household surveys are not simple random samples. Sampling weights, stratification, non-response, and multiple imputation affect the uncertainty of the estimated coefficients, so a naive independent household likelihood would not correctly capture their precision. Instead, we estimate Σ_j directly from the survey design using replicate weights or design-based bootstraps, allowing the state-space model to weight each coefficient according to its actual sampling uncertainty. This modular approach also prevents errors in the dynamic model from affecting the cross-sectional estimates (Plummer 2015; Jacob et al. 2017).

3.3 Using the Data and Using the Method

Given these mode-conditional smoothed factors, we can reconstruct the sequence of joint distributions, $\hat{\Xi}_t^j$, from the implied standardized series of polynomial coefficients. Appendix D outlines the procedure. Importantly, this reconstruction provides estimates for all quarters, including those without direct microdata observations, by combining information from aggregate variables, the estimated dynamics, and microdata available in other periods.

Using the data This estimation approach allows practitioners to use the above data in three principal ways: First, in its factor representation, $\hat{F}_t$, second, in its functional representation, $\hat{\theta}_t^j$, or, third, in its observational representation, e.g., quantile functions and copula—$\left(\{\hat{\Xi}_{jmt}^{-1}\}_{m=1}^d,\, d\hat{C}_t^j\right)$. All three representations are provided for download at distributionaldynamics.com.

For example, if one is interested in replicating the exercise of Chang and Schorfheide (2024), one can use the second representation and use the coefficients related to the marginal distributions of interest in their *distribution*-augmented VAR. If one is only interested in augmenting a VAR with a low-dimensional summary containing virtually all variation in the joint distribution of consumption, income, and wealth—in the style of these factor-augmented VARs—one can use the first representation.

Lastly, one can use the coefficients to generate economically interpretable data and work with this economically interpretable/meaningful data directly. One can generate the quantile functions and copula densities, but also other common statistics of interest for the marginal distribution: e.g., Gini coefficients of consumption, income, or wealth; levels, quantiles, shares, etc. and objects that arise from the correlational structure (the copula density): e.g., correlation measures, conditional probabilities (e.g., population shares), and Kendall's tau.

For the application in Section 5, we use this last representation to generate a time series of synthetic cross-sections from the quantile functions and copula densities. Appendix D explains in more detail how to generate these time series from the resulting distributional estimates of the model.

Using the method The method can flexibly accommodate various economically relevant joint distributions beyond the application considered here. These include subcomponents of consumption, income, and wealth, such as different asset classes or income sources, or distributions over other economic agents, such as firms. Given micro- and macrodata inputs, researchers can adapt the method to these settings, combining datasets with different frequencies, coverage, and precision to construct high-frequency distributional estimates.

The method also enables greater accessibility to confidential microdata. Central banks and similar authorities with access to restricted microdata could apply our procedure internally to construct estimates of joint distributions without releasing the underlying household records. From these estimates, they could generate synthetic microdata samples or publish distributional statistics, enabling research without direct access to confidential records. In this way, pending legalities, the method could facilitate the dissemination and use of distributional information that would otherwise remain inaccessible.

4 Credibility Checks

Before turning to the application, we evaluate the procedure in three steps, using actual data (Section 4.1) and data simulated from a heterogeneous-agent model (Section 4.3.2). First, we show that the factor representation preserves the information in the distributional data (Section 4.2). In Appendix E we show that the estimated distributional paths stay within the bounds implied by sampling uncertainty. Second, in three exercises, we ask whether the state-space model predicts data it has not seen (Section 4.3): (i) predicting SCF survey waves one at a time (ii) re-estimating the model while withholding larger blocks of microdata, and (iii) comparing our model estimates to a simulated economy whose true dynamics are known (Bayer, Born, and Luetticke 2024). In all three, the predictions closely track the withheld or true distributions. Appendix F provides a different form of validation, finding that the cyclical fluctuations of our estimates, where applicable, also agree with those in the World Inequality Database, Real-

Table 1: Micro Data Sources and their Sample Periods

Object	CEX	CPS	SCF	PSID	SIPP
Consumption quantiles	1984Q4–2021Q4	-	-	1999Q2–2021Q2	-
Income quantiles	1984Q4–2021Q4	1967Q4–2022Q4	1962Q4–2021Q4	1967Q4–2020Q4	1982Q4–2022Q4
Wealth quantiles	-	-	1962Q3–2022Q3	1984Q2–2021Q2	1983Q3–2022Q4
Copula densities	1984Q4–2021Q4	-	1962Q3–2022Q3	1984Q2–2021Q2	1983Q3–2022Q4

Notes: The table reports sample periods for different micro datasets across the different objects.

Time Inequality, and the Distributional Financial Accounts.[27] Finally, we examine whether these estimates are supported by the data, documenting which micro- and macro-data sources contribute to the estimates and how strongly the available observations pin them down relative to the model's dynamics. The section begins by covering the data used, with each concern addressed in turn.

4.1 Data

To test and apply our method, we estimate the joint distribution of consumption, income, and wealth at the household level for the United States from 1962 to 2024. We rely on commonly used microdata for the U.S.: the *Consumer Expenditure Survey* (CEX), *Current Population Survey* (CPS, Flood et al. (2023)), *Panel Study of Income Dynamics* (PSID), *Survey of Consumer Finances* (SCF); including the historical backfiles (SCF+), and the *Survey of Income and Program Participation* (SIPP).[28] Table 1 lists the distributional objects from each dataset, together with their respective sampling periods. Appendix G visualizes more elaborately the underlying scarcity, charting each source's reference periods, variable coverage, sample sizes, and publication lags.

We do not impose any demographic household restriction and only mild sample selection for consistency (see Appendix G) in all of these datasets. For the CEX consumption and SIPP income, which are at quarterly frequency, we remove seasonality using X-13 ARIMA-SEATS (Appendix G.6). Wealth is dated to the interview quarter (quarter 2 for the PSID, quarter 3 for the SCF). Income in the PSID, SCF and CPS refers to the previous calendar year and is dated to its fourth quarter, so that the model's annual average of the quarterly states covers that year. SIPP income data are aggregated to quarterly level and then naturally assigned to the respective quarter; wealth assignment depended on the survey vintage.[29] Further details on the microdata and in particular how the respective measures are defined can be found in Appendix G.

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[27] Trends are fitted by construction; see Section 3.2.1.

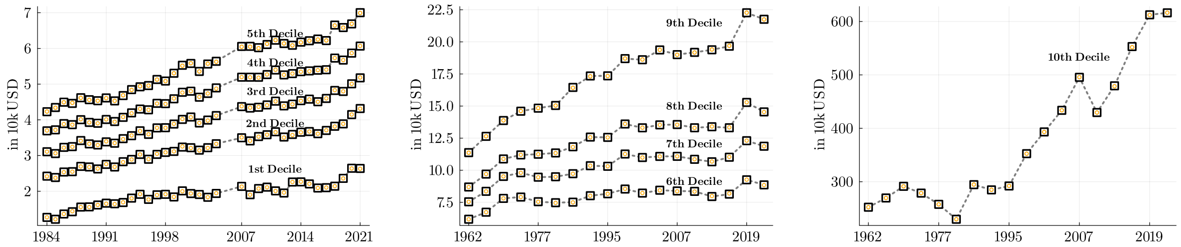
[28] The methodology accommodates permanent and time-varying differences in measurement, but major survey redesigns require intervention. For the CPS (1994) and SIPP (1996, 2013), we treat pre- and post-break data as separate surveys—i.e., distinct measurements of the distribution to avoid spurious dynamics from seam bias.

[29] Documentation on the timing of the CPS can be found here, pg. 6. For the timing of the SCF, see here, pg. 33. For the PSID, interview dates are provided as variables, with the vast majority of interviews falling in quarter 2. Income refers to the calendar year before the interview and enters the measurement equation as the average of the four quarterly states of that year (Section 3.2.3). Wealth is dated to the interview quarter. For data on income, which is for the year prior, it is scaled by the growth rate of their respective aggregate to proxy the contemporaneous level.

Figure 4: Raw Data vs. Approximation Using Important Factors

Quantile Functions Across Measures

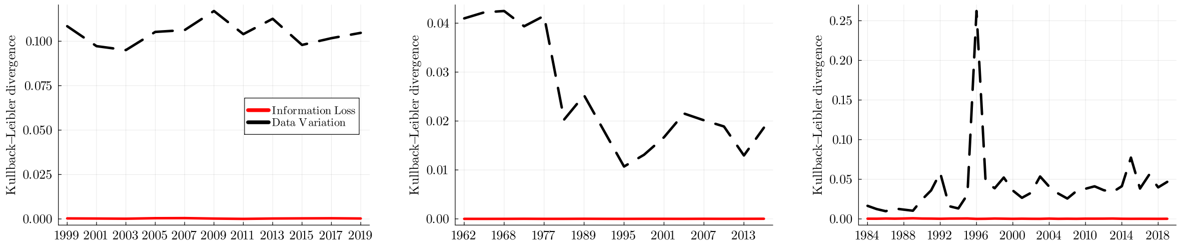
Consumption	*Income*	*Wealth*



(a) 1st to 5th decile (b) 6th to 9th decile (c) Top decile

○ Approximation □ Data

Copula Divergence Across Measure Combinations



(d) C, Y, and W (e) Y and W (f) C and Y

Information loss - - - - Copula dynamics

Notes: Panels (a)–(c) show quantile functions (mean within decile) for consumption, income, and wealth deciles from the survey data (squares) and from an approximation using only the fluctuations in the most important factors in (19) (circles). The consumption panel uses CEX consumption, the income panel uses SCF income, and the wealth panel uses SCF wealth. Dotted lines show linear interpolation between survey waves. Panels (d)–(f) show the Kullback–Leibler divergence between the raw-data copula and two reference copulas, by survey year. The black dashed line gives the divergence between the time-averaged copula and the raw-data copula, while the red solid line gives the divergence between the copula implied by allowing only the most important factors in (19) to fluctuate and the raw-data copula. Panel (d) uses PSID data (consumption, income, and wealth), panel (e) SCF data (income and wealth), and panel (f) CEX data (consumption and income).

In terms of aggregate data, we use a wide range of standard business cycle data (GDP, consumption, employment, etc.) as well as data on household balance sheets, expectations, asset prices, and interest rates from McCracken and Ng (2021). We include data from 1962Q3 to 2024Q1. The starting point of the aggregate data determines the earliest date for the sample periods of the microdata used. From these aggregate time series, we extract the 11 leading factors.[30] More details on the macro-data can be found in Appendix G.

4.2 Precision of the Factor Model

The first step in our procedure is to estimate Legendre polynomial coefficients for the functional representation of the distribution. Then, we estimate the factor structure in these coefficient data. Since we only retain "important" factors, we potentially introduce an approximation error (relative to the data) resulting from forcing "unimportant" factors f_t to take time-averaged values. The size of the approximation error can be controlled by choosing how many factors to keep. We choose to retain the eight most important factors, which explain 99% of the (business cycle frequency) variation of the distribution (i.e., of $\tilde{\boldsymbol{\theta}}$ to be precise).[31] The different rows in Figure 4 visualize the approximation error in our application. The first row compares the observed conditional decile means for consumption (bottom five), income (next four), and wealth (top) (squares) with their approximated counterparts (circles). The circles are typically centered around the midpoint of the squares, indicating the factor model with its eight main factors approximates well the dynamics inherit in the marginal distributions. The second row shows that the important factors also approximate copula dynamics well across all three combinations of consumption, income, and wealth. The solid red line measures the Kullback–Leibler divergence between the approximated and raw-data copulas (*information loss*), while the dashed black line measures the divergence of the raw-data copula from its long-run average, indicating how much variation there is to capture. The latter ranges from 0.01 to 0.11 in most cases, and the relatively small information loss suggests that the factors capture much of this variation. Put simply: There are significant fluctuations in the copulas over time, and the factors are able to capture them well.

4.3 Predictability of Distributional Data

To validate how well the method can predict distributional dynamics, we perform three experiments. The first experiment fixes the model parameters at their baseline estimates and removes some waves of microdata as inputs when running the Kalman smoother. This exercise is performed using the SCF, where we omit one wealth survey wave at a time. This experiment answers the question how informative the model structure, aggregate data, and remaining microdata are for pinning down the distributional series when some survey waves are missing, holding parameters fixed. The second experiment repeats this exercise but *re-estimates* the model on the reduced data. This experiment takes into account that changes in the data also imply changes in the model parameter estimates. In the final experiment, we evaluate our model's ability to recover distributional dynamics generated from a heterogeneous agent model. We show the results of these experiments in Figures 5 and 6.

---

[30] Comparing marginal data densities $p(\tilde{\tilde{\theta}}_t, \hat{Y}_t)$ across candidate aggregate blocks is not appropriate because each specification changes the measurement vector. Instead, we compare the likelihood contributions from the microdata alone, holding each candidate aggregate block fixed. The baseline achieves the highest survey-only likelihood, although differences across specifications are small; Appendix B, Table 5 reports them.

[31] A PCA over other distributional data (World Inequality Database / Distributional Financial Accounts) is in the same neighborhood of number of factors.

4.3.1 Predicting Omitted Survey Waves

First, we run 18 mini experiments, each time dropping one SCF wave and generating new smoother estimates. This gives us 18 smoothed distributional data series alongside the one using all SCF waves. Figure 5, Panels (a) to (c), illustrate the first experiment across three different groups: the average of the top 10%, next 40% and bottom 50%. The solid lines are estimates from using all the SCF data. The squares represent the smoothed predictions corresponding to the timing of each omitted survey wave. The circle shows the direct empirical estimate of the corresponding survey (with its confidence limits). For example, for Panel (a), the square for 1992 is the prediction for the wealth distribution where the 1992 SCF survey *did not* enter the smoother.

Two readings should be kept apart: the proximity of the squares to the all-data line shows that a single wave has little leverage on the smoother, while accuracy is assessed by comparing the squares with the withheld survey estimates (black dots) and their confidence bounds: the predictions fall within these bounds in 70–92% of cases (Appendix Table 7). An exception to highlight happens in 2004 for Panels (b) and (c), where during the housing boom, the model (squares) is more cyclical than the baseline estimate (red line—with all data).

For the second experiment, instead of predicting omitted data with fixed parameters, we now predict omitted data while re-estimating the model. The resulting estimates from these re-estimations with less data are shown in Figure 5, Panels (d) to (l). The first row withholds all *wealth* microdata during the housing cycle (2004Q1–2009Q4), a period with large swings in wealth inequality (Kuhn, Schularick, and Steins 2020), to evaluate the "nowcast" performance for the wealth distribution. This asks whether the model can recover distributional dynamics using aggregate data and microdata observations.[32]

The second exercise includes only every fourth CEX survey year in the model and re-estimate—mimicking the fact that countries in Europe only survey consumption every four years and reducing the number of CEX survey years entering the smoother from 38 to 10. We show the average of the top 10%, next 40% and bottom 50% for the CEX consumption by consumption percentiles (Panels (g) to (i)), which corresponds to the marginal distribution, and consumption by income (Panels (j)–(l)), which corresponds to the correlation captured by the copula.

Across all exercises, the reconstructed series closely track the baseline estimates, even when substantial blocks of microdata are withheld. Importantly, the predictions from the omitted-wave experiments also closely track the withheld survey observations and generally fall within their sampling confidence intervals. These results demonstrate that our method recovers distribution dynamics even during periods without direct microdata observations by exploiting aggregate information and surrounding microdata through the state-space model. This finding is consistent with results in Section 4.4 and Appendix I, but also earlier evidence that shows that indeed aggregate factors can explain a sizeable share of the distributional fluctuations (Kuhn, Schularick, and Steins 2020; Bayer, Born, and Luetticke 2024).

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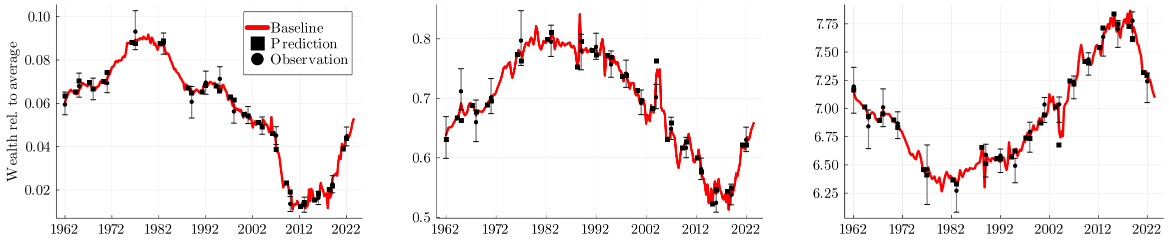
[32] Appendix H, Table 9 reports correlations between the baseline estimates and all missing-data estimates; over the entire sample, they lie between 0.96 and 0.98; restricted to the withheld periods themselves, they range from 0.94 to 0.99.

Figure 5: Predictability of Distributional Data

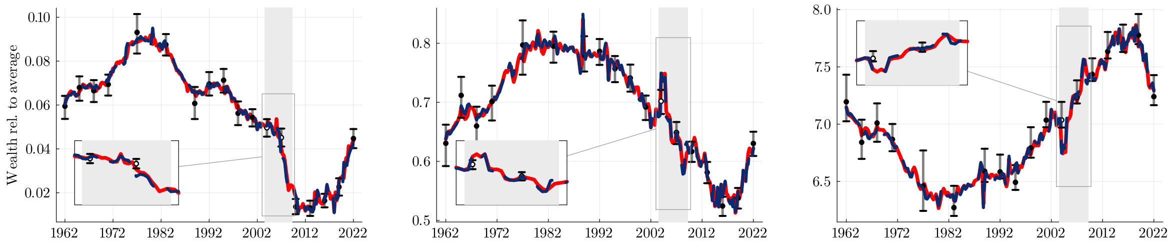
Wealth by Wealth Percentile

No Re-estimation: Removing one SCF wave at a time



(a) Bottom 50 Percent (b) 50–90 Percent (c) Top 10 Percent

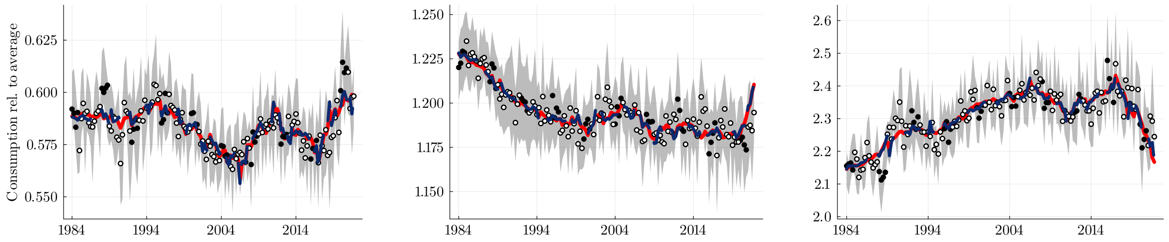
With Re-estimation: Removing microdata from the housing cycle



(d) Bottom 50 Percent (e) 50–90 Percent (f) Top 10 Percent

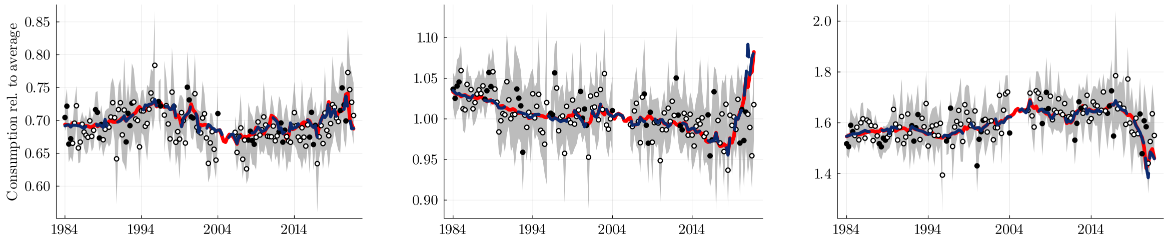
With Re-estimation: CEX data vs. estimates from using CEX every 4 years

Consumption by Consumption Percentile



(g) Bottom 50 Percent (h) 50–90 Percent (i) Top 10 Percent

Consumption by Income Percentile



(j) Bottom 50 Percent (k) 50–90 Percent (l) Top 10 Percent

Baseline Less Data

Notes: Solid red lines show baseline estimates for wealth (Panels (a)–(f)) and consumption (Panels (g)–(l)). In Panels (a)–(c), black squares show Kalman-smoothed predictions obtained by excluding individual SCF waves, while black dots show the corresponding survey observations. In Panels (d)–(f), dashed blue lines show estimates after re-estimating the model without wealth observations from 2004Q1 to 2009Q4. Panels (g)–(l) repeat these exercises for consumption, retaining CEX observations only every fourth year (filled black dots). Empty dots indicate excluded observations. Error bars denote survey-specific 95% confidence intervals.

4.3.2 Predicting Dynamics from a simulated HANK model

Thus far, we have examined the model's ability to predict intentionally omitted data. While informative, these tests do not fully resolve a central concern: in empirical settings, the underlying distributional dynamics are only imperfectly observed and agreement with these sparse microdata (or external benchmarks, see Appendix F) is not definitive proof that the model recovers the true latent process. To assess this, we study the model in a controlled environment with a known data-generating process. For this purpose, we simulate a heterogeneous agent business cycle (HANK) model (concretely, from Bayer, Born, and Luetticke 2024) as a data-generating process. This model is well suited to our setting, as it generates business-cycle fluctuations in general equilibrium in response to a set of aggregate shocks and their interaction with the joint distribution of households, also characterized by a copula and a set of marginals. The simulation produces functional distributional data, which we can use to draw microdata comparable to ones actually available.

The simulation procedure is only summarized here, but elaborated on in more detail in Appendix J. We simulate 2,200 quarters of a HANK economy, driven by aggregate shocks, and discard the first 1,200 as burn-in. From the resulting joint distribution of consumption, income, and wealth, we draw five representative survey-style samples (*A*–*E*) mimicking the PSID, CPS, CEX, SCF, and SIPP, with different sample sizes, frequencies, and variable coverage. Sampling preserves the dependence structure across variables, and we abstract from measurement error. We extract aggregate factors from the simulated structural shocks and divide the remaining 1,000 quarters into ten non-overlapping 25-year estimation windows. For these exercises, our method selects five distributional and eight aggregate factors.

We use the five survey-style samples (*A*–*E*) as microdata inputs to our dynamic factor method alongside the factor representation of the aggregate shocks from the HANK model. The resulting estimates are labeled as *Distributional Dynamics Factor Model (DDFM)* and compared to two benchmarks. The first benchmark consists of moments obtained from the HANK economy's full cross-sectional distribution, calculated each quarter from the model rather than from a sample. This is the distribution unobserved in practice and is denoted as *Truth*. The second benchmark is a linear interpolation of the same moments, but instead computed from a sample of households *drawn* from the *Truth*. Since these are draws from the *Truth*, their moments will vary due to sampling error, which depends on the sample size and sampling procedure. The linear interpolation is over the survey release dates for some variable and dataset X. This *naive* Sample X benchmark ignores the fact that these samples (*A*–*E*) are from the same underlying distribution with some dynamic structure.

Table 2 compares two benchmarks to the HANK simulated truth. We find that the DDFM generally improves on simple interpolation, with particularly large gains when the underlying distributional statistics are observed only infrequently or indirectly through other variables. In most cases, the correlation between the truth and DDFM implied distributional summary statistics are around 0.90. Sample *C* would be a demanding benchmark, observing consumption every quarter, making its interpolation almost the truth itself (0.89–0.99). The DDFM comes

Table 2: Time-Series Correlations: Model Estimates vs. HANK Simulated Truth

	Consumption		Income		Wealth	
	DDFM	Sample C	DDFM	Sample A	DDFM	Sample D
			By Consumption groups			
Bottom 40%	0.91	0.99	0.96	0.63	0.91	–
Middle 40–80%	0.90	0.94	0.60	0.34	0.85	–
Top 20%	0.92	0.91	0.95	0.64	0.90	–
Top 10%	0.93	0.90	0.93	0.64	0.90	–
			By Income groups			
Bottom 40%	0.91	0.99	0.95	0.57	0.74	0.67
Middle 40–80%	0.80	0.90	0.81	0.29	0.73	0.64
Top 20%	0.90	0.89	0.95	0.61	0.77	0.74
Top 10%	0.78	0.90	0.94	0.64	0.65	0.71
			By Wealth groups			
Bottom 40%	0.90	–	0.94	0.39	0.93	0.93
Middle 40–80%	0.86	–	0.84	0.36	0.90	0.81
Top 20%	0.92	–	0.89	0.46	0.92	0.37
Top 10%	0.92	–	0.85	0.57	0.91	0.48

Notes: The table shows correlations between two benchmarks and the HANK simulated truth. Column *DDFM* shows correlations between the HANK simulated truth and our model estimates (*DDFM*). Column *Sample X* shows correlations between the HANK simulated truth and a linearly-interpolated Sample $X \in \{C, A, D\}$. As in the figure, conditional moments are averages by *Percentile Group* (Bottom 40%, Middle 40–80%, Top 20%, and Top 10%) of the distribution of consumption, income, and wealth and all time series are relative to the economy-wide average. Sample B, which contains only income, is omitted for brevity. Sample E as well. Correlations are averages over the ten simulated economies.

close to this benchmark and does better for the top consumption and income groups. This suggests overall that the model approximates well the marginal distributions and correlational structure across variables, as well as their tails.[33]

Figure 6 provides the results of this exercise visually (for economy 1), plotting the evolution of consumption, income, and wealth by groups, relative to the economy-wide average. The figure compares the *DDFM* (solid red) estimates to the HANK simulated truth (dotted blue). For comparison, we plot *naive* estimates using Sample A in a light-pink line; simply a linear-interpolation of the black triangles which mark the observations in Sample A. This also visualizes the sparseness of the data that enters our estimation. In general, the movements across the DDFM and the HANK simulated truth are in sync, i.e. they are highly correlated, and around the same magnitudes. Using the naive linear-interpolation would miss key distributional movements, especially for consumption and income.

4.4 The Role of Observed Data in the Distributional Estimates

The experiments above have assessed the reliability of our method by examining how well it predicts deliberately withheld microdata observations. We now turn to a complementary

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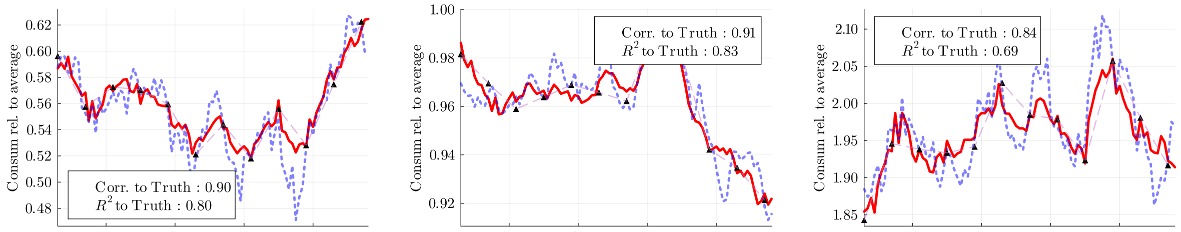
[33] Wealth by income is the exception. It moves considerably throughout the time horizon, especially the wealth of the middle 40%-80% income, and its comparatively lower correlation also reflects the steep income profile of wealth, which the truncated basis captures less well. Truncating at order 20 or higher improves correlations in exchange for computation time.

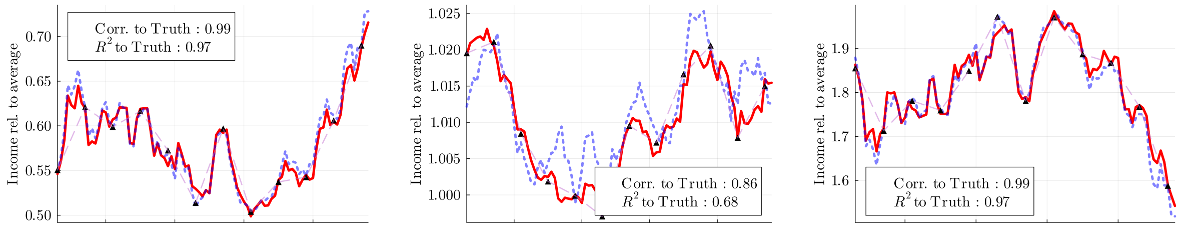
Figure 6: Model Estimates (in Red) vs. HANK Simulated Truth (in Blue)

Consumption by Percentile Group



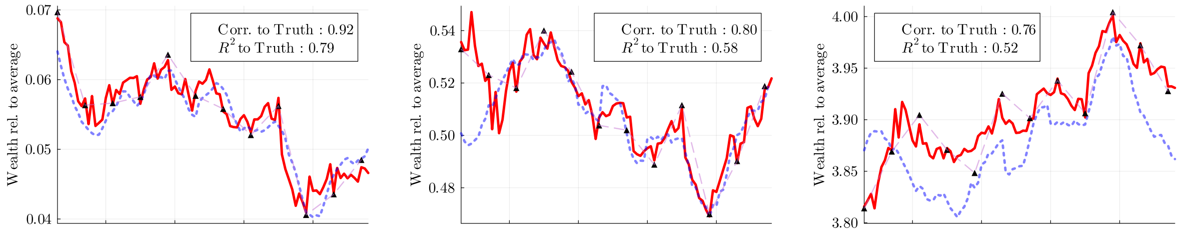
(a) Bottom 40% (b) 40–80% (c) Top 20%

Income by Percentile Group



(d) Bottom 40% (e) 40–80% (f) Top 20%

Wealth by Percentile Group



(g) Bottom 40% (h) 40–80% (i) Top 20%

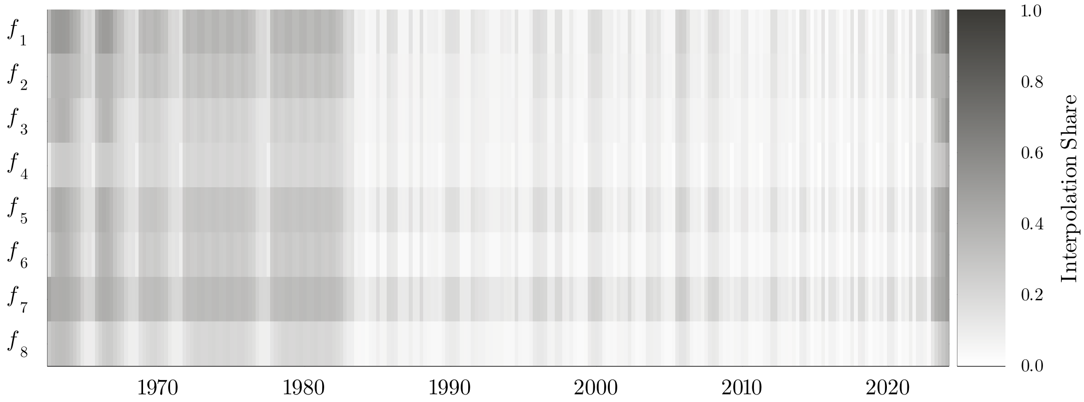
DDFM HANK Simulated Truth ▲ HANK Data Used

Notes: The figure reports estimates for consumption, income, and wealth relative to their economy-wide average. Estimates are shown for three groups: the bottom 40%, the next 40%, and the top 20%. Each panel compares model estimates with the HANK simulated truth. Model estimates (*DDFM*) from incomplete samples are shown as a solid red line; *HANK Simulated Truth* is shown as a dotted blue line. The black triangles labeled *HANK Data Used* mark the observations of Sample *A* (calculated based on the polynomial quantile estimator)—one of the five samples entering the DDFM estimates—and serve as the interpolation nodes of the naive benchmark (light-pink dashed line); the objects displayed follow Sample *A*'s measurement. *Corr. to Truth* reports the correlation between the red and blue lines. R^2 *to Truth* is the coefficient of determination from regressing detrended HANK simulated truth on detrended model estimates.

question about the resulting distributional estimates: how strongly are they pinned down by micro- and macro-data rather than by the model's dynamics? In Appendix I, we show how different data sources contribute to the smoothed estimates. However, knowing which sources contribute most does *not* tell us how strongly the observations constrain the estimates. Even when one dataset dominates, sparse observations may leave the estimates largely dependent on model dynamics.

To answer this question, we compare the uncertainty implied by the estimated dynamics alone with the uncertainty *remaining* after incorporating all available data. Let P^{∞}_{kk} denote the stationary variance of factor $f_{k,t}$ and $\sigma^2_{k,t|T}$ its Kalman-smoothed variance.

Figure 7: How Much Do the Observations Inform Each Factor?



Notes: Each cell reports the interpolation share $\iota_{k,t}$, the fraction of a factor's unconditional uncertainty remaining after conditioning on all observations. Dark cells indicate greater reliance on model dynamics; light cells indicate stronger information from observed data. The measure uses past and future observations and thus describes the historical panel rather than real-time estimates.

Figure 7 reports the interpolation shares for the eight factors, from f_1 at the top to f_8 at the bottom, defined as

$$\iota_{k,t} = \frac{\sigma^2_{k,t|T}}{P^{\infty}_{kk}} \in [0,1]. \tag{25}$$

On average, observations reduce the factors' unconditional uncertainty by 86 percent (interpolation share of 0.14), with shares below 0.2 in nearly three quarters of all factor-quarters. Darker areas indicate greater reliance on model dynamics, while vertical striping reflects individual survey waves. Shares are highest before 1984 (0.25 on average), when only SCF, PSID, and CPS are available; decline to 0.08 during 1984–2019 with the arrival of CEX and SIPP; and rise again after 2020, particularly near the sample end as fewer future observations are available to the smoother. Sparse observations, however, do not necessarily imply poor estimation performance: validation exercises above and in Appendix F show that estimates from data-sparse periods can nevertheless capture cyclical movements consistent with external evidence.

5 Distributional Dynamics over the Business Cycle

The focus of our paper is to provide a method for synthesizing dynamic distributional data from micro- and macro data inputs. The previous section showed that the method does this reliably: it recovers the underlying dynamics, it is consistent with the data it draws on, and it continues to produce dependable estimates when microdata are sparse or missing altogether. We can then proceed with two useful applications of the synthetic data. First, with both aggregate and *autonomous* distributional innovations in the model, we revisit the question studied by Bayer, Born, and Luetticke (2024): what role do aggregate shocks play in driving distributional dynamics? Second, we return to the paper's original motivation—empirical measure-

Table 3: Forecast Error Variance Decomposition on the Distributional Factors

Distributional Factor #	1	2	3	4	5	6	7	8
Spanned by aggregates	33%	32%	47%	42%	44%	42%	48%	59%
Autonomous	67%	68%	53%	58%	56%	58%	52%	41%

Notes: Table reports the forecast error variance decomposition (FEVD) for the eight estimated distributional factors over a five-year horizon. Entries are posterior medians over 2,000 parameter draws. The rows show the percentage of variation in each factor spanned by innovations to the aggregate block versus the autonomous remainder. Identification is via a blocked Cholesky with aggregates ordered first, which attributes all common variation to the aggregate block.

ment should both discipline existing economic theories—and uncover new facts that guide future ones. Using our quarterly joint distributions of consumption, income, and wealth, we examine how consumption evolves across income and wealth during the last three U.S. recessions. We address each application in turn.

5.1 Variance Decomposition of the Distributional Dynamics

Since our dynamic factor model is linear, we can examine how much of the variation in the distributional factors is spanned by aggregate dynamics and how much is autonomous—that is, variation in the cross-section not accounted for by the modeled aggregate block, potentially reflecting changes in risk-sharing, idiosyncratic risk, or the incidence and timing of transfers.[34]

We revisit this question within our much less structural dynamic factor model. Because the reduced-form innovations of the state equation are correlated (Ω is non-diagonal), the exercise requires an orthogonalization, and we deliberately choose one that avoids structural labels: a *blocked* Cholesky factorization with the aggregate block ordered first, formalized in Appendix I. The FEVD then asks how much of the forecast-error variance of each distributional factor is spanned by innovations to the aggregate block, and how much is orthogonal to all of them—autonomous in the sense above. No assumption is made about causal ordering within either block, and we do not interpret the orthogonalized innovations as structural shocks; where we write "shocks" below, orthogonalized innovations are meant.

Table 3 reports the results for the distributional factors. Aggregate innovations span 32 to 59 percent of the five-year forecast-error variance of the *cyclical* movements in the distributional factors, 44 percent on average (posterior medians). The autonomous component occupies the larger part for seven of the eight factors, 41 to 68 percent. Appendix I, Table 10 reports the aggregate share for more interpretable moments (e.g., the top 20 percent wealth, consumption by low income and wealth), using the linear mapping from factors to observables and observes the same pattern.[35]

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[34] Formally, the autonomous component is the part of the one-step-ahead forecast error of the distributional factors that is orthogonal to the contemporaneous forecast errors of the entire aggregate block. It is defined relative to our aggregate factor set: forces imperfectly captured by that block, or reaching the cross-section before the measured aggregates, are attributed to the autonomous component—biasing its share upward—while ordering aggregates first in the orthogonalization attributes all common variation to aggregates—biasing it downward. Autonomous innovations may still affect aggregates at later horizons through the feedback matrix C.

[35] We refer the reader to Appendix D for the transformations from factors to observables. Means and trends enter

The FEVD is an unconditional, average attribution. In Appendix I, using historical decompositions, show that the aggregate share is state-dependent: across different NBER recessions, we show that the aggregate share of the smoothed factor paths is about 55 percent on average, higher than the 44 percent average share here, indicating distributional dynamics track aggregate innovations more closely around recessions than over the cycle as a whole.

5.2 Consumption Dynamics over the last three recessions

The relative distribution of consumption losses in recessions has received considerable attention in the macroeconomic literature (e.g., Coibion et al. 2017; Cloyne, Ferreira, and Surico 2020; Bilbiie et al. 2025; Holm, Paul, and Tischbirek 2021; Fagereng, Holm, and Natvik 2021). The motivation is partly normative, in the sense of identifying groups that benefit or are adversely affected, and partly theoretical because a greater cyclicality of incomes of high-MPC households can destabilize the economy (Bilbiie 2020). Recessions are also where the balance between the aggregate and autonomous forces vary the most in our estimates: while autonomous innovations account for somewhat more than half of the fluctuations in the distributional factors and marginals on average, a historical decomposition of the factor paths shows that their role differs sharply across NBER recession windows—with aggregate innovations accounting for more than half of the factor variation in five of the seven recessions since 1969 (Appendix I).

Using the newly constructed synthetic data, we trace consumption dynamics along both the marginal and joint distributions of income and wealth across the last three U.S. recessions. We find that their distributional consequences are state-dependent: income and wealth position shaped the Dot-com bust, balance-sheet distress the financial crisis, and again the combination of income and wealth the COVID-19 recession. The most affected households are not always those the marginal distributions single out—in the financial crisis, the largest and most persistent relative decline is among asset-rich, low-income households, a group defined only in the joint distribution—challenging research strategies that rely on marginal distributions and unconditional cyclical elasticities.

What is known about these episodes comes from a variety of settings and econometric approaches. For the Financial Crisis, county-level variation ties consumption declines to collapsed housing net worth and leverage (Mian, Rao, and Sufi 2013), and household surveys show the wealth-poor cutting expenditure the most (Krueger, Mitman, and Perri 2016; Petev, Pistaferri, and Saporta-Eksten 2011). For COVID-19, card and bank-account data show high-income households cutting spending the most while transfers propped up spending at the bottom (Chetty, Friedman, and Stepner 2024; Cox et al. 2020). For the Dot-com recession, the evidence is limited to boom-era wealth-effect estimates for stockholders (Dynan and Maki 2001; Maki and Palumbo 2001) and to annual consumption-inequality series spanning the episode (Krueger and Perri 2006).

Relative to this literature, our contribution is threefold. First, the synthetic data reproduce the observations above, but within the lens of a single model—an additional, external validation

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the mapping as deterministic terms and therefore do not affect the decomposition.

of the method. Second, we extend the data to the *joint* income–wealth position: consumption groups like "low-income but indebted" or "asset-rich but low-income," which drive our results below, are unobservable in spending data (which lack wealth) and hard to isolate in single-survey studies. Third, our approach covers all episodes within a common framework, at quarterly frequency—and reaches back to episodes that predate modern spending data—the cross-recession comparison itself becomes possible, and it is this comparison that documents variation across recession episodes. Existing distributional evidence spanning comparably long horizons concerns secular trends (Heathcote, Perri, and Violante 2010; Meyer and Sullivan 2023); our estimates supply the business-cycle counterpart.

To begin, for the three recessions, we first consider the consumption dynamics of households along the marginal distributions of income and wealth, creating four groups: (1) the bottom 20%, (2) the 20% to 40%, (3) 40% to 80%, and (4) the top 20%.[36] These are the first two rows of Figure 8. For the final row, we then consider the joint distribution and estimate consumption dynamics for four key household groups. Group (i) consists of liquidity-constrained hand-to-mouth households, defined as the bottom 20% income *and* bottom 20% wealth; group (ii) is a population of asset-rich, low-income households which offers some resemblance to the wealthy hand-to-mouth, defined here as the bottom 40% of income but top 30% of wealth; group (iii) is the middle class, presumably prudent, patient "buffer-stock" households, located between the 40th and 80th percentiles in both income and wealth; and group (iv) comprises the richest households in terms of income and wealth, largely unconstrained, presumably Permanent Income Hypothesis households, residing in the top 20% in both income and wealth.[37]

For all groups, absolute consumption (not displayed) falls in a recession and rises in a recovery; therefore, we compare the business cycle dynamics of consumption for the different income and wealth groups, in each period t, *relative* to the household-wide consumption average at period t and then index these relative consumption dynamics to the quarter preceding the recession (the peak).[38]

In this sense, our goal is to understand how a group's consumption shifts relative to the aggregate (the average household) over a business cycle. The observable deviations from the aggregate provide a reduced-form summary of the distributional channels at work, which may in turn amplify or dampen the corresponding macroeconomic shocks. Figure 8 summarizes how these deviations are distributed across households depending on the nature of recessions.

Dot-com recession (Panels a, d, g). This recession resembles a financial wealth shock, though

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[36] The income measure we use includes transfers, but is not net of taxes.

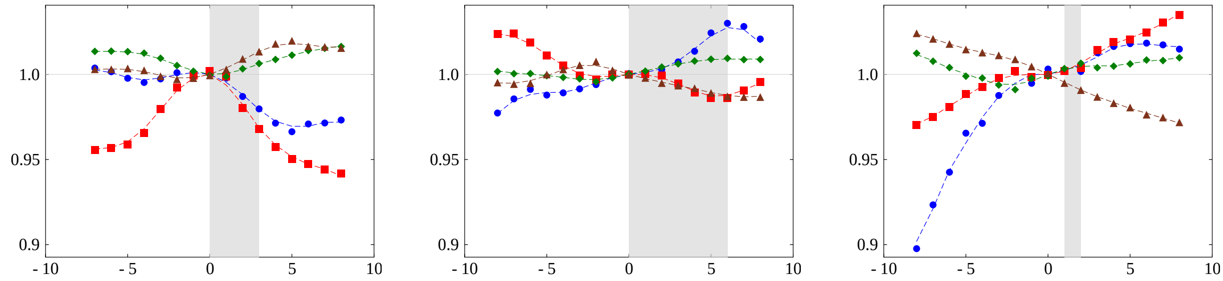
[37] For our analysis, groups are defined by contemporaneous ranks, so membership can change from quarter to quarter. This implies some *churning* relative to a true panel. Absent large asset price movements that affect portfolios, compositional changes within our broad bins can be expected to be small (cf. Kuhn, Schularick, and Steins 2020).

This interpretation is consistent with evidence that mobility is limited: positive shocks required for upward mobility are rare at the bottom, while large negative shocks are infrequent at the top (Guvenen et al. 2021); hand-to-mouth status is persistent (Aguiar, Bils, and Boar 2025); wealth positions are especially sticky due to persistent heterogeneity in returns (Fagereng et al. 2020); and earnings mobility is largely confined to adjacent deciles (Ettmeier, Kim, and Schorfheide 2024). Using coarse groups mitigates these concerns, though they become more salient when conditioning on multiple dimensions.

[38] We construct symmetric 3-quarter moving averages and use this moving average for the normalization to the pre-recession quarter.

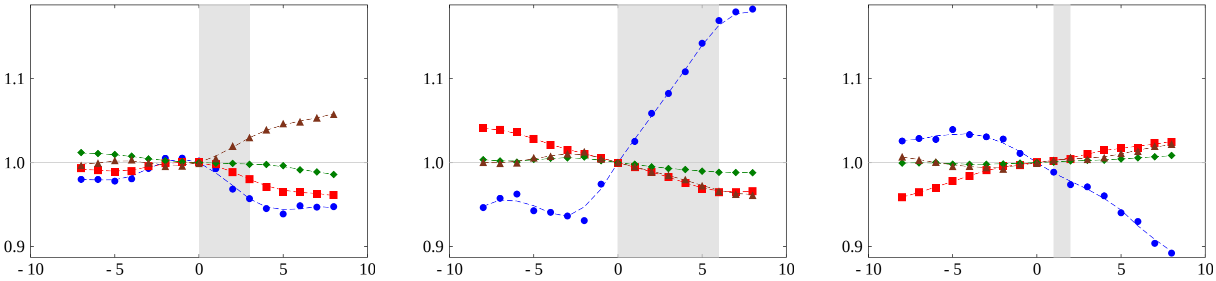
Figure 8: Comparison of Consumption Dynamics during Recessions

Consumption Dynamics by Income



(a) Dot-com (b) Global Financial Crisis (c) COVID-19

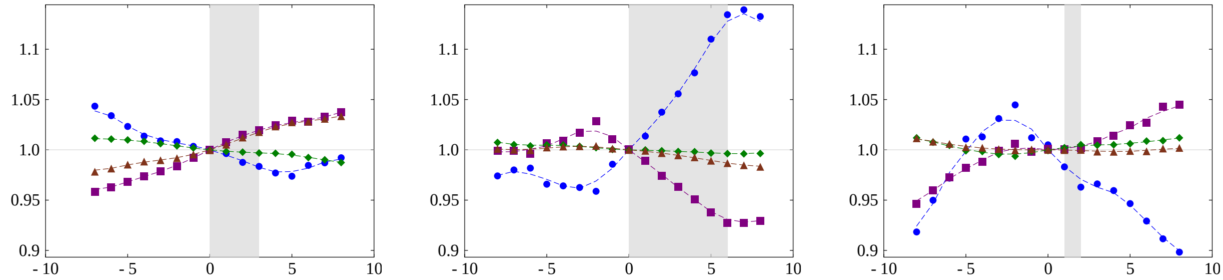
Consumption Dynamics by Wealth



(d) Dot-com (e) Global Financial Crisis (f) COVID-19

● Bottom 20% ■ 20% to 40% ◆ 40% to 80% ▲ Top 20%

Consumption Dynamics by Income and Wealth



(g) Dot-com (h) Global Financial Crisis (i) COVID-19

● Bottom 20% ◆ 40% to 80% ▲ Top 20% ■ Top 30% Wealth, Bot. 40% Income

Notes: Relative consumption dynamics during recessions along the income and wealth distribution. Consumption dynamics of each income/wealth group are shown relative to average household consumption. These relative consumption time series for each group are indexed to the peak. The vertical axis shows changes of consumption over time relative to the change of average consumption over time. The horizontal axis shows the time relative to the peak (= 0). The recessions are the Dot-com recession in 2001Q1, the Global Financial Crisis in 2007Q4, and the COVID-19 recession in 2019Q4. Shaded areas mark the NBER peak-to-trough quarters of each recession; the thin horizontal line marks the aggregate benchmark.

its relative burden falls on the bottom of the wealth distribution rather than on the wealthy: along income (Panel a), the bottom two income groups lose a moderate 3–6% relative to average while the top half of the distribution gains slightly. Along wealth (Panel d), the top 20% pull away, gaining about 6% relative to average, while all other wealth groups decline by 1–6%. The sharpest relative losses accrue to the bottom 20% wealth group—likely indebted households with high MPCs (Havranek and Sokolova 2020)—whereas the wealthiest, whose consumption is least tied to current income, absorb the equity correction without cutting relative consumption. Income position also matters, with also the bottom 40 percent suffering comparable relative losses to wealth. The asset-rich, low-income group (purple, Panel g) climbs steadily throughout the window, from below its pre-recession level to about 4% above it two years later, and the joint bottom group falls below the average. To our knowledge, no episode-specific account of consumption dynamics along the income and wealth distribution exists for this recession—the contemporaneous literature instead measured consumption wealth effects for stockholders during the preceding boom (Dynan and Maki 2001; Maki and Palumbo 2001)—so these estimates provide the first such account.

Global Financial Crisis (Panels b, e, h). The defining feature of that recession is its balance-sheet character. Correspondingly, consumption shares by income groups do not show much dynamics, within 2–3 percent in each direction. Yet by wealth, this differs. Compared to the onset of the crisis, the consumption share of the wealth poorest was lower before and increases afterwards (Panel e). It is tempting to interpret this as a result of lower interest payments freeing up resources for net borrowers. Yet, the rise is mainly compositional: the correlation of income and wealth weakens as leveraged home-owners with relative high incomes lose their net worth; the center of the balance-sheet channel of Mian, Rao, and Sufi (2013).[39] Among the remaining groups, the wealthiest 20% and the 20–40% wealth group cut relative consumption by 3–4%, and the asset-rich, low-income group (purple, Panel h) declines most persistently, by about 8%, consistent with wealth-dependent consumers bearing the asset-price collapse. Low-income households fared relatively well (Panel b), possibly supported by fiscal transfers.

COVID-19 recession (Panels c, f, i). During this last recession, the top 20% of the income distribution is the only income group that cuts relative consumption, by about 3–4% two years out (Panel c), in line with spending-data evidence that high-income households cut the most (Chetty, Friedman, and Stepner 2024). The other income groups gain 1–4% relative to average, the bottom 20% about 2%, likely due to the large government transfers (cf. Cox et al. 2020). By wealth (Panel f), the dispersion is even larger: low-wealth households lose persistently, about 9% relative to average with no recovery two years out, while every other wealth group holds at or above the average. The asset-rich, low-income group (purple, Panel i) gains about 3% relative to average, more than any other group in the joint panel, perhaps benefiting from transfers, asset-price rebounds, and insulation from labor-market shocks. The bottom 20% in income and wealth are the most exposed.

The key finding is *state-dependence*: the channels through which recessions affect households—

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[39] A fixed-composition (Laspeyres) decomposition (by income rank) attributes essentially all of the group's relative rise to this reshuffling (results available upon demand). Household-panel evidence that the wealth-poor cut expenditure the most (Krueger, Mitman, and Perri 2016; Petev, Pistaferri, and Saporta-Eksten 2011) is not in conflict: those studies follow fixed households, whereas our groups are wealth ranks whose membership churns precisely in balance-sheet recessions.

financial, balance-sheet, or labor-market—vary with the nature of the shock. Our findings suggest that during the Dot-com bust, wealth and income mattered; during the Global Financial Crisis, wealth mattered; and during COVID-19, it is again a combination of both income and wealth. Neither income nor wealth alone is a sufficient statistic for understanding consumption dynamics. This state-dependence cautions against research strategies that rely on uniform unconditional cyclical elasticities. Our new synthetic distributional data, spanning more than six decades, reveal these differences and provide empirical targets for developing and evaluating heterogeneous-agent models. By making these data publicly available, we enable future research to extend the analysis to additional recessions and explore alternative, more granular groupings within the joint distribution of income and wealth. The findings also underscore that stabilization policy needs to diagnose the primary channel through which a crisis operates and take a holistic view of the sum of all stabilization measures.

6 Conclusion

We develop a new general method for estimating high-frequency distributional dynamics across multiple economic variables. Our approach treats joint distributions as functional time series and represents their dynamics through a factor structure embedded in a Bayesian state-space model. This allows us to combine microdata sources that differ in frequency, variable coverage, sampling uncertainty, and measurement concepts, together with aggregate time series. The method thereby recovers the evolution of joint distributions even when direct microdata observations are sparse or unavailable, generating synthetic microdata for studying distributional dynamics at high frequency.

We apply the method to the joint distribution of consumption, income, and wealth in the United States, constructing the first quarterly estimates of this distribution from 1962 to 2024. A series of validation exercises demonstrates that our method reliably recovers distributional dynamics, both when observations from actual surveys are deliberately withheld and when the true underlying distribution is known from a simulated HANK economy. We make the new distributional data publicly available at distributionaldynamics.com, together with measures of estimation uncertainty and the contributions of individual data sources.

To illustrate one use of these new data, we examine how consumption evolves across the joint distribution of income and wealth during the last three U.S. recessions. We document substantial differences across episodes in which household groups experience the largest relative consumption changes, highlighting that neither income nor wealth alone is sufficient to characterize distributional consumption dynamics and that the importance of different dimensions of household heterogeneity varies across recessions. A complementary analysis shows that innovations in aggregate variables account for roughly 30 to 60 percent of fluctuations in the distributional estimates, about 45 percent on average, rising to more than half in recessions.

Beyond these initial applications, the new method and data allow researchers to examine specific episodes, study more granular groups within the joint distribution, and construct alternative distributional statistics to evaluate and discipline quantitative models. This includes estimating other joint distributions over different economic outcomes and populations. It can also be extended to short panels to study the high-frequency evolution of individual-level dynamics (Almuzara et al. 2025). More broadly, our contribution strengthens the feedback between economic theory and empirical measurement by providing new tools and data to study the distributional dynamics at the heart of modern heterogeneous-agent macroeconomics.

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Appendix

A Economic Fit

The series estimator fits a single set of coefficients $\theta_t^j = \left(\xi^j_{m,o_1,t}, \ldots, \kappa^j_{(o_1,\ldots,o_d),t}\right)$ by projecting the entire distribution of the variable(s) of interest onto an orthogonal polynomial basis; it is therefore a *global* method in the sense that one parameter set governs all quantile/copula regions simultaneously. If distributions were exactly composed of finitely many orthogonal polynomials, the global estimator would be exact. In practice, however, the truncation we impose may systematically over- or underestimate certain segments, potentially smoothing over economically relevant features.[40] Also, even without truncation, we might fit polynomials to regions with zero mass in the measure—generating, potentially, spurious variation. Both concerns are naturally important in our setting, where our task is to recover the cross-sectional and time-series variation in the entire distribution, *Economic Fit*, not just segments best approximated by our estimator.

Two natural questions arise: (1) where does our estimator systematically fall short and (2) can these weaknesses be accommodated. For (2), this means (a) can the current (global) polynomial weights be adjusted to better capture specific regions (without compromising other regions), such as the upper tail of the wealth distribution or do they already carry enough flexibility to capture these specific regions (and the others)? and (b) can the estimator accommodate distributions with point-zero mass? In this spirit, the appendix addresses two points. First, it evaluates the degree to which we miss local variations. Second, it shows how an economist with a theoretical prior about which parts of the distribution to emphasize (e.g., the top of the wealth distribution) could adjust the coefficients accordingly. As a byproduct, for a given estimator, it also provides detailed evidence of variation within the distribution via observing weight changes within the distribution. More on this latter point later.

A.1 Performance of the Estimator at the Tails

To investigate the first question, we begin with Figure 9, which compares two estimators: (1) the polynomial approximation as in the paper and (2) the empirical percentile function, shown with 99% interval bands. Each panel repeats the same percentile curve but varies its upper-tail cutoff, using final grid points $p \in \{0.995, 0.999, 0.9995, 0.9999\}$. This addresses, in one panel, where (1) falls short, with particular focus on the tails, motivated by the footnote. As such, an inset for each panel is provided, zooming into the top 5% of the respective distribution. For wealth, we observe near coincidence of the two estimators up to about $p = 0.9995$ in this diagnostics. Only beyond the 99.95th percentile—the top one-twentieth of one percent—does the series estimate begin to pull away from the empirical percentile function (itself subject to sampling error). Hence, any inference inside the top 0.05 percent of the distribution should be treated with particular caution.[41]

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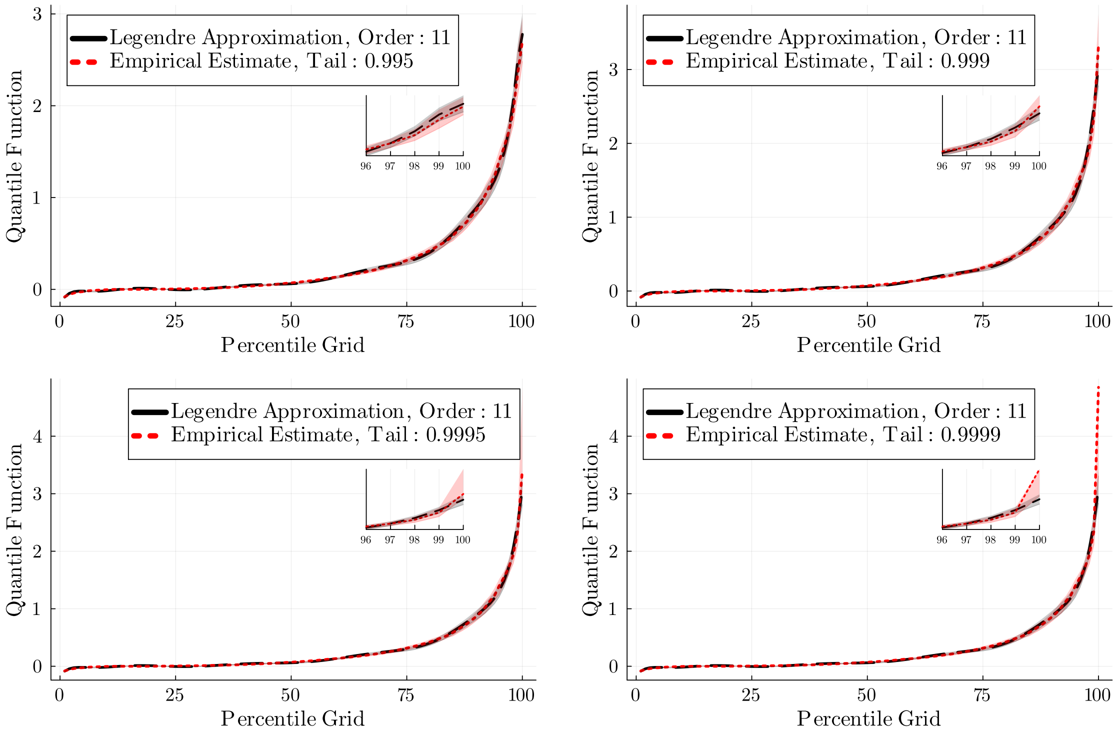
[40] Our projection forces parts of the distribution onto the square-integrable function space $\mathcal{L}^2$, even though those parts themselves do *not* belong to $\mathcal{L}^2$. Good examples are the upper tails of the consumption, income, and wealth, which are often modeled by Pareto laws. Vermeulen (2018) shows that wealth displays extremely heavy tails—so heavy that its variance is undefined—because the Pareto "alpha" parameters fall below 2. Likewise, Atkinson, Piketty, and Saez (2011) document that the U.S. income-tail Pareto beta rose from 1.82 to 3.42 between 1976 and 2007, implying a shift in the corresponding alpha from 2.22 down to 1.41. As Toda and Walsh (2015) emphasize, analyses involving fat-tailed data require great care: when higher-order moments fail to exist, the Central Limit Theorem breaks down and conventional confidence intervals—for instance, those on the top-10 percent wealth share—lose validity. We are grateful to a referee for highlighting this issue.

Figure 9: Wealth Tails



Notes: Figure shows four panels plotting the quantile function for wealth, evaluated on a grid $G = \{0.01, \ldots, 0.99,\ p\}$. Wealth, as defined, is scaled by its respective aggregate and transformed via an inverse hyperbolic sine function. The four panels differ in the plotted estimate of the tail, denoted by p, which appears in the legend. The inset of each panel zooms in on the tail, covering the last five percentiles. No estimator is evaluated for $p = 1.0$.

A.2 Local vs. Global Estimation

To answer the second question, we introduce a *local* estimator that computes a set of coefficients within a moving kernel window. By sliding the window along a percentile grid G, we obtain $|G|$ coefficient vectors, each tailored to the local variation within that window. The local estimator is flexible and well-suited for capturing local structure.[42] By comparing the global estimator's single set of coefficients with the collection of locally estimated coefficients, we can assess where and how they differ—especially around the tails of the distribution. Below, we describe the comparison procedure in detail and show that, for the economic data analyzed in this paper, our global estimator performs comparably to the local approach with minimal loss.

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[41] Results on consumption and income are readily available and can be provided upon request.

[42] Naturally, the estimator will perform poorly outside the window, but we make clear that we use the corresponding set of coefficients for a certain point when evaluating the percentile function at this point, i.e., the optimal coefficients in that window in a least squares sense.

Data Let $w_i \in \mathbb{R}$ denote the economic variable of interest for household i as described in the main body of the text. The variable, as before, is first scaled by the corresponding average (quarterly, per-household level) and stabilized with an inverse hyperbolic sine transform. Results presented here rely on the 2019 wave of the PSID and are presented for the different measures consumption, income, and wealth. For this reason, we drop all time indices t and measure indices m.

Global Legendre-series estimator To have the appendix section self-contained, we provide again our estimator. Truncating the polynomial order O to 11, the shifted basis of orthogonal Legendre polynomials, evaluated on the empirical cumulative distribution u_i^m, is $Q_o(u_i^m)$ for $o = 0, \ldots, 11$. Explicitly, $u_i^m = \frac{s_i}{\sum_i s_i}$ for s_i the survey weight corresponding to household i after sorting households based on variable w. Projecting w_i on the basis yields coefficients

$$\hat{\xi}_o^w := N^{-1} \sum_i w_i Q_o(u_i^w) \qquad \text{for } o = 0, \ldots, 11.$$

The estimated quantile function with these coefficients is

$$\hat{\Xi}_{glob}^{-1}(p) = \sum_{o=0}^{11} \hat{\xi}_o \, Q_o(p), \qquad p \in [0, 1], \tag{26}$$

where we use *glob* to denote that this is a global estimator and p some grid point in $[0, 1]$ for which we can evaluate the percentile function $\hat{\Xi}_{glob}^{-1}(p)$.

Local kernel Legendre estimator For the local estimation, let $p_g \in G$ be some probability grid point and $G = \{0.01, 0.02, \ldots, 0.995\}$. Let $K_h(u)$ denote a Gaussian kernel with bandwidth h[43]

$$K_h(u) = \frac{1}{h\sqrt{2\pi}} \exp\Big\{-\tfrac{1}{2}\big(u/h\big)^2\Big\}.$$

For a given survey weight of the PSID, s_i, we can generate the local weight $s_i^{(g)} = s_i \, K_h(u_i - p_g)$. Let $W_g = \operatorname{diag}(s_1^{(g)}, \ldots, s_n^{(g)})$ and $\Phi_{io} = Q_o(u_i)$. Solving the weighted least-squares system

$$(\Phi^\top W_g \Phi) \, \boldsymbol{\xi}^{local}(p_g) = \Phi^\top W_g \mathbf{y}$$

produces the *local* coefficient vector $\boldsymbol{\xi}^{local}(p_g) = (\xi_0(p_g), \ldots, \xi_{11}(p_g))^\top$, which is estimated separately for each grid point. The estimated *local* quantile function then uses, for each grid point evaluation, a new set of coefficients $\xi_o^{local}(p_g)$, generating

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[43] Results below are unchanged for various smaller bandwidths $h \in \{.005, .01, .05, .10\}$

$$\hat{\Xi}_{local}^{-1}(p_g) = \sum_{o=0}^{11} \xi_o^{local}(p_g) Q_o(p_g).$$[44]

Thus, we have the two extremes: the global approach of one set of coefficients to estimate the percentile function and the local approach of many sets of coefficients—localized throughout the entire distribution—to estimate the percentile function.

To see how the coefficients change across the distribution, independently of overall scale, each coefficient vector (from the different estimators) is transformed to unit ℓ^1 length:

$$\tilde{\xi}_o^{local}(p_g) \;=\; \frac{|\xi_o^{local}(p_g)|}{\sum_{o=0}^{11} |\xi_o^{local}(p_g)|}, \qquad \tilde{\xi}_o^{glob} \;=\; \frac{|\widehat{\xi}_o^{glob}|}{\sum_{o=0}^{11} |\xi_o^{glob}(p_g)|}.$$

One can then average over the vector of coefficients (over intervals of G) to assess how these weights fluctuate for, for example, the bottom of the wealth distribution versus the top. Figure 10 shows precisely this (results are analogous for consumption and income). In short, our estimator assigns weights almost identically to a local estimator that focuses on the 40th to 60th percentile. In every other case, it tends to assign more weight to the polynomials of order 4 to 8 in exchange for less weight on the first 3 polynomials. One might therefore conclude that we should instead use a different set of coefficients. Figure 11 provides evidence to the contrary.

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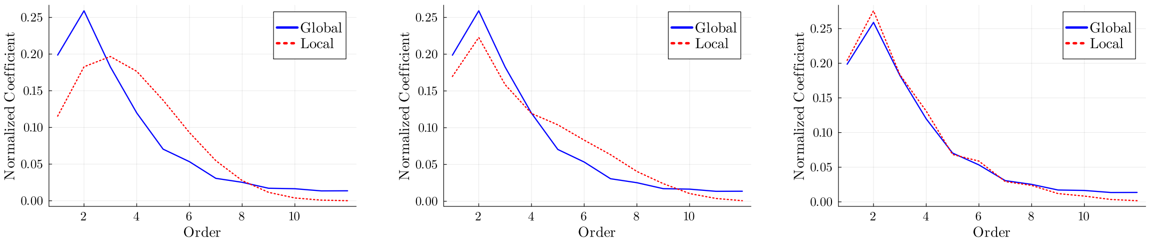
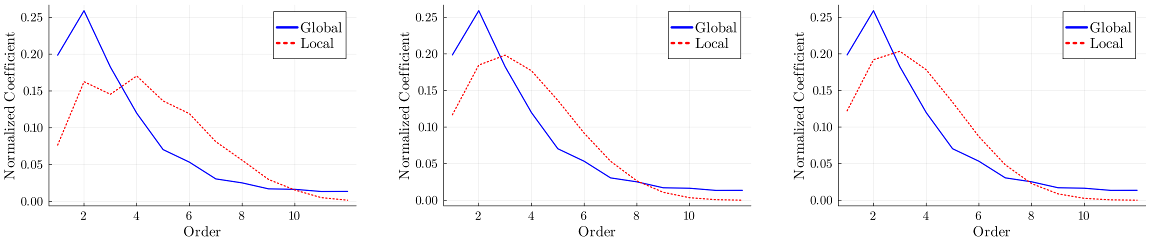
[44] Alluding to our comment before, one can alternatively estimate $\hat{\Xi}_{local}^{-1}(p) = \sum_{o=0}^{11} \xi_o^{local}(p^*) Q_o(p)$, where $\xi_o^{local}(p^*)$ are the weights associated for a selected grid point p^* e.g., the 99th percentile.

Figure 10: Local vs. Global Estimator: Coefficients



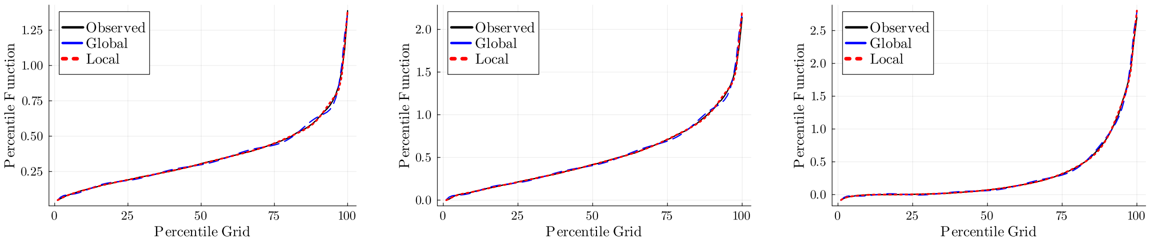
(a) Bottom 20 (b) 21st Percentile to 40th (c) 41st Percentile to 60th



(d) 61st Percentile to 80th (e) 81st Percentile to 100th (f) 91st Percentile to 100th

Notes: Figure shows panels comparing the global estimator to the local estimator for different segments of the wealth distribution. Each panel compares the contribution of each weight across the two estimators. Weights for the global estimator are the same across panels. Weights for the local estimator are calculated by averaging over the set of coefficients in that segment e.g., the Bottom 20 has 20 vectors of coefficients to average over.

Figure 11: Local vs. Global Estimator: Percentile Function



(a) Consumption (b) Income (c) Wealth

Notes: Figure shows three panels of percentile functions, comparing the global estimator to the local estimator, along with the empirical percentile function estimator. Percentile functions are evaluated over a set of grid points $G = \{0.01, \ldots, 0.99,\ 0.995\}$. The data are from the 2019 PSID wave.

A.3 Treatment of non-continuity of the marginal CDF

Some margins (e.g., earnings or financial assets) exhibit an atom at zero in addition to a continuous distribution over $\mathbb{R}_+$. Let Z_{mt} denote marginal m at time t and

$$\pi_{mt} := P_t(Z_{mt} = 0).$$

Write the CDF as a mixture of the non-differentiable part and (c)ontinuous part

$$\Xi_{mt}^{Z}(z) = \pi_{mt}\mathbf{1}\{z \geq 0\} + (1 - \pi_{mt})\,\Xi_{mt}^{Z,c}(z), \qquad \Xi_{mt}^{Z,c}(z) := P_t(Z_{mt} \leq z \mid Z_{mt} > 0).$$

The unconditional quantile function is

$$\Xi_{mt}^{Z,-1}(u) = \begin{cases} 0, & u \leq \pi_{mt}, \\ \Xi_{mt}^{Z,c,-1}\Big(\dfrac{u - \pi_{mt}}{1 - \pi_{mt}}\Big), & u > \pi_{mt}, \end{cases} \tag{27}$$

where it is 0 when the household's rank is below π_{mt}, positive otherwise, where the positive part is rescaled to mass $1 - \pi_{mt}$. Any strictly increasing transform satisfies these cases e.g., the inverse hyperbolic sine—the transformation we use in practice. Empirically, we estimate the mixed distribution in two steps: (i) estimate the point mass

$$\hat{\pi}_{mt} = \frac{\sum_i s_{it}\mathbf{1}\{Z_{mit} = 0\}}{\sum_i s_{it}},$$

and (ii) using only $Z_{mit} > 0$, estimate the conditional quantile function of $X_{mit} = T(Z_{mit})$ via the Legendre expansion

$$\Xi_{mt}^{X,c,-1}(u) \approx \sum_{o=0}^{O} \xi_{mot}^{c,X} Q_o(u).$$

The coefficient vector entering the factor extraction carries only the conditional coefficients $\xi_{mot}^{c,X}$, so its dimension—and the entire state-space apparatus—is unchanged relative to a continuous marginal. The point mass $\hat{\pi}_{mt}$ is precisely measured by the participation surveys and is treated as observed: it enters at the reconstruction stage through (27), where the flat segment and the rescaling of the conditional quantile function reassemble the mixed distribution.

A.4 Treatment of the Copula with semi-continuous margins

Let Z again denote the semi-continuous margin. For the three-dimensional copula, this means there is a flat region along the Z margin, which renders differentiability in that neighborhood infeasible. To deal with this, there are three sets of dependence coefficients to keep track of: those among holders ($Z > 0$), those among non-holders ($Z = 0$), and those of the population copula. This may seem daunting to estimate, but due to properties of the estimator, this is estimated quite efficiently. Throughout, let u_Y, u_W and u_Z denote population ranks (computed on the full sample, with the zeros of Z occupying $[0, \pi_t]$), and let $u'_Z = (u_Z - \pi_t)/(1 - \pi_t)$ denote the corresponding rank within holders.

For example, when expanding the copula in the orthonormal shifted Legendre basis, the coefficients would be $\kappa_{jkl,t} = E_t[Q_j(u_Y)\, Q_k(u_W)\, Q_l(u_Z)]$. Since $Q_0 \equiv 1$, the coefficients with $l = 0$ *are* the population (Y, W)-copula coefficients κ_t^{YW}—the same objects measured by surveys that do not observe Z. This is exemplified in our Section 3.

The holder and non-holder objects $\kappa^{YWZ}|_{YW}$ and $\kappa^{YW,0}$ are therefore coefficients of the densities of population ranks conditional on holder status, not copulas of their own, and (28) below is the law of total expectation applied to $E_t[Q_j(u_Y)Q_k(u_W)]$. The population, holder, and non-holder (Y, W) coefficients satisfy the following identity

$$\kappa_t^{YW} = \pi_t\, \kappa_t^{YW,0} + (1 - \pi_t)\, \kappa_t^{YWZ}\big|_{YW}, \tag{28}$$

where $\kappa_t^{YWZ}|_{YW}$ collects the (Y, W) coefficients among holders and $\kappa_t^{YW,0}$ those among non-holders.

Estimation therefore splits the coefficient array by the order of the marginal Z: the $l = 0$ entries are estimated on the full sample (the population copula), while the $l \geq 1$ entries are estimated on holders only, with Z entering through the within-holder rank u'_Z — the same rescaling as in (27). Like $\hat{\pi}_{mt}$, the holder (Y, W) slice $\kappa_t^{YWZ}|_{YW}$ is measured directly from the participation surveys and treated as observed. At reconstruction, we can recover the non-holder rank density, which follows from (28),

$$\kappa_t^{YW,0} = \frac{\kappa_t^{YW} - (1 - \pi_t)\, \kappa_t^{YWZ}\big|_{YW}}{\pi_t},$$

and the copula density on the unit cube is reassembled as the mixture

$$c_t(u_Y, u_W, u_Z) = \begin{cases} c_t^{YW,0}(u_Y, u_W), & u_Z \leq \pi_t, \\ c_t^{YWZ}\Big(u_Y, u_W, \dfrac{u_Z - \pi_t}{1 - \pi_t}\Big), & u_Z > \pi_t, \end{cases}$$

with values on the Z-margin from the conditional quantile function. Within the zero block, u_Z is assigned uniformly on $[0, \pi_t]$, independently of (u_Y, u_W); this is the standard randomized probability-integral transform for an atom. The non-holder coefficients are formed only when $\hat{\pi}_t > 10^{-10}$ (otherwise the zero block carries no mass), and, like all truncated series densities in the paper, the recovered density need not be pointwise nonnegative; we report its exact integrals over cells.

B Estimation of Factor Structures and Marginal Data Densities

A key input to our procedure is the mapping from factors to observables, defined by the projection matrix Γ. To estimate Γ, one method would be based on complete data alone. Complete here will mean there is no missing data in the spatial sense i.e., the data reports consumption, income, and wealth. For our case, this means using exclusively the PSID data to estimate Γ

based on the PCA of $\tilde{\boldsymbol{\theta}}^{PSID}$. We consider different scenarios where we retain a different number of factors each time. Specifically, for this approach, we estimate the model with 3, 6, 7, and 8 distributional factors, obtained from the 12 PSID waves; however, this approach runs the risk of not fully representing the entire subspace that characterizes the evolution of the joint distributions because of the infrequent releases of the PSID data.

The literature has suggested alternatives for taking into account *incomplete* data in factor models. Here, this would mean surveys that only collect data on some, but not all marginals e.g., only income and wealth as in the SCF. In particular, we explore two alternative estimators for Γ: we consider the Tall-Wide algorithm of Bai and Ng (2021) and an EM algorithm. We compare the quality of the model estimates under the alternative methods using marginal data densities for model comparison.

B.1 Tall-Wide Algorithm of Bai and Ng (2021)

An alternative approach to incorporating an additional block of data to the complete data is provided by Bai and Ng (2021). The paper tackles this problem by identifying two blocks of data within some larger T by N matrix—a tall block (data observed for all periods) and a wide block (time periods for which entire distribution is observed). For our setting, this means we add a tall block to our estimation of Γ, while the wide block would be what we call the complete data.[45] The tall block we add incorporates data on lower dimensional copulas of (income, consumption) and (income, wealth), as well as additional data on the marginals. This would be variation coming from both the CEX and SCF.

This alternative estimator for Γ, since it relies on more data, may improve model fit. For our application, we extract the factors that explain 80% of the summable variation, as well as 85%. This corresponds to 10 and 12 factors; however, as mentioned before, it is unclear the degree of variation shared among this set of factors relative to the original set of factors. Furthermore, keeping more factors from this estimator runs the risk of over-parameterizing the model. Table 4 presents the marginal likelihoods of these *TW* projection matrices. Results suggest these models have inferior fit than models that only rely on complete data.

B.2 EM-completed loadings.

As a second alternative to the complete-case projection matrix, we estimate the loadings on the full, unbalanced panel of survey coefficients. Missing entries are filled by the expectation-maximization (EM) algorithm for principal components: starting from the unconditional mean, the algorithm alternates between imputing the missing coefficients from the current low-rank fit and re-estimating the factors and loadings by principal components on the completed panel, until the imputations converge. Banbura and Modugno (2014) develop the likelihood-based counterpart for arbitrary patterns of missing data. Unlike the complete-case estimator, which uses only the survey waves in which every coefficient is observed, the EM-completed loadings draw on all waves of all surveys. The data reject this alternative clearly. With eight distributional and eleven aggregate factors, the EM-completed specification attains a log marginal data density worse than that of the complete-case baseline, both evaluated on the common observation vector (Table 4).

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[45] It is important to note that, given some data, such an estimator can consistently estimate the common component without making any assumptions on the nature of missingness. We refer the reader to the paper for more details on how the exact procedure is performed.

B.3 Marginal Data Densities and Optimal Factor Structures

We estimate various versions of the state space model for different factor loadings Γ, which differ both in the number of factors and in the estimation of Γ itself, and for different numbers of aggregate factors. For each estimated model, we calculate the marginal data density (MDD) in order to discriminate between these alternatives. The models considered rely on mostly the same data, but differ in the size of the parameter space.[46] The MDD effectively internalizes these two features: it selects the model for which we can expect the best forecasting performance, while penalizing models with larger parameter spaces through a lower density.

The marginal data densities $p(\boldsymbol{y})$ are estimated using Geweke's modified harmonic mean estimator. Although standard, this estimator raises two concerns. First, given the dimensionality of the model, the posterior may not be well approximated by a Gaussian distribution. Second, the estimator may not be numerically stable, as it requires the inversion of large matrices. We therefore also estimate the density using a bridge sampler (Meng and Wong 1996), which is numerically more stable and may better internalize the shape of the posterior.

Table 4 covers the span of proposed models that were ultimately assessed, as well as the marginal data densities over the different estimators. First, we find that the MDD is most elastic to the number of distributional factors, as opposed to the number of aggregate factors. Among models with 11 aggregate factors, the model with the projection matrix from the complete PSID data, labeled *Standard Projection Matrix*, achieves the highest MDD with the maximum number of 8 factors.[47] Using the factor loadings Γ from the TW-algorithm of Bai and Ng (2021), or loadings completed by the EM algorithm, yields a lower MDD and thus a worse model fit.

B.4 Alternative Aggregate Blocks

The baseline summarizes the aggregate data by the first eleven principal components of all series (Section 4.1). Because each candidate aggregate block changes the measurement vector, marginal data densities are not comparable across blocks. Table 5 therefore compares the candidates by the posterior mean of the log-likelihood contribution of the microdata alone, which holds the survey observations fixed. We consider two alternatives: eleven group-specific first

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[46] Since aggregate factors enter as measurements every quarter, their number changes the observation vector and hence the likelihood contributions. Figure 16 provides suggestive evidence on the strength of this influence. We therefore also report MDDs evaluated on a common observation vector; see below.

[47] Models with different numbers of aggregate factors have different observation vectors, so their MDDs are not directly comparable. The columns labeled *Common observation vector* add, for a model with $k < 15$ aggregate factors, the log marginal density of the aggregate scores $k+1, \ldots, 15$ under independent AR(1) processes, so that all models are evaluated on the same data. On this common basis, the model with 3 aggregate factors attains a higher MDD than the model with 11.

Table 4: Model Comparison

	Own observation vector		Common observation vector	
Model	Harmonic Mean	Bridge	Harmonic Mean	Bridge
		TW Projection Matrix		
12 distributional factors, 11 agg. factors	−88397.5	−88300.7	−89745.8	−89649.0
10 distributional factors, 11 agg. factors	−88057.4	−87971.0	−89405.6	−89319.3
		Standard Projection Matrix		
3 distributional factors, 3 agg. factors	−84585.6	−84577.4	−88508.1	−88499.8
6 distributional factors, 11 agg. factors	−73085.8	−73032.0	−74434.0	−74380.3
7 distributional factors, 11 agg. factors	−64587.4	−64527.8	−65935.7	−65876.1
8 distributional factors, 3 agg. factors	−51880.3	−51846.2	−55802.7	−55768.6
8 distributional factors, 11 agg. factors	−55575.8	−55506.9	−56924.1	−56855.1
8 distributional factors, 15 agg. factors	−59747.3	−59620.5	−59747.3	−59620.5
		EM-completed Projection Matrix		
8 distributional factors, 11 agg. factors	−88029.0	−87949.1	−89377.3	−89297.4

Notes: The table reports the log of the marginal data densities (MDD) across different model specifications. The MDD is estimated using two estimators: (1) Harmonic Mean and (2) a Bridge sampler. TW Projection Matrix are models estimated using the Tall-Wide algorithm of Bai and Ng (2021). Standard Projection Matrix is from a PCA estimator on complete cases; EM-completed uses EM-completed loadings. Since the aggregate block is observed through its first k principal-component scores, models with different numbers of aggregate factors are densities over different observation vectors (*Own observation vector*). The *Common observation vector* columns add, for a model with $k < 15$ aggregate factors, the log marginal density of the scores $k+1, \ldots, 15$ under independent AR(1) processes with conjugate priors, so that all rows are densities over the same data. Higher values indicate greater marginal likelihood.

principal components, one per category of aggregate series, and the first two components of each of five broad categories (ten factors). The baseline attains the highest survey-only likelihood, but the differences are small relative to the spread across chains.

Table 5: Survey-only Log-likelihood for Alternative Aggregate Blocks

Aggregate block	Aggregate factors	Survey log-likelihood
Principal components of all series (baseline)	11	−50,446
One component per category	11	−50,452
Two components per category, five categories	10	−50,598

Notes: Posterior mean of the log-likelihood contribution of the survey (distributional) measurements, which is comparable across aggregate blocks because the survey observation vector is the same in all three models. Each model is estimated with the same priors, seed and number of sampler iterations as the baseline.

C On the Bayesian Implementation

We need to estimate seven objects: from the state equation, the factor-autoregression matrix A, the loading matrix B from aggregate controls to distributional factors, the loading matrix C from distributional factors to aggregate factors, the diagonal of D, which governs the autocorrelation of the aggregate factors, and the variance-covariance matrix of the innovations to the factors Ω; and from the measurement equation, the variance-covariance matrices $\Sigma_j^{1/2} S^j \Sigma_j^{1/2\prime}$ and Σ_Y of the measurement errors. In a first step, the covariance structure $\Sigma_j^{1/2}$ is estimated outside the time-series model, as noted above, leaving only the scaling matrices S^j to be estimated within the model. Given the size of the A, B, C, D, Ω, S^j, and Σ_Y matrices, we use a Bayesian approach to estimate the system. This imposes regularization via priors on the parameters, as well as hyperpriors on the hyperparameters, to avoid overfitting the data.

This results in a hierarchical Bayesian state-space estimation with mixed-frequency data, conditional on the first-stage estimates. We collect all parameters and hyperparameters into a single parameter vector $\boldsymbol{\psi} = (\boldsymbol{\psi}_{\text{par}}, \boldsymbol{\psi}_{\text{hyper}})$, estimated jointly. The remainder of this section details the specific prior distributions on $\boldsymbol{\psi}$, first for the state equation parameters and subsequently for those of the measurement equation.

Priors on A, B, C, D For the state equation, we stack the parameters as $\psi_\Phi := (\text{vec}(A)', \text{vec}(B)', \text{vec}(C)', \text{diag}(D)')'$ and impose the following priors:

$$\begin{aligned}
&\psi_\Phi \sim \mathcal{MN}(\mu_{\text{Minn}}, V_{\text{Minn}}), \\
\mu_{\text{Minn}}: \quad & A_{ij} = \begin{cases} \kappa_{2,i}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad B = \mathbf{0}, \quad C = \mathbf{0}, \quad D_{ii} = \kappa_3, \\
& V_{\text{Minn},ii} = \begin{cases} \kappa_0 / l^2, & \text{own lags}, \\ \frac{\kappa_0 \kappa_1}{l^2} \frac{\hat{\sigma}_{ii}^2}{\hat{\sigma}_{jj}^2}, & \text{cross-lags}, \end{cases}
\end{aligned} \tag{29}$$

First, we impose a Minnesota prior on A, B, C, and $\text{diag}(D)$. We set the vector of expected values μ_{Minn} so that all *but* the autocorrelation terms in A and D have an expected value of zero. This implies that B and C, as well as the off-diagonal elements of A, are shrunk toward zero if they are not needed to describe the data. The expected values for the autocorrelations (the main diagonal of A and D) are governed by hyperparameters.

For the state equation, there are $|\boldsymbol{\psi}_{\text{hyper}}| = 5 + r$ hyperparameters that are jointly estimated with $\boldsymbol{\psi}_{\text{par}}$. First, the specification of μ_{Minn} requires hyperparameters $\kappa_{2,1}, \ldots, \kappa_{2,r}$, which sets the prior persistence of *each* of the r distributional factors separately, and κ_3 that of the aggregate factors.[48] A single persistence parameter shared by all factors would imply either a set of fast (e.g., those loading on quarterly income) or slow factors (e.g. those loading on quarterly wealth). A separate target per factor lets each find its own persistence and potentially fit better income and wealth dynamics. We work with unconstrained parameters and map them to $(-0.99, 0.99)$ via a scaled inverse-logit. The hyperprior on the aggregate target is diffuse, $\kappa_3 \sim \mathcal{N}(0, 5)$. The factor-specific targets have $\kappa_{2,i} \sim \mathcal{N}(3, 1.5)$ on the unconstrained scale, which centers the implied persistence near 0.9 and puts 0 about two standard deviations below.

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[48] Applying the same factor flexibility to the aggregate factors did not change results. Results available upon request.

Prior on Ω For the prior on Ω, we impose a *separation strategy* (see e.g., Barnard, McCulloch, and Meng 2000) and decompose Ω into a diagonal matrix of standard deviations Ω_σ and a correlation matrix Ω_{corr}, i.e., $\Omega = \Omega_\sigma \times \Omega_{corr} \times \Omega_\sigma{}'$. This avoids the a priori dependence between variances and covariances normally imposed in Inverse-Wishart setups (Barnard, McCulloch, and Meng 2000). For the innovations, Ω_σ is further decomposed into Ω_F and Ω_Y, allowing different priors on the standard deviations of distributional innovations and aggregate innovations. We specify these priors on the diagonal elements $\Omega_{\sigma,ii}$.

For the parameters $\Omega_{\sigma,ii}$, we first draw from a set of Normal distributions and map these parameters to $\mathbb{R}^+$ using the soft-plus transformation, denoted as $T(\cdot)$. This provides flexible exploration for the optimizer by avoiding exponential jumps from small perturbations in these parameters, while also preventing acceptance rates from being driven by out-of-bounds draws (Chapter 55, Stan User Guide). For $\Omega_{F,ii}$, the mean μ_{Ω_F} is set such that the post-transformation entry—the innovation standard deviation—is centered at $1 - \kappa_{2,i}^2$. Together with the prior on the autocorrelation (in the matrix A), this centers the a priori long-run variance of each distributional factor at $1 - \kappa_{2,i}^2$, below the unit-variance normalization of the factors: a conservative centering that shrinks factor volatility rather than presuming the normalization holds a priori. The same construction applies to the standard deviations of the innovations to the aggregate factors, $\Omega_{Y,ii}$, using κ_3. Together, there priors are:

$$\begin{aligned} T^{-1}(\Omega_{F,ii}) &\sim \mathcal{N}(\mu_{\Omega_F}, \sigma_F^2), \\ T^{-1}(\Omega_{Y,ii}) &\sim \mathcal{N}(\mu_{\Omega_Y}, \sigma_Y^2). \end{aligned} \tag{30}$$

For the estimation of the correlation matrix Ω_{corr}, several constraints must be satisfied simultaneously, which may be difficult to enforce in optimization: the matrix must be symmetric, have a unit diagonal, and be positive definite. To circumvent this, we employ a reparameterization strategy:

$$L_{\text{chol}} \sim \text{LKJChol}(r + q, \eta), \tag{31}$$

The key insight is to work with the Cholesky factor L_{chol}, where $\Omega_{corr} = L_{chol} \, L_{chol}'$. Following the methodology of Lewandowski, Kurowicka, and Joe (2009, hereafter: LKJ), we construct L_{chol} from an unconstrained vector of real numbers, $v \in \mathbb{R}^{(r+q)(r+q-1)/2}$, and evaluate our prior on this Cholesky factor. The LKJ-Cholesky prior is governed by a single shape parameter, $\eta \geq 1$, which controls the expected correlation strength between aggregate and distributional innovations.

Prior on S and Σ_Y For the distributional measurements, we specify independent priors for the measurement error variances for each dataset and object, s_ζ^j. Recall that estimating the diagonal of S^j would imply one variance parameter per coefficient n. As such, we impose the restrictions explained in the previous section, assigning one variance parameter s_ζ^j per object ζ and dataset j. This means a maximum (since some measures are unobserved) of $J \times (d+1)$ variance parameters. To ensure strict positivity of s_ζ^j while maintaining efficient sampling, we again parameterize each variance using a softplus transformation and place Normal priors on the unconstrained parameters $T^{-1}(s_\zeta^j)$. For the distributional data, we assign a weakly informative prior centered such that the median variance is 1.0, with a standard deviation of 2.0. This implies, first, that on average there is only sampling uncertainty and no additional measurement error reflecting conceptual differences, but also that $\hat{\Sigma}_j$ is itself an estimate, allowing each s_ζ^j to deviate from 1.[49]

$$T^{-1}(s_\zeta^j) \sim \mathcal{N}(0.54, 2^2), \qquad\qquad T^{-1}(\Sigma_{Y,ii}) \sim \mathcal{N}(-7.6, 0.02^2). \tag{32}$$

For the aggregate factors, we impose highly informative priors on the variances $\Sigma_{Y,ii}$, reflecting that the data underlying these factors are precisely observed aggregates. The priors are centered at a near-zero variance, with a very narrow standard deviation of 0.02.

Likelihood and Sampling With these priors on $\boldsymbol{\psi}$, we obtain the model likelihood $p(\tilde{\tilde{\boldsymbol{\theta}}}, \hat{Y}|\boldsymbol{\psi})$ of the stacked observation vector in (24) using a Kalman filter. The log posterior density is then calculated as the sum of the hyperprior log-probability, the prior log-probability, and the data log-likelihood. As previously mentioned, the standard approach of using quasi-maximum likelihood with EM algorithms may converge to a local optimum in our high-dimensional setting. We instead initialize with a global differential-evolution search and, to sample from the potentially complex, multimodal, high-dimensional posterior distribution, employ the Differential-Independence Mixture Ensemble (DIME) sampler from Boehl (2024), whose proposals are designed to move between modes of the posterior. No method guarantees locating the global mode in finite time, but this reduces the risk of reporting a local optimum and, unlike a point estimate, characterizes parameter uncertainty.

The sampling procedure consists of an ensemble of $4n$ chains (for n the size of the parameter vector). To speed up adaptation, 10% of chains are initialized at a tentative mode from a separate optimization procedure; the remainder are initialized from the priors though it should remarked that this is unnecessary. For the baseline, the size of the parameter space is $|\boldsymbol{\psi}| \approx 500$, which implies $4(500) \times 1600 = 3,200,000$ chain iterations, of which 25 percent are retained. The sampler uses a single tuning parameter χ that governs the mixture between the local and global transition kernels; we set $\chi = 0.1$. For stability, we check that draws of Φ have a spectral radius below 0.995; draws outside this radius are rejected. Figure 12 shows the traces of the (scaled) log-likelihood for all chains and the stabilization of the sampled log-likelihood.[50] Additional diagnostics are reported in the figure notes.

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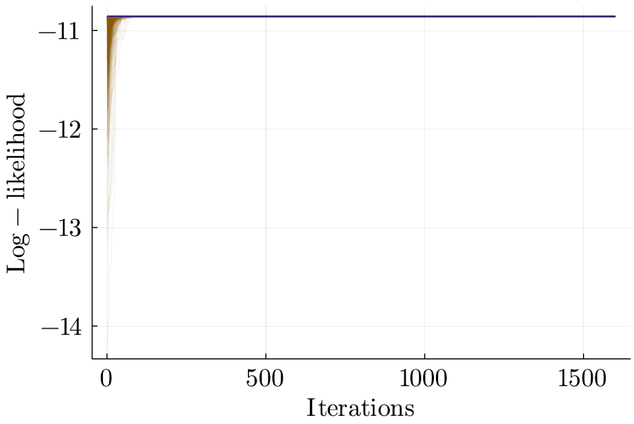
[49] Identical priors across datasets mean that we do not a priori prioritize a conceptual measure for object ζ in one dataset over another. Setting different priors is generally possible if a particular measurement concept should be prioritized on theoretical grounds.

Figure 12: Converging Chains



Notes: Figure shows the evolution of the ensemble over approximately 3,200,000 chain iterations in terms of (scaled) log-likelihood; the last 25% of draws of each chain are retained and shown. The blue line shows the mode of an ensemble at a given iteration. The sampler also reports the log-weight on the history of the proposal distribution and the standard deviation of likelihoods. Early on, log-weights are positive, indicating adaptation; after convergence they are close to zero (around 10^{-7}). Standard deviations are about 1% of the mode log-likelihood. Acceptance rates are in the 40%–45% range, consistent with a good initialization. Refer to the text for model specifications.

D Reconstructing Distributional Data

This appendix details the reconstruction step of Section **??**: how the smoothed factors are turned into quantile functions, copulas and, from these, the synthetic cross-sections used in Section 5. Each observation X_{it}^j corresponds to a representative household in the synthetic microdata and contains its probability weight and mean levels of consumption, income, and wealth. Each representative household i stands for a group of households defined by a cell of ranks $\mathcal{U}_i \equiv \prod_{m=1}^d U_i^m$, with one range $U_i^m \subseteq [0,1]$ per variable (e.g., deciles). Integrating the copula over the cell gives the representative household's probability weight. Integrating each quantile function, after undoing the inverse-hyperbolic-sine transformation, against the copula over the same cell and dividing by this weight gives the mean level of that variable for the representative household.[51] Below is precisely this combination, for the example of the distribution of consumption, income, and wealth, which together forms a single row of the time series of weighted synthetic cross-sections:

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[50] Traces of all estimated models can be provided upon request.

[51] Here $\bar{w}_{mt}$ denotes the scale used in the transformation of variable m in Section 3.2.1, the aggregate per-household level of consumption, income, or wealth. Because the copula is a polynomial, the integrals factor into one-dimensional integrals over the cell's rank ranges. Cells with negligible mass carry the average of each variable over its own rank range instead.

$$X^j_{it} = \begin{pmatrix} \omega^j_{it} \\ c^j_{it} \\ y^j_{it} \\ w^j_{it} \end{pmatrix} = \begin{pmatrix} \int_{\mathcal{U}_i} d\hat{C}^j_t(\mathbf{u}) \\ \frac{1}{\omega^j_{it}} \int_{\mathcal{U}_i} \bar{w}_{ct} \sinh\big(\hat{\Xi}^{-1}_{jct}(u_c)\big)\, d\hat{C}^j_t(\mathbf{u}) \\ \frac{1}{\omega^j_{it}} \int_{\mathcal{U}_i} \bar{w}_{yt} \sinh\big(\hat{\Xi}^{-1}_{jyt}(u_y)\big)\, d\hat{C}^j_t(\mathbf{u}) \\ \frac{1}{\omega^j_{it}} \int_{\mathcal{U}_i} \bar{w}_{wt} \sinh\big(\hat{\Xi}^{-1}_{jwt}(u_w)\big)\, d\hat{C}^j_t(\mathbf{u}) \end{pmatrix}. \tag{33}$$

In this example, we have a synthetic (representative) household i, where $(U^c_i \times U^y_i \times U^w_i)$ is the quantile combination that defines household i, e.g., the first decile in consumption c, the third decile in income y, and the seventh decile in wealth w. The mass, ω^j_{it}, of the households in that cell defines a weight for that synthetic household; the weights sum to one in each quarter. This implies that, without the need for sampling, we obtain an output that can be immediately interpreted as synthetic microdata. The group means in the application (Figure 8) are the same object, integrated over larger groups. As illustrated above, X^j_{it} is dataset specific, but one can obtain a consensus estimate across datasets by averaging the trend-adjusted coefficients $\hat{\theta}^j_{nt}$ of step 2 below over the datasets j that observe coefficient n, either as a simple average or as a weighted average using, e.g., the inverse measurement error standard deviations (by object and dataset) as weights, and reconstructing the quantile functions and copula from the averaged coefficients.

Following the procedure below, one can obtain the quarterly distributional data. Given the estimated coefficients $\left(\hat{\xi}^j_{mot}, \hat{\kappa}^j_{o_1\cdots o_d,t}\right)$ and the associated polynomials, the user only needs to specify $\mathbf{u} \in [0,1]^d$, the domain of integration. In our application, we integrate to construct deciles. In practice, for the marginal group means the integration bounds exclude the exact endpoints 0 and 1, which are replaced by 10^{-6} and $1-10^{-6}$, respectively, in accordance with our analysis in Appendix A and properties of our estimator. Two features of the procedure deserve mention. First, the loading $\hat{\Gamma}^{MF,j}$ of step 1 is dataset specific: for quarterly measurements it equals $[\hat{\Gamma}\ \ 0\ \ 0\ \ 0]$ and returns the quarterly distribution, whereas for annual flow measurements and the copulas of annual surveys it averages the four quarters the survey measures (Section 3.2.3), so that the dataset-specific reconstruction of an annual survey is the four-quarter average that survey would report. Second, no monotonicity or non-negativity constraint is imposed at this stage; we verify ex post that the reconstructed quantile functions are increasing, and copula masses that turn negative because of the truncation at order O are set to zero before the masses are renormalized (step 5).

Procedure:

$$\hat{\tilde{\theta}}^j_t = \hat{\Gamma}^{MF,j}\,\hat{F}_t \qquad \text{(Project factors)}$$

$$\hat{\theta}^j_{nt} = \sigma_{\zeta(n)} \times \hat{\tilde{\theta}}^j_{nt} + \mu^j_n + g^j_n(t) \qquad \text{(Unstandardize and add trend)}$$

$$\hat{\theta}^j_t = (\hat{\xi}^j_{m,o,t}, \ldots, \hat{\kappa}^j_{(o_1,\ldots,o_d),t}), \quad \hat{\kappa}^j_{0\cdots 0,t} = 1 \qquad \text{(Decomposition)}$$

$$\hat{\Xi}^{-1}_{jmt}(u_m) = \sum_{o=0}^{O} \hat{\xi}^j_{mot}\, Q_o(u_m) \qquad \text{(Reconstruct quantile function)}$$

$$\hat{C}_{jt}(u_1,\ldots,u_d) = \sum_{o_1,\ldots,o_d=0}^{O} \hat{\kappa}^j_{o_1\cdots o_d,t} \prod_{m=1}^{d} \int_0^{u_m} Q_{o_m}(v)\,dv \qquad \text{(Reconstruct copula)}$$

$$\omega^j_{it} = \Delta_{\mathcal{U}_i}\hat{C}_{jt}, \qquad \omega^j_{it} \leftarrow \max(\omega^j_{it},0)\Big/\sum\nolimits_{i'}\max(\omega^j_{i't},0) \qquad \text{(Cell masses)}$$

In the decomposition step, the coefficients that are fixed by the uniform margins of the copula and therefore not part of θ—the constant $\hat{\kappa}_{0\cdots 0,t} = 1$ and all $\hat{\kappa}$ with at most one non-zero order, which are zero—are re-inserted before the copula is evaluated. $\Delta_{\mathcal{U}_i}\hat{C}_{jt}$ denotes the increment of the copula over the cell $\mathcal{U}_i$. The quantile function is on the transformed scale; the level at rank u is $\bar{w}_{mt}\sinh\big(\hat{\Xi}^{-1}_{jmt}(u)\big)$, and cell or group means integrate this level over the rank range, against the copula for cells, before averaging.

Table 6: Description of Equation Components

Symbol	Description
$\hat{F}_t$	Smoothed factor state, stacking $\hat{f}_t, \ldots, \hat{f}_{t-3}$
$\hat{\Gamma}^{MF,j}$	Dataset-specific loadings from the factor state to the standardized coefficients of dataset j
$\hat{\tilde{\theta}}^j_t$	Projected (standardized) coefficient vector of dataset j
$\sigma_{\zeta(n)}$	Standard deviation term, varies by object $\zeta(n)$ (copula or margin) to which coefficient n bel
μ^j_n	Mean adjustment for coefficient n in dataset j
$g^j_n(t)$	HP trend ($\lambda = 1600$) of coefficient n in dataset j, fitted on the interpolated quarterly series
$\hat{\theta}^j_{nt}$	Unstandardized, trend-adjusted coefficients
$\hat{\xi}^j_{m,o,t}$	Marginal polynomial coefficients
$\hat{\kappa}^j_{(o_1,\ldots,o_d),t}$	Copula polynomial coefficients
$Q_o(u_m)$	Shifted orthonormal Legendre polynomial of order o, $O = 11$
$\hat{\Xi}^{-1}_{jmt}(u_m)$	Reconstructed quantile function (inverse CDF) for margin m, transformed scale
$\hat{C}_{jt}(u_1,\ldots,u_d)$	Reconstructed copula (dependence structure)
ω^j_{it}	Probability mass of cell $\mathcal{U}_i$, the weight of synthetic household i
$\bar{w}_{mt}$	Aggregate per-household level of variable m used in the transformation

Notes: Table summarizes the role of each symbol in the above equations, moving from factor projections to reconstruction of marginal quantiles and copula.

E Validation of Hyperparameter Choices

To evaluate our hyperparameter choices on the Bayesian estimation priors for the measurement error variances, we compare the series resulting from the Kalman smoother after estimation with the actual point estimates and their confidence bounds from the survey data.

Intuitively, if the prior mean for the measurement error variance is too low, it will force the estimator to exactly match each survey estimate of the distribution, despite the fact that each survey estimate is itself subject to measurement error. Thus, we should expect the smoother estimate to fall within the confidence bounds of each sample estimate at most with the corresponding confidence level of the bounds. The fact that the confidence level is an upper bound reflects that the estimated measurement error captures not only the sampling uncertainty that the confidence bounds capture, but also conceptual differences.

Choosing narrow measurement errors would overstate precision and potentially limit comovement with aggregates. Another reason not to be conservative with the measurement errors is that allowing for measurement error also accounts for the fact that we combine data from different sources to produce a consensus estimate. These different data sources, despite their individual detrending, may produce some temporarily divergent estimates of the distributions. Without sufficient measurement error, the consensus estimate is then forced to oscillate between these different distribution estimates over short time intervals, rather than capturing their co-movements.

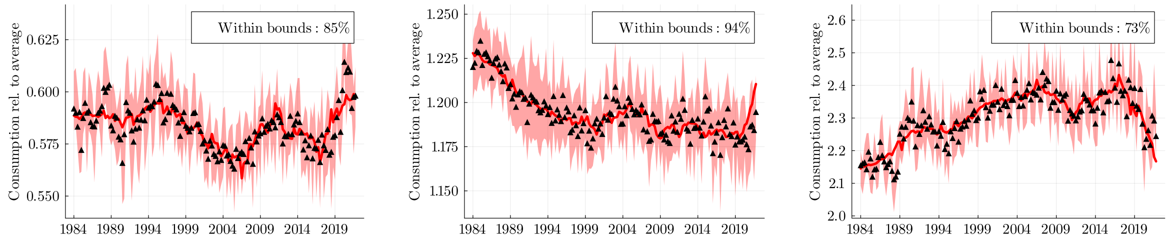
On the other hand, if the prior mean for the measurement error variance is too high, the estimator will treat the data as uninformative, and the smoother will miss the survey estimates more often and to a much greater extent than implied by its confidence bounds. We validate the choice of hyperparameters graphically for income and wealth in the SCF data and provide comprehensive summary statistics across all datasets and estimates. This visual inspection helps confirm that our smoothed series respect the noise inherent in the available microdata.

Figure 13 shows average consumption (top row), income (middle row) and wealth (bottom row) for the bottom 50 percent (first column), the next 40 percent (second column), and the top 10 percent (last column) of the respective distributions. It shows the point estimates from the surveys (black triangles), along with their 95% confidence limits, and the results from the Kalman smoother based on our estimates of the parameters of Equation (21). Overall, the smoothed estimates fall inside their respective confidence bounds in 87 out of the 108 observations (from (d) to (i)).

Table 7 provides a comprehensive summary of this validation approach. For all quantile functions and copulas, we report for each dataset and survey year how often the respective smoother estimate is within the confidence limits. Again, we use a confidence level of 95%. For the copulas and most quantile functions, the fraction of smoothed estimates within the confidence bounds is close to the confidence level (81–100 percent). The exceptions are the SIPP (66–69 percent), the SCF income quantiles (70 percent), and the CEX consumption quantiles (78 percent), for which our estimator suggests measurement error beyond sampling uncertainty, reflecting differences in sample design and income/wealth measurement.

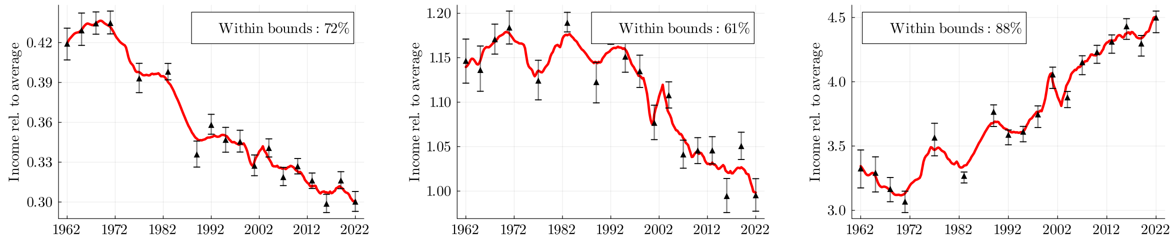
Figure 13: Comparison of Smoothed Distributional Data and Direct Survey Estimates

Consumption by Consumption



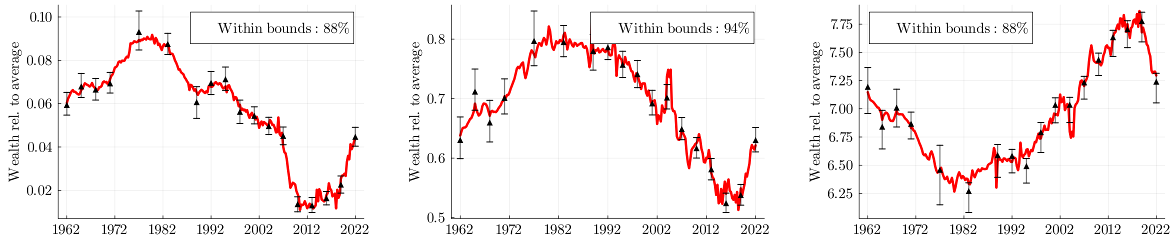
(a) Bottom 50 Percent (b) 50–90 Percent (c) Top 10 Percent

Income by Income



(d) Bottom 50 Percent (e) 50–90 Percent (f) Top 10 Percent

Wealth by Wealth



(g) Bottom 50 Percent (h) 50–90 Percent (i) Top 10 Percent

— Model ▲ Data

Notes: Figure shows the average consumption, income, and wealth for the bottom 50 percent, 50–90 percent, and top 10 percent of households of the respective distribution. Triangle markers show the estimates from the individual survey waves together with 95% confidence bounds. The solid red line shows the baseline estimate from the Kalman smoother at the posterior mode. Consumption shows CEX data and reconstruction. Income and wealth show SCF data and reconstruction. The legend reports for each Panel the share of smoothed estimates within the confidence bounds of the survey waves.

Table 7: Deviations of Smoothed Estimates and Microdata: Fraction within Confidence Bounds

Measure	CEX	CPS	SCF	SIPP	PSID	Overall
Consumption quantiles	78%	—	—	—	100%	80%
Income quantiles	89%	89%	70%	69%	97%	83%
Wealth quantiles	—	—	81%	66%	99%	79%
Copula densities	95%	—	92%	89%	100%	96%

Notes: The table reports, by microdata and object, the fraction of estimates from the Kalman smoother at the posterior mode that fall within the 95% bootstrapped confidence intervals for the respective microdata. Quantile and copula estimates are defined on a decile grid.

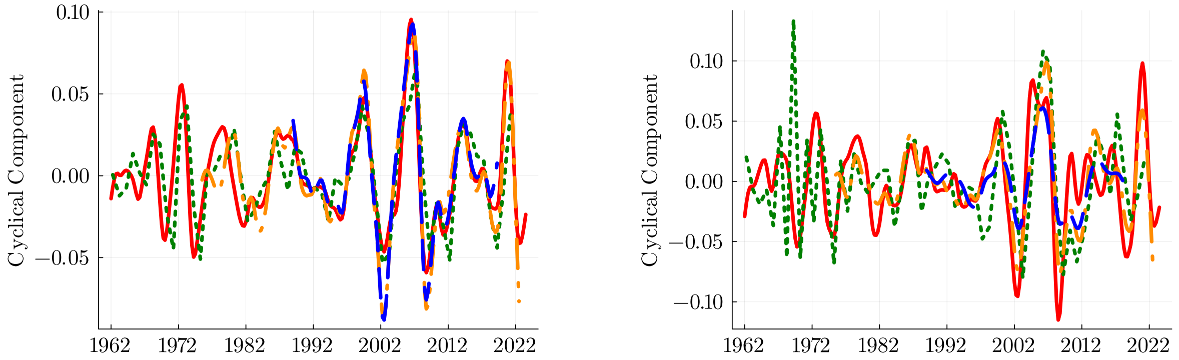
F Comparison with External Estimates

This section assesses whether the resulting high-frequency income and wealth dynamics align with external benchmarks. Since true distributional dynamics are unobservable, we compare our cyclical estimates to the *Distributional Financial Accounts (DFA)* (Batty et al. 2020), the *World Inequality Database (WID)* (Piketty, Saez, and Zucman 2018), and the Real-time Inequality (WID-Q) database (Blanchet et al. 2026). The DFA produces quarterly wealth distributions from the SCF using a different estimation approach, providing an "SCF flavor" benchmark. The WID provides annual estimates from administrative tax data without a time-series model, offering a robustness check for our assumption of stable and approximately linear distributional dynamics and their link to macro factors. The RTI is a public database of monthly and quarterly distributional national accounts for the United States that allocates aggregate income and wealth growth across income and wealth groups by combining high-frequency macro data with the latest survey and tax microdata. To assess dependence dynamics (copula), we additionally compare our implied wealth-by-income series to the DFA.

Figure 14 reports the cyclical components (in logs). Panels (a)–(b) compare wealth by wealth group, focusing on the top 10% and the next 40% since the bottom half has wealth close to zero (Kuhn and Ríos-Rull 2016). Panels (c)–(d) compare wealth by income groups (informative about the joint distribution). Panels (e)–(f) compare income by income group (available in the WID). Across all cases, our estimates comove closely with RTI, DFAs and WID and fall within the range of the external series.

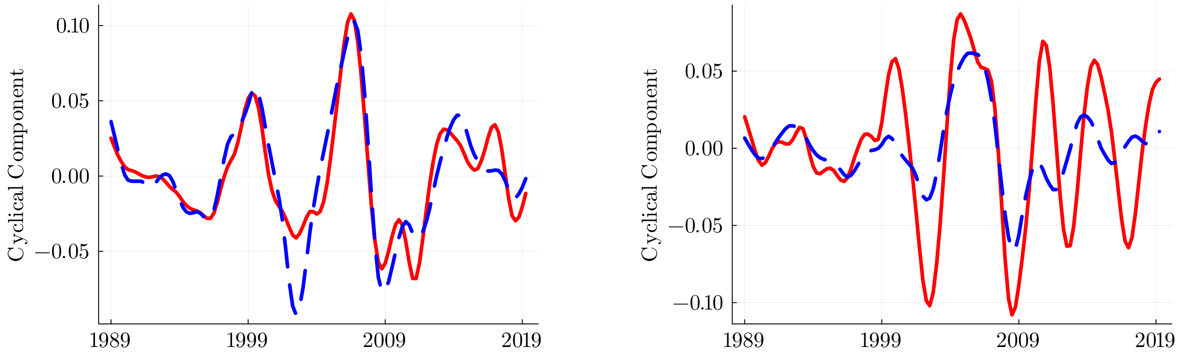
Figure 14: Cyclical Component of Distributional Data vs. External Sources

Wealth (Average)



(a) top 10 Percent in Wealth

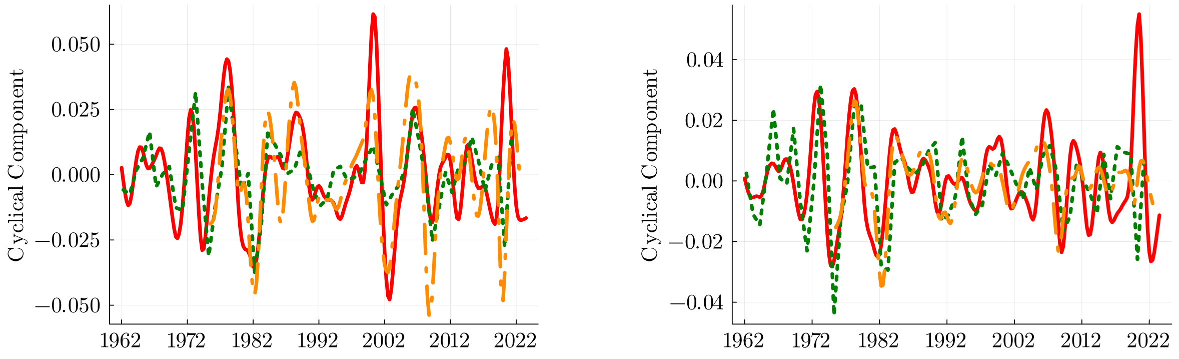
(b) 50th–90th Percentiles in Wealth



(c) top 20 Percent in Income

(d) 40th–80th Percentiles in Income

Income (Average)



(e) top 10 Percentile in Income

(f) 50–90 Percentile in Income

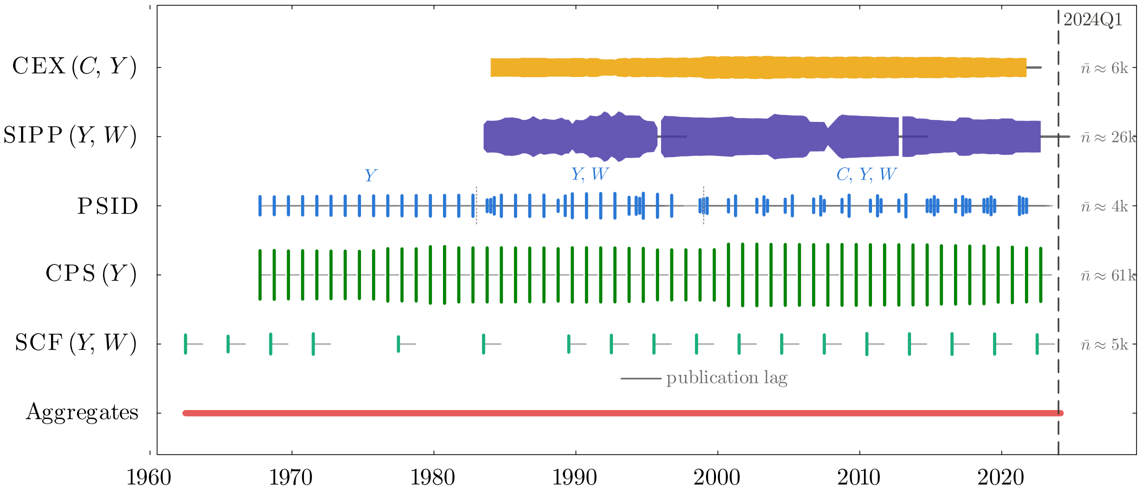
Model
DFA
WID-A
WID-Q

Notes: Figure shows the cyclical component of (log) average wealth of (a) the wealthiest 10 percent, (b) the next wealthiest 40 percent, (c) the 20 percent income-richest households, and (d) the next 40 percent income-richest households. Bottom row shows the cyclical component of (log) average income of (e) the income-richest 10 percent and (f) the next income-richest 40 percent. Red lines show cyclical components from baseline model at quarterly frequency. Dotted green lines show annual data from the *World Inequality Database* (WID-A). Dashed orange lines show the quarterly data from Real-time Inequality (WID-Q). Dashed blue lines show quarterly data from the *Distributional Financial Accounts* (DFA). Cyclical components are obtained by an HP-filter with smoothing parameter $\lambda = 6$ for annual data and $\lambda = 1600$ for quarterly data.

G Data

The construction of these estimates relies on a great deal of data. An advantage of our method, however, is that it can incorporate these different microdata and their various differences in generating consensus estimates of the distributional data. Below, we describe the data, all expressed in 2019 dollars, and explain the mappings across data to ensure measures are broadly comparable (Curtin, Juster, and Morgan 1989; Czajka, Jacobson, and Cody 2003; Pfeffer et al. 2016). See Table 1 for information on their availability, and Figure 15 for a graphical summary of reference periods, variable coverage, sample sizes, and publication lags across sources.

Figure 15: Coverage, Sample Sizes, and Publication Lags across Microdata



Notes: Each row is one data source; parentheses list the variables it contributes (consumption C, income Y, wealth W). Colored marks show reference periods: one vertical bar per survey wave, or a solid band for sources available at least quarterly. The vertical extent of the marks is proportional to the square root of the average sample size per release, reported at the right margin. Gray whiskers indicate the typical lag between the end of a reference period and the publication of the data. For the PSID, the covered variables change over time: income throughout, wealth from the 1984 wave (a stock measured at the interview date, which we date to 1984Q2), and consumption from 1999. For the SIPP, only income is continuous at (sub)quarterly frequency; wealth is collected at topical-module waves before 2013 and once per reference year (as of December 31, dated to Q4) thereafter. Sample sizes are household counts computed from our estimation inputs (for the SCF, per implicate); $\bar{n}$ reports the average per release. Publication lags are approximate. The dashed line marks the end of our estimation sample (2024Q1).

Sample selection and weights. We do not impose any demographic household restriction and only mild sample selection for consistency across datasets (Section 4.1). In every survey the unit of observation is the household (the family in the PSID and the consumer unit in the CEX), records with a missing sampling weight are dropped, and all values are expressed in 2019 dollars using the CPI-U. We do not equivalize by household size. Each observation enters the estimation of the Legendre coefficients with its survey weight: ranks are cumulative weight shares, and the coefficients are weighted means of the basis functions evaluated at these ranks. The bootstrap that provides the sampling covariance of the coefficients resamples households with replacement within each survey period, carrying their weights (999 draws; 500 for the

SIPP; 100 for the CPS, where households are resampled in clusters within four census-division strata). The survey-specific rules are as follows. *SCF:* the SCF+ sampling weight; all five implicates are fit separately and their coefficients averaged, and the implicates are pooled in the bootstrap; pensions are excluded from wealth; no trimming. *PSID:* the family weight; income below one dollar is set to one dollar, negative values of consumption, asset and debt components are set to missing while negative net worth is kept, and families with all measures missing are dropped. *CPS:* the ASEC household weight, one record per household, with top-coded incomes replaced by the IPUMS component sums. *CEX:* the consumer-unit weight (FINLWT21, divided by four for the quarterly cross-sections) from the interview survey, 1983 onwards; income is taken from the interviews in which it is collected and non-positive income is set to missing; consumption is set to missing when food expenditure is zero or total expenditure non-positive; the bottom one percent of the rental-equivalence measures is set to missing in each quarter. *SIPP:* the household weight for the 1984–2008 panels and the within-household average of the person weight for the 2014 and later panels; quarterly income is formed only when all three months are observed; zero wealth and zero debt are set to missing and the bottom one percent of income and of wealth are dropped in each quarter; residents outside the states and households of ten or more members are dropped in the 2014 and later panels; the seam quarters 1985Q4, 1988Q1, 1995Q4, 1996Q1, 1999Q4 and 2003Q4 are dropped.

G.1 SIPP

The SIPP panel is a nationally representative, individual-level survey known for providing high-frequency dynamics on employment, earnings, wealth, household composition and program participation in the U.S. For the data cleaning, the data are aggregated to the household-level, at quarterly frequency. Due to structural breaks induced by changing survey designs, we treat the SIPP as if it were three separate surveys (1983Q3–1995Q4, 1996Q1–2012Q4, 2013Q1–2022Q4). This is to minimize spurious fluctuations (e.g., seam bias) that obscure actual economic phenomena.

Income. We use total household income as reported by the Census Bureau in all releases: `THTOTINC` from the 1996 panel onward and its predecessor `H*TOTINC` (one variable per reference month) in the 1984–1993 panels. Income is reported monthly; we sum it to a quarterly figure only when all three months of the quarter are observed, treating the quarter as missing otherwise, and then annualize (multiplying by four) and deflate to real dollars.

Wealth. For the 2014 releases and onward, we use the `THNETWORTH` variable for wealth. For data releases prior, wealth is defined as total assets (`hhtwlth`) net total liabilities (`hhusdbt`, `hhscdbt`).

Note: The SIPP has undergone several major redesigns, and ignoring these "seams", even with our prescribed data treatment, would lead to spurious swings in measured inequality—an issue documented by, among others, Moore (2008) and Czajka, Jacobson, and Cody (2003) and confirmed in our own diagnostics. To deal with this issue, we treat the SIPP as if it were three different surveys: (1) SIPP before 1996 (2) SIPP 1996–2012 and (3) SIPP 2013–2022. We found that around these seams, estimates tend to exhibit large jumps, which was at odds with their aggregate counterpart and general business cycle conditions. Additional observations were dropped as well ($1985Q4, 1988Q1, 1995Q4, 1996Q1, 1999Q4, 2003Q4$): all six are panel-boundary quarters in which mean income dips 11–20% below neighboring quarters while the sample collapses to as little as a third of its usual size (only exiting rotation groups remain in the field), with no counterpart in aggregate income at any of these dates. For the post-2013 SIPP, assets and liabilities are measured once per reference year (as of December 31) and replicated by the Census across all monthly records; we accordingly date the wealth-side variables to the fourth quarter of the reference year—the quarter the measurement refers to—rather than carrying the identical value in all four quarters, while income, which is genuinely monthly, remains quarterly. Fortunately, the estimation framework allows each of the three SIPP "sub-surveys" to carry its own measurement-error process, so we retain almost all available information while preventing seam effects from distorting the results.

G.2 SCF+

The SCF provides detailed household income and wealth data; SCF+ extends the modern surveys with historical waves compiled by Kuhn, Schularick and Steins (2020). The SCF+ is also augmented with the Forbes 400, from Forbes, for the years 2021 to 2023. For the years 1985 to 2020, we use the per capita dataset of Fernholz and Hagler (2023), whose observations originate from families of the Forbes 400.[52] Below we describe the measurements we used from the SCF+.

Income. Our definition of income follows Kuhn, Schularick, and Steins (2020), which consists of the following components: (1) labor income (i.e., earnings) (2) income from public transfers (3) income from professional practice and self-employment (4) income from rents (5) dividend income and (6) business/farm income. A different taxonomy that illustrates these components are taxable and transfer income.

Assets. Total assets include (1) liquid assets such as a household's checking and savings account, CDs, call/money market accounts, short-term government bonds, and mutual funds (2) illiquid assets such as housing and other real estate, automobiles (3) defined-contribution retirement plans (4) the cash value of life insurance (5) stocks and (6) business equity.

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[52] Accessing the data via the usual link (e.g., `http://www.forbes.com/ajax/list/data?year=2005&uri=forbes-400&type=person`) returns Forbes 400 data, but unfortunately incomplete from the 1990s until the late 2000s—with only around 200 of the 400 observations there. Using Fernholz and Hagler (2023), with the same procedure imposed by Bricker, Hansen, and Volz (2019), generates the same result as in Bricker, Hansen, and Volz (2019) (increases the top wealth share by 1.5%), but is more complete (has almost all 400 every year). Implementation details: Forbes 400 observations are dated to the third quarter, matching the timing convention of the SCF; each Forbes 400 member is entered into all five SCF implicates and assigned the smallest sampling weight observed in the SCF, so that each represents (approximately) a single household. To avoid double counting the top tail, we follow Bricker, Hansen, and Volz (2019): within year-specific wealth bins defined by multiples of the Forbes 400 minimum $F_y^{\min}$ ($[1, 1.5)$, $[1.5, 2.5)$, $[2.5, 5)$, and $[5, \infty)$ × $F_y^{\min}$), SCF and Forbes observations are reweighted in proportion to their relative frequencies; observations below $F_y^{\min}$ keep their original weights. Forbes records carry net worth only; their income and consumption are treated as missing.

***Debt*.** We define debt of a household as the sum of personal (mostly unsecured) debt and housing (mortgage) debt. Housing debt includes debt from all properties and any loans made against the housing e.g., through HELOCs. Personal debt includes car loans, education loans, any loans from relatives, credit card debt, medical debt and legal debt.

Wealth. Wealth is total assets net of total debt of a household.

G.3 PSID

The Panel Study of Income Dynamics complements the SCF+ extraordinarily well, as they take our estimations over more than half a century. In comparison to the post-1983 SCF, a deeper analysis of their similarity can be found in Pfeffer et al. (2016).

Income. The PSID has collected family income annually from 1968 to 1996 and then biennially from 1997 to 2021. Its measure of income is the sum of taxable income, transfers, and social security for the reference person, the spouse/partner (if any) and other members of the family.[53]

Assets. Data collection on household wealth took place in 1984, 1989, 1994, and then every wave beginning in 1999. The data on assets is split into liquid and illiquid assets. Although minor, the definition of liquid assets will vary between datasets, so careful attention is warranted here. Liquid assets for the PSID include checking and savings accounts, short-term instruments such as money-market accounts, certificates of deposit, and treasury bills. Illiquid assets include business equity, financial assets held in mutual funds, stocks, bond funds, investment funds; real assets held in real estate, vehicles like motor homes, boats, trailers, and cars; and retirement wealth in private annuities or IRAs.

Debt. For the PSID, we achieve the same debt split: personal and mortgage debt. This includes all kinds of real-estate debt, and unsecured debt such as credit card debt, student loans, medical debt, legal debt, and loans from relatives.

Wealth. Wealth is total assets net total debt of a household.

***Consumption*.** Studying papers such as Skinner (1987), Cutler et al. (1991), Flavin and Yamashita (2002), Attanasio, Hurst, and Pistaferri (2014), and Attanasio and Pistaferri (2014), we define consumption as the sum of these expenditures: food, rent (for renters), housing rental equivalence (for home-owners), utilities, health, public transport, education, and childcare. We set the housing rental equivalence to be 6% of the home market value reported by households in the PSID. Consumption data is only available from 1999 in a biennial interval.

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[53] In the PSID, a family is a group of people living together who are economically interdependent.

G.4 CPS

We use the Current Population Survey (CPS) Annual Social and Economic Supplement (ASEC). The sample is designed primarily to produce estimates of the labor force characteristics and runs from 1967 to 2022. Similar to the SIPP, we treat the CPS as a combination of two separate surveys to avoid fluctuations driven by survey design: 1967Q4–1992Q4 and 1993Q4–2022Q4 (income years, dated to the fourth quarter).

Income. Income data are collected as part of the ASEC for the months of February, March and April as a supplement to the regular CPS monthly labor force interviews. The ASEC asks each person in the sample who is 15 years old and over about the amount of income received from a list of sources in the previous calendar year. We treat these observations as being observed in quarter four of the previous calendar year. For details on top-coding, see https://cps.ipums.org/cps/topcodes_tables.shtml.

G.5 CEX

The Consumer Expenditure Survey (CEX) is the most comprehensive household survey in the U.S. for recording the consumption habits of households. The CEX has two components: the interview survey (IS) and the diary survey (DS). The interview survey has sufficiently rich data on what we need, so we only use data from this component. Within this component, there are several files, each of which pertains to a topic, from which we can extract information. The following table breaks down each category of consumption, defining which `UCCs` belong to which category and which file it can be found in. All of these categories will combine to make the consumption variable. The table will also define wealth concepts of the CEX we use in our study. Since each household consumption record is with respect to a `UCC`, we find this presentation most apropos.

Item	UCCs / FMLI label	File
	Consumption	
Food	`190904, 790220, 190901, 190902, 190903, 790410, 790430, 200900, 790330, 790420, 800700, 790230, 790240`	MTBI
Rent	`210110, 800710`	MTBI
Utilities	`250111, 250112, 250113, 250114, 250211, 250212, 250213, 250214, 250221, 250222, 250223, 250224, 250901, 250902, 250903, 250904, 250911, 250912, 250913, 250914, 260111, 260112, 260113, 260114, 260211, 260212, 260213, 260214, 270211, 270212, 270213, 270214, 270411, 270412, 270413, 270414,270101, 270102, 270104, 270105, 270310, 270311, 690116, 270901, 270902, 270903, 270904`	MTBI
Health	`570110, 570111, 570210, 570220, 570230, 560110, 560210, 560310, 560330, 560400, 340906, 540000, 550110, 550320, 550330, 550340, 570901, 570903, 570240, 580111, 580112, 580113, 580114, 580311, 580312, 580901, 580903, 580904, 580905, 580906, 580400, 580907`	MTBI
Public Transport	`520531, 520532, 530311, 530312, 530501, 530902, 530210, 530411, 530412, 520511, 520512, 520521, 520522, 520542, 520902, 520903, 520904, 520905, 520906, 520907, 530110, 530901, 520110, 520310`	MTBI
Education	`210310, 370903, 390901, 660110, 660210, 660310, 660900, 670110, 670210, 670901, 670902, 800802, 800804, 690111, 690112, 660410, 660902, 670410, 670903, 690114, 690310`	MTBI
Child care	`340210, 340211, 340212, 670310, 660901`	MTBI
Rental Equivalence	`910050, 800721 (market value of home), SIMHOUSX, RENTEQVX`	FMLI, MTBI

Gas & Vehicle Repairs	`470111, 470112, 470113, 470220, 470211, 470212, 480110, 480212, 480213, 480214, 490110, 490211, 490212, 490221, 490231, 490232, 490311, 490312, 490313, 490314, 490318, 490319, 490411, 490412, 490413, 490501, 490502, 490900, 520410, 480215, 620113`	MTBI
	Other Concepts	
Housing Debt	`QBLNCM1X, QBLNCM2X, QBLNCM3X, QBLNCM1G, PRINAMTX`	MOR
Personal Debt	`6001, 6002 (1990-2013), 5400, 5500, 5600, CREDITX, STUDNTX, OTHLONX, CREDITX1, CREDITX5, QBALNM1X`	MTBI, ITBI, FMLI, FN2
Liquid Assets	`SAVACCTX, CKBKACTX, USBNDX, 920010, 920020, 920030, 5100, LIQUIDX`	FMLI, ITBI
Financial Assets	`5800, 920040, STOCKX, SECESTX, OTHASTX`	FMLI, ITBI
Income	`FINCBTAX`	FMLI

Table 8: Table shows, by item, the identifiers necessary to construct each component of consumption, income and wealth for the CEX. The location of these identifiers can be found under the *File* column.

G.6 Observation frequency and seasonal adjustment

Not every series we extract from the microdata is a quarterly flow, and only quarterly flows can carry a seasonal component. In the SIPP, income is genuinely quarterly (monthly reports aggregated to quarters), while assets and liabilities come from topical modules: in the post-2013 panels they are measured once per reference year and dated to the fourth quarter (see the SIPP note above), and in the earlier panels the asset modules rotate across interview waves, so the calendar quarter of a wealth observation reflects rotation-group composition rather than a seasonal cycle. In the CEX, consumption is a genuine quarterly flow (three-month recall), whereas income is collected as a twelve-month recall—a rolling annual concept that is season-free by construction. The PSID, CPS, and SCF+ are annual or lower frequency; their moment series are never seasonally adjusted. In summary:

Series	Concept
SIPP (all eras), income	quarterly flow
SIPP 1983–2012, wealth	rotating topical module
SIPP 2013–2022, wealth	annual, dated Q4
CEX, consumption	quarterly flow
CEX, income	twelve-month recall

Seasonal adjustment of the quarterly moment series (X-13ARIMA-SEATS, X-11 adjusted component) is therefore applied series by series, subject to two screens. First, an observation-pattern screen: a series qualifies only if it is observed in at least three distinct calendar quarters; a series measured only in the fourth quarter (post-2013 SIPP wealth) has no identifiable seasonal component, and adjustment would merely perturb the genuine observations. Second, an identifiability test: X-13's M7 statistic must indicate identifiable, stable seasonality ($M7 < 1$); the diagnostic is a by-product of the same X-13 run that produces the adjusted series, so the test is costless. Series failing either screen enter the estimation unadjusted. The screening decisions are made once on the point data and reused in the bootstrap that constructs the measurement-error distributions and confidence intervals.

G.7 Aggregates

Together with the microdata, we specify a model component that captures the various aggregate shocks that potentially buffet the joint distribution of consumption, income, and wealth. This is represented in the state equation of the state-space model. The aggregate data we rely on to extract this information come from the FRED-QD (McCracken and Ng 2021). This has various macro-data that has already been seasonally-adjusted on industrial production, employment, housing, inventories, prices, earnings, productivity, household expectations, household balance sheets, interest rates, credit, etc. You can find more information on https://research.stlouisfed.org/econ/mccracken/fred-databases/.

Before performing the PCA on the aggregates, we transform each series to stationarity following the codes proposed by McCracken and Ng (2021), with one exception: price indices enter as inflation rates (Δlog) rather than changes in inflation.[54] We stack the series with four lags of each series, standardize each column, and extract 11 principal components, a number suggested by marginal data density. The loadings are estimated excluding 2020Q1–2021Q1, so that the pandemic quarters do not dominate the components, and all quarters, including the pandemic ones, are then projected onto them. The 11 components enter the state vector as latent states with their own dynamics and measurement error, rather than as exogenous inputs.

H Out-of-Sample Correlations

This appendix complements the missing-data experiments of Section 4.3, in which the model is re-estimated after withholding blocks of microdata (Figure 5, Panels (d)–(l)). Table 9 reports correlations between the cyclical estimates of the baseline model, estimated with all data, and those of each missing-data model, for the entire estimation period and for the withheld periods alone. Over the entire sample, they lie between 0.96 and 0.98; restricted to the withheld periods themselves, they range from 0.94 to 0.99. The estimates with and without the withheld data are thus nearly identical, which is what the panels of Figure 5 show visually: the model recovers the withheld dynamics from the remaining surveys and the aggregates.

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[54] Under the second-difference code, the price block is dominated by high-frequency noise. Inflation is treated as stationary by assumption; ADF rejects a unit root for most but not all inflation series, a standard finding given its high persistence.

Table 9: Out-of-Sample Correlations with the Baseline Estimates

Condition		**Bottom**			**Middle**			**Top**	
	C	**I**	**W**	**C**	**I**	**W**	**C**	**I**	**W**
					Excluding Housing Cycle Wealth				
Entire Series	-	0.98	0.96	-	0.98	0.97	-	0.98	0.96
Specific Timeframe	-	0.97	0.97	-	0.95	0.99	-	0.95	0.98
					Every 4 Years				
Entire Series	0.97	0.98	-	0.97	0.96	-	0.97	0.97	-
Specific Timeframe	0.98	0.98	-	0.98	0.96	-	0.94	0.97	-

Notes: Table reports correlations between the baseline cyclical estimates and the missing-data models, split by panel. Entire Series reports correlations for all estimates in the estimation timeframe. Specific Timeframe reports correlations of estimates in periods where data was intentionally left out of the estimation of the specific missing-data model. **Bottom**, **Middle**, and **Top** are the bottom 50, next 40, and top 10 of the respective distribution, denoted by **C, I,** and **W**. **C** is for consumption, **I** is for income, and **W** is for wealth. For the first panel, correlations are made between SCF model estimates (no consumption); the second panel presents correlations from CEX model estimates (no wealth). The missing-data models (in headers) are described in Section 4.3.

I Information Shares, Variance and Historical Decompositions

This appendix collects the derivations and additional results behind Sections 4.4 and 5.1. The state-space structure permits complementary, exact answers to two further questions, for every quarter at once: whose data is an estimate resting on, and how much does it rest on data at all? Section 4.4 addresses the second question through interpolation shares; Appendix I.1 addresses the first by attributing each smoothed factor to the data sources that inform it. Appendix I.2 then gives the formal identification of the forecast error variance decomposition of Section 5.1, Appendix I.3 maps it to observable moments, and Appendix I.4 describes the historical decomposition of the smoothed factor paths across NBER recessions.

I.1 Information Shares across Data Sources

To quantify the contribution of each data source to the estimated distributional factors, we exploit the fact that, conditional on the estimated model parameters and observation pattern, the Kalman smoother is an affine function of the observed data (Koopman and Harvey 2003). Collect the state of (21) as $x_t = (F_t', Y_t')'$ and group observations by source b. Let y_s^b denote the transformed coefficient observations $\tilde{\tilde{\theta}}_s^j$ from datasets j belonging to source b at date s, with aggregate observations $\hat{Y}_s$ forming a separate source. The smoothed state can then be written as

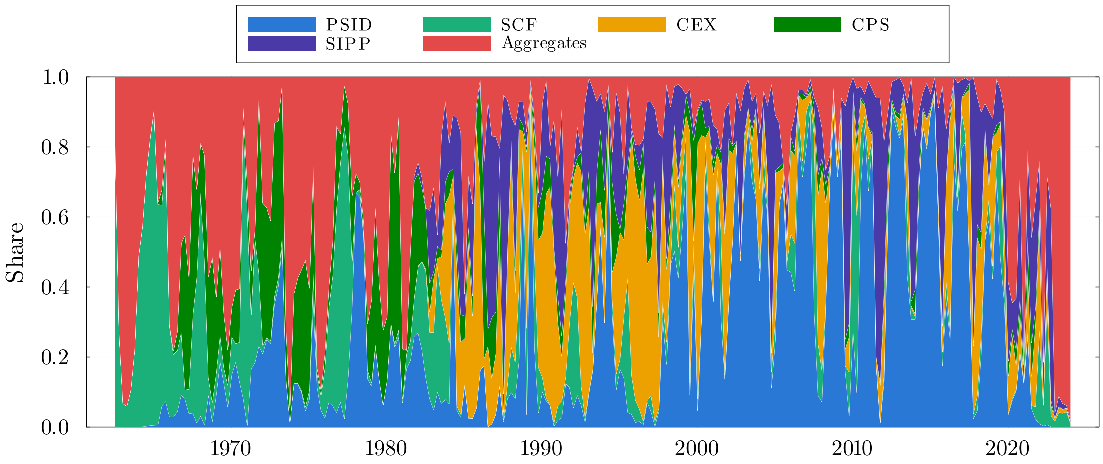
$$\hat{x}_{t|T} = c_t + \sum_b \sum_{s=1}^{T} W_{t,s}^b y_s^b, \tag{34}$$

where $W_{t,s}^b$ maps observations from source b at date s into their contribution to the estimate at date t, and c_t collects deterministic terms and the contribution of the prior. The weights depend on both the state and measurement equations, allowing observations to inform estimates in other quarters through the model's dynamics.

This representation allows us to isolate each source's contribution exactly. We set observations from all other sources to zero while leaving the selector matrices H_t in (24) unchanged, keeping the Kalman gains fixed. Re-running the smoother then isolates the selected source's contribution together with c_t, which we subtract.[55] Let $\hat{x}_{k,t|T}^b$ denote the resulting contribution of source b to factor k at date t. We define its *information share* as

$$s_{k,t}^b = \frac{|\hat{x}_{k,t|T}^b|}{\sum_{b'} |\hat{x}_{k,t|T}^{b'}|}. \tag{35}$$

Figure 16: Information Shares



Notes: Stacked information shares $s_{k,t}^b$ of each data source in the smoothed factor, shown for $k = 1$.

Figure 16 illustrates these shares for the first distributional factor. Before the mid-1980s, when microdata sources are sparse, the estimate relies more heavily on macroeconomic aggregates. Once CEX and SIPP enter the sample, their contributions rise and that of aggregates declines. The other factors rely less on aggregate data and differ in the importance assigned to individual surveys, reflecting the different aspects of the joint distribution they capture.

I.2 Identification of the Variance Decomposition

The variance decomposition of Section 5.1 separates aggregate from distributional innovations with a blocked Cholesky factorization, aggregates ordered first. This subsection formalizes it.

---

[55] Running the Kalman smoother with all observation values set to zero returns c_t. This term is negligible in our application because the coefficient observations are detrended and the prior mean of the state is zero.

Companion form. Let $f_t \in \mathbb{R}^r$ denote the vector of estimated distributional factors and let $Y_t \in \mathbb{R}^q$ denote the latent aggregate factors. We work with a companion-state representation

$$x_t \;=\; \Phi x_{t-1} + \epsilon_t, \qquad \epsilon_t \sim (0, \Omega), \qquad x_t \in \mathbb{R}^n, \;\; n = rp + q, \tag{36}$$

where x_t stacks $(f_t, f_{t-1}, ..., f_{t-p+1}, Y_t)$ for the p lags and $\Phi \in \mathbb{R}^{n\times n}$ is the corresponding companion transition matrix.

Define $J \in \mathbb{R}^{r\times n}$ as the selector that extracts the contemporaneous factors from the state,

$$f_t \;=\; J x_t, \qquad J = \begin{bmatrix} I_r & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{37}$$

Let $S_f \in \mathbb{R}^{n\times r}$ and $S_y \in \mathbb{R}^{n\times q}$ embed factor and aggregate innovations, respectively, into the companion innovation vector:

$$S_f = \begin{bmatrix} I_r \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad S_y = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ I_q \end{bmatrix}. \tag{38}$$

We collect shocks using $S \in \mathbb{R}^{n\times(r+q)}$. Under the blocked ordering with aggregates first,

$$S \;=\; \begin{bmatrix} S_y & S_f \end{bmatrix}, \tag{39}$$

and define the selected innovation subvector $\epsilon_t^{(S)} \equiv S^\top \epsilon_t \in \mathbb{R}^{r+q}$ with covariance

$$\Omega_{\text{sub}} \;\equiv\; S^\top \Omega S \;\in\; \mathbb{R}^{(r+q)\times(r+q)}. \tag{40}$$

Identification. Partition Ω_{sub} conformably as

$$\Omega_{\text{sub}} = \begin{bmatrix} \Omega_{YY} & \Omega_{YF} \\ \Omega_{FY} & \Omega_{FF} \end{bmatrix}, \tag{41}$$

where the Y block corresponds to aggregate shocks and the F block to distributional shocks. The blocked Cholesky factorization constructs an impact matrix

$$P_{\text{sub}} = \begin{bmatrix} P_{YY} & 0 \\ P_{FY} & P_{FF} \end{bmatrix}, \qquad \Omega_{\text{sub}} = P_{\text{sub}} P_{\text{sub}}^\top, \tag{42}$$

with

$$\begin{aligned} P_{YY} &= \mathrm{chol}(\Omega_{YY}), && (43)\\ P_{FY} &= \Omega_{FY} P_{YY}^{-\top}, && (44)\\ P_{FF} &= \mathrm{chol}\Big(\Omega_{FF} - P_{FY} P_{FY}^{\top}\Big)\,. && (45) \end{aligned}$$

This yields orthogonalized innovations $e_t \in \mathbb{R}^{r+q}$ satisfying

$$\epsilon_t^{(S)} = P_{\mathrm{sub}} e_t, \qquad \mathbb{E}[e_t e_t^{\top}] = I_{r+q}, \qquad e_t = P_{\mathrm{sub}}^{-1} \epsilon_t^{(S)}. \tag{46}$$

Embedding back into the full innovation space gives

$$P \;\equiv\; S P_{\mathrm{sub}} \;\in\; \mathbb{R}^{n \times (r+q)}. \tag{47}$$

I.3 Variance Decomposition of Observable Moments

Table 10 maps the five-year FEVD of the factors (Table 3) to observable moments through the linear mapping from factors to observables of Appendix D: conditional means of consumption, income, and wealth by percentile group, and population shares of the cells of the copula. The aggregate share is similar across the group means of consumption (33–37 percent) and somewhat lower for income (19–36 percent); it is highest for the wealth of the bottom 20 percent (55 percent) and for the high-income, high-wealth cells of the copula (43–50 percent).

Table 10: FEVD of group means and population shares. Percent of forecast-error variance

	Conditional means of ...					
	Consumption		*Income*		*Wealth*	
Group	Auton.	Agg.	Auton.	Agg.	Auton.	Agg.
Bottom 20%	67	33	66	34	45	55
Bottom 20–40%	63	37	75	25	58	42
Middle 40–80%	63	37	81	19	71	29
Top 20%	67	33	64	36	66	34

	Population shares with ...			
	Low Consumption		*High Consumption*	
Low Income, Low Wealth	57	43	55	45
Low Income, High Wealth	61	39	68	32
High Income, Low Wealth	81	19	62	38
High Income, High Wealth	50	50	57	43

Notes: Table reports the five-year FEVD of Table 3 mapped to observable moments of consumption, income, and wealth, evaluated at the posterior mode. The top panel covers group means for the bottom 20%, 20–40%, 40–80%, and top 20% of each distribution; the bottom panel covers population shares defined by all combinations of above (high) and below median (low) consumption, income, and wealth. A blocked Cholesky orders aggregates first, attributing all common variation to aggregates; autonomous shares (Auton.) are therefore lower bounds.

I.4 Historical Decomposition

To characterize the historical drivers of distributional dynamics, we implement a historical decomposition that attributes the time path of each estimated factor to aggregate versus autonomous distributional shocks. The decomposition is computed from the state transition equation in companion form and uses the same blocked-Cholesky identification as in the FEVD, ensuring that the two shock blocks are orthogonal and that the aggregate block is ordered first.

Let ϵ_t denote the smoothed reduced-form innovations from the Kalman smoother. Because $\hat{\Omega} = \mathbb{E}[\epsilon_t \epsilon_t^\top]$ is non-diagonal, we first map ϵ_t into orthonormal structural innovations e_t via the blocked-Cholesky impact matrix. We then propagate the contribution of each "shock" block through the transition equation to obtain time-t contributions to each factor. This yields an additive decomposition of the smoothed factor path into aggregate and distributional components (plus an initial zero condition term).

Formally, given smoothed reduced-form innovations $\{\epsilon_t\}_{t=1}^T$ (e.g., from the Kalman smoother), compute structural innovations

$$e_t \;=\; P_{\text{sub}}^{-1} S^\top \epsilon_t. \tag{48}$$

Let $e_{Y,t}$ and $e_{F,t}$ denote the aggregate and distributional blocks of e_t under the aggregate-first ordering. Define the block-attributed innovations

$$\epsilon_t^{\text{agg}} \;=\; P_{[:,Y]} e_{Y,t}, \qquad \epsilon_t^{\text{dist}} \;=\; P_{[:,F]} e_{F,t}, \tag{49}$$

and propagate contribution states

$$z_t^{\text{agg}} = \Phi z_{t-1}^{\text{agg}} + \epsilon_t^{\text{agg}}, \qquad z_t^{\text{dist}} = \Phi z_{t-1}^{\text{dist}} + \epsilon_t^{\text{dist}}. \tag{50}$$

The contributions to the factors are then

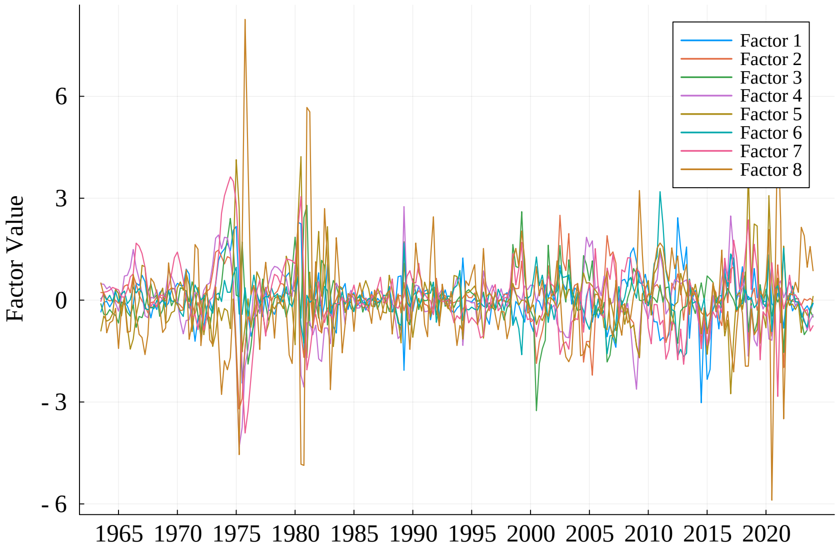
$$c_t^{\text{agg}} = J z_t^{\text{agg}}, \qquad c_t^{\text{dist}} = J z_t^{\text{dist}}, \qquad \hat{f}_t = J x_t^{\text{smoothed}}, \tag{51}$$

so that the smoothed factors admit the additive decomposition

$$\hat{f}_t = c_t^{\text{agg}} + c_t^{\text{dist}}. \tag{52}$$

Figure 18 summarizes the historical decomposition by NBER recession window (peak to trough) by reporting *absolute contribution shares*—the share of the total absolute contribution accounted for by the aggregate block versus the distributional block. Level paths of the factors are shown in Figure 17. While recessions naturally vary in duration, magnitude, and context, a clear pattern emerges: in five of the seven recession windows, contributions from the aggregate block exceed those of the distributional block, averaging 55 percent across the eight factors and windows (66 percent in 1981–82 and 73 percent in 2020; below one half only in 1973–75 and 2001). Over the full sample, by decade, the two blocks are closer to even (49 percent aggregate on average), so recessions are periods where aggregate innovations are relatively more active, though we cannot comment (yet) on whether the two channels are possibly offsetting one another.

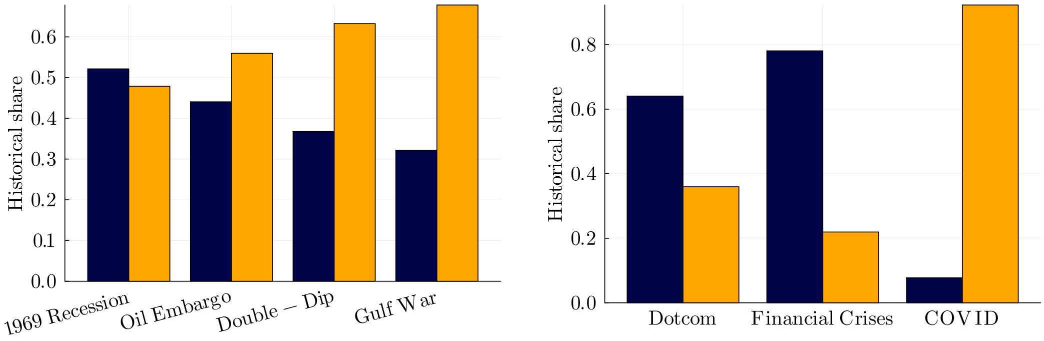
Figure 17: Distributional Factors



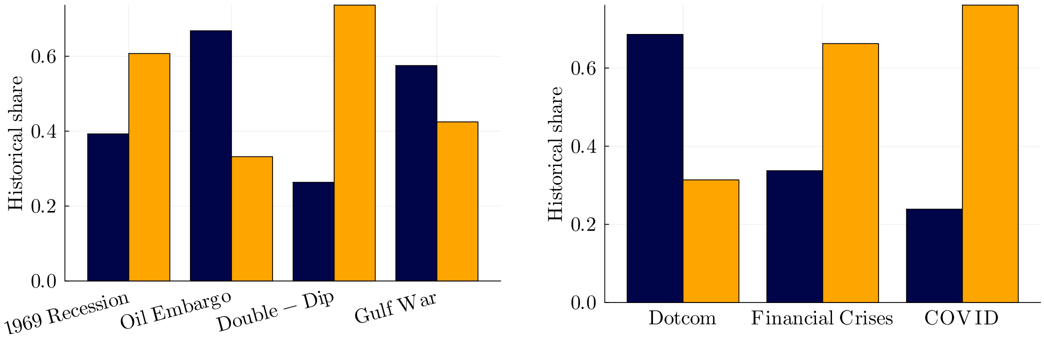
Notes: Raw factor estimates from the baseline estimation of the model. Estimates run from 1962Q3 until 2024Q1.

This is not at odds with the variance decomposition above: the FEVD measures unconditional forecast-error variance at a five-year horizon, whereas the recession windows condition on particular episodes and compare absolute contributions to the realized paths. Taken together, the two results say that autonomous innovations contribute substantially to distributional fluctuations on average, and that aggregate innovations take the larger share during recessions.

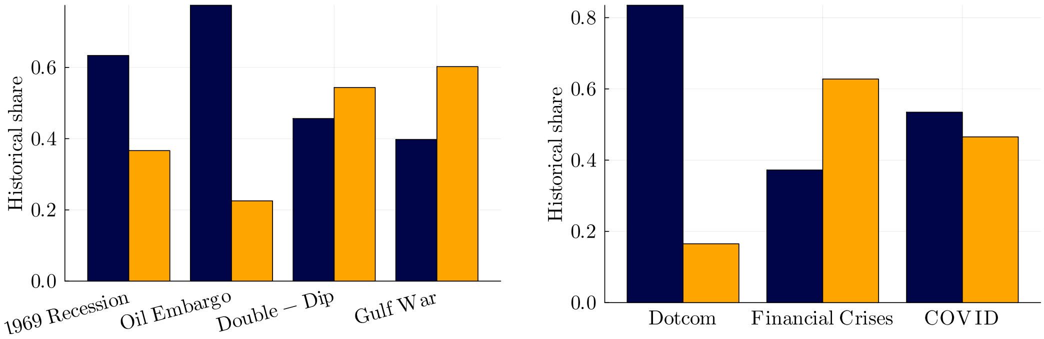
Figure 18: Historical Decomposition of Distributional Factors



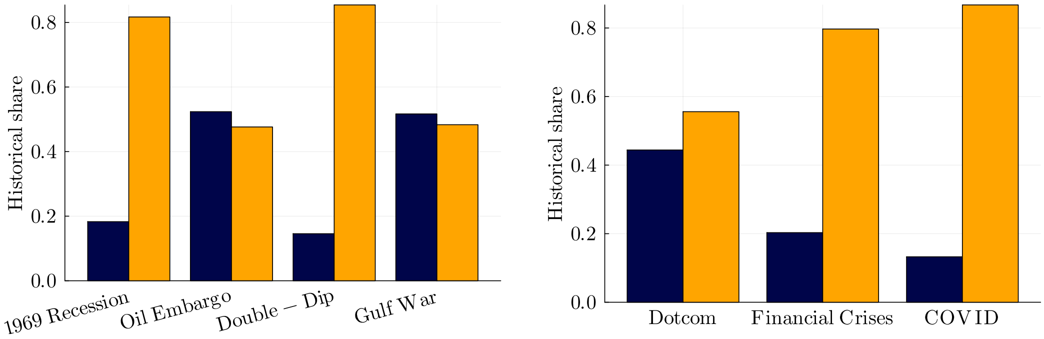
Factor #1



Factor #2

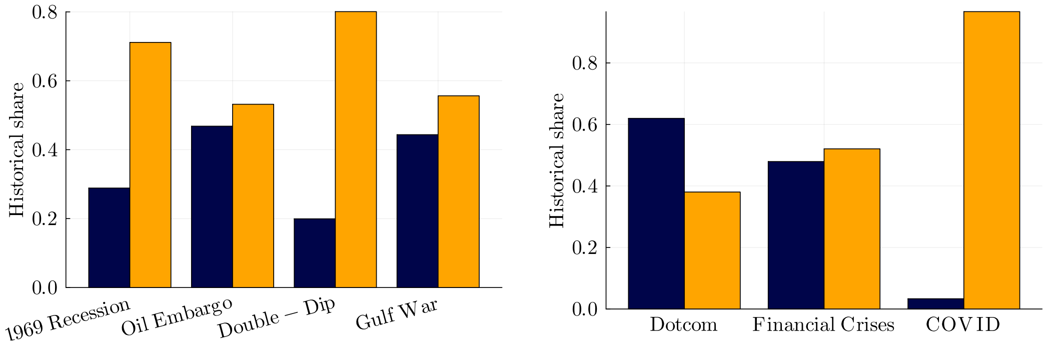


Factor #3

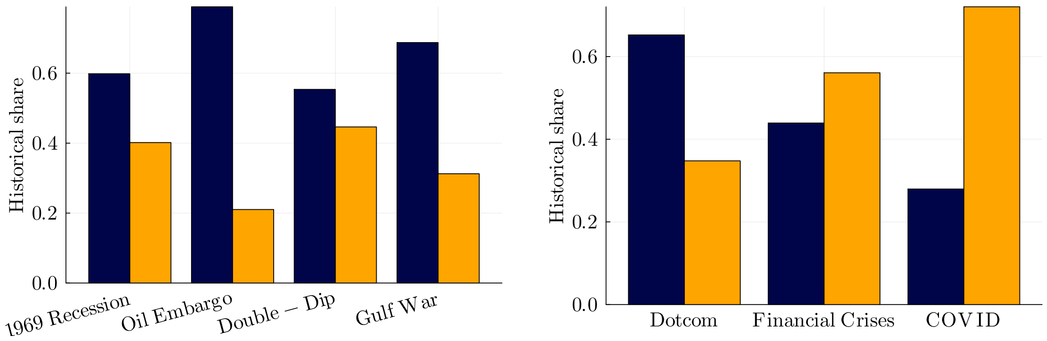


Factor #4

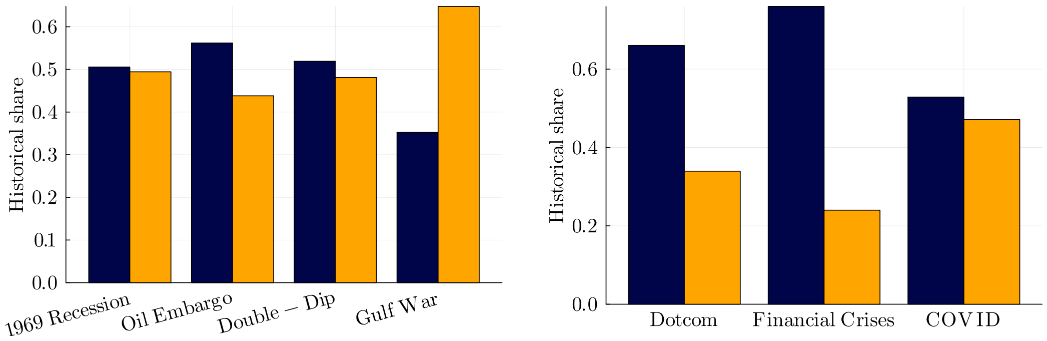
■ from Distribution 🟧 from Aggregates



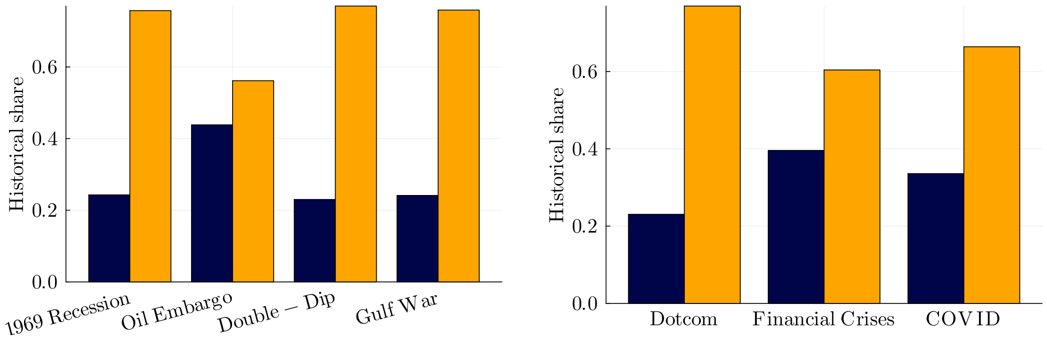
Factor #5



Factor #6



Factor #7



Factor #8

Notes: Event windows are from the official NBER recession dates, from peak to trough (1969 Unemployment Recession: 1969Q4–1970Q4, Oil Embargo: 1973Q4–1975Q1, Double-Dip Recession: 1981Q3–1982Q4, Gulf War: 1990Q3–1991Q1, Dot-com Recession: 2001Q1–2001Q4, Global Financial Crisis: 2007Q4–2009Q2, and COVID-19: 2019Q4–2020Q2).

J Predicting HANK Dynamics: Simulation Procedure

We simulate one economy for $T = 2200$ periods (quarters). For each period t, we buffet the HANK model economy with a series of aggregate shocks, whose effects propagate through linearized policy functions. From the resulting joint distribution, we generate **five** samples: A, B, C, D and E. Each sample is drawn from the copula histogram, ensuring that the samples preserve the correlational structure between economic measures. Sampling is stratified along each dimension to ensure coverage of the marginal distributions. Each simulated household defines a unit of observation and is characterized by its consumption, income, and wealth and a sampling weight. We abstract from unit-level measurement error.

Samples A, B, C, D and E are designed to mimic one dataset from the U.S., each representative, but with differences in their degree of missingness and precision. Dataset A can be considered a modern PSID-like dataset: 9,000 household observations per survey, containing consumption, income, and wealth, and released every two years. Dataset B can be considered CPS-like: 60,000 observations per survey (very precise), with only income, released every year. Dataset C can be considered a CEX-like dataset: 6,000 observations per survey, with income and consumption, released every quarter. Dataset D can be considered an SCF-like dataset: 5,000 observations per survey, with only wealth and income observed, released every three years. Dataset E can be considered a SIPP-like dataset: 25,000 observations per survey, with income observed every quarter, and wealth measured only in the fourth quarter of each year. We abstract from differences in operationalization across the different sample measurements.

We also record the structural shocks driving the simulated economy and extract principal components from their time series. The resulting factors provide the aggregate information used in the state-space model.[56] We discard the first 1,200 periods as burn-in to allow the simulated economy to converge toward its stationary distribution and reduce the influence of initial conditions. We divide the remaining 1,000 periods into 10 non-overlapping samples of 100 quarters each. This allows us to conduct 10 separate estimation exercises, each covering 25 years (100 quarters), comparable to the time span of the PSID consumption data. For these exercises, our estimation method selects five distributional factors and eight aggregate factors.

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[56] The HANK economy is subject to shocks to TFP, investment-specific technology, consumption wedges, labor wedges, monetary policy, government spending, fiscal policy, and household preferences. For a full description of the model, see Bayer, Born, and Luetticke (2024) and the associated `BASEtoolbox.jl` code repository.

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